Class 9
MATHEMATICS
Solutions

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1. QUADRATIC EQUATIONS

WORKSHEET – 1
Level – I

1. B

$15x^2 + 14x - 8 = 0 \Rightarrow 15x^2 + 20x - 6x - 8 = 0 \Rightarrow 5x(3x + 4) - 2(3x + 4) = 0$

$(5x - 2)(3x + 4) = 0 \Rightarrow x = \frac{2}{5}$ or $x = -\frac{4}{3}$

2. A

$ix^2 - 2(1 + i)x + (2 - i) = 0$

given $\alpha = 2 - i\ \beta = ?$

$\alpha \cdot \beta = \frac{2-i}{i} \Rightarrow \alpha \cdot (\beta) = \frac{(2-i)}{i} \times \frac{i}{i} \Rightarrow (2 - i)\beta = (2 - i) \times \frac{i}{-1} \Rightarrow \beta = -i$

3. B

$3x^2 + 4x - 7 = 0 \Rightarrow a = 3\ b = 4\ c = -7$

$\Delta = b^2 - 4ac = 16 + 84 = 100 \Rightarrow \Delta > 0$ and perfect square

$\therefore$ Roots are rational and not equal

4. C

$16x^2 + kx + 1 = 0$ has equal roots $\Rightarrow \Delta = 0 \Rightarrow k^2 - 64 = 0 \Rightarrow k = \pm 8$

5. C

$2x^2 + (k - 3)x + 3k - 5 = 0 \ \because \alpha + \beta = \alpha\beta$

$\Rightarrow \frac{-b}{a} = \frac{c}{a} \Rightarrow b + c = 0 \Rightarrow \mathrm{k} - 3 + 3\mathrm{k} - 5 = 0 \Rightarrow 4\mathrm{k} - 8 = 0 \Rightarrow \mathrm{k} = 2$

6. D

$\alpha + \beta = -1\ \alpha\beta = \frac{1}{4}$

$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \Rightarrow x^2 - (-1)x + \frac{1}{4} = 0 \Rightarrow 4x^2 + 4x + 1 = 0$

7. B

$\alpha = 2 + 3i; x^2 + ax + b = 0$

If $\alpha = 2 + 3i$ then $\beta = 2 - 3i\ (\because a, b \in R)$

$\alpha + \beta = 4 \Rightarrow -a = 4 \Rightarrow a = -4$

$\alpha \cdot \beta = 4 + 9 = 13 \Rightarrow b = 13$

$(\mathrm{a}, \mathrm{b}) = (-4,13)$

8. C

$\alpha + \beta = -k \Rightarrow \alpha - \beta = 1 \Rightarrow 2\alpha = 1 - k$

$\alpha = \frac{1-k}{2} \Rightarrow \left(\frac{1-k}{2}\right)^2 + k\frac{(1-k)}{2} + 2 = 0 \Rightarrow 1 + k^2 - 2k + 2(k - k^2) + 4(2) = 0$

$\Rightarrow 1 + k^2 - 2k + 2k - 2k^2 + 8 = 0 \Rightarrow 9 - k^2 = 0 \Rightarrow k = \pm 3$

Level – II

1. C	2. C	3. D	4. D	5. C	6. D
7. D	8. A	9. C	10. A	11. B	12. C
13. A	14. C	15. 2	16. A	17. A	18. B
19. B	20. A	21.D			

Level – III

22. B	23. A	24. B	25. C
26. D	27. a-t; b-q; c-s; d-t		28. 3
29. ABC	30. a-r; b-p; c-s; d-q		31. A
32. B	33. C		

SOLUTIONS

Level – II

1. $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0 \Rightarrow \sqrt{3}x^2 + 12x - 2x - 8\sqrt{3} = 0$

$\Rightarrow \sqrt{3}x(x + 4\sqrt{3}) - 2(x + 4\sqrt{3}) = 0 \Rightarrow (\sqrt{3}x - 2)(x + 4\sqrt{3}) = 0$

$\Rightarrow \sqrt{3}x - 2 = 0$

or

$x + 4\sqrt{3} = 0$

$\Rightarrow x = \dfrac{2}{\sqrt{3}}$

or

$x = -4\sqrt{3}$

2. α, β are roots of $ax^2 + bx + c = 0$

$\therefore \alpha + \beta = -\dfrac{b}{a}\ \alpha\beta = \dfrac{c}{a}$

Now $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2+\beta^2}{\alpha^2\beta^2} = \frac{(\alpha+\beta)^2-2\alpha\beta}{(\alpha\beta)^2} = \frac{b^2-2ac}{c^2}$

3. $\alpha \cdot \beta = 1:3$ then $(1)(3)b^2 = (1+3)b^2 = (1+3)^2ac \Rightarrow 3b^2 = 16ac$

4. Let α, β be the roots are given equation.

Given that, $\beta = \alpha^2$

$\alpha + \alpha^2 = -1 \Rightarrow \alpha^2 + \alpha + 1 = 0 \Rightarrow \alpha = \omega($ or $)\omega^2$ and $\alpha^3 = -k \Rightarrow k = -1.$

5. α, β are roots of $3x^2 - 5x + 7 = 0$

$\alpha + \beta = \frac{5}{3}\ \alpha\beta = \frac{7}{3}$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta\ = \frac{25}{9} - \frac{14}{3} = -\frac{17}{9}$

6. α, β are roots of $ax^2 - ax + b = 0$ then $\alpha + \beta = 1\ \alpha\beta = \frac{b}{a}$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{1 - \frac{2b}{a}}{\frac{b}{a}} = \frac{a}{b} - 2$$

7. α, β are roots of $x^2 - p(x + 1) + c = 0 \Rightarrow x^2 - px - p + c = 0$

$\alpha + \beta = p\ \alpha\beta = -p + c$

$(1 + \alpha)(1 + \beta) = 1 + (\alpha + \beta) + \alpha\beta = 1 + \text{p} - \text{p} + \text{c} = 1 + \text{c}$

8. $mnb^2 = (m + n)^2 ac$

9. $\alpha : \beta = 1 : 6 \Rightarrow (1)(6)(-14)^2 = (1 + 6)^2 \times p \times 8 \Rightarrow p = \frac{6 \times 14 \times 14}{7 \times 7 \times 8} \Rightarrow \text{p} = 3$

10. α, β are roots of $ax^2 + bx + c = 0$

$\alpha + \beta = -\frac{b}{a}\ \alpha\beta = \frac{c}{a} \Rightarrow \alpha\beta^2 + \alpha^2\beta + \alpha\beta = \alpha\beta(\alpha + \beta) + \alpha\beta$

$= \frac{c}{a}\left(-\frac{b}{a}\right) + \frac{c}{a} = \frac{c}{a}\left(-\frac{b}{a} + 1\right) = \frac{ac - bc}{a^2}$

11. α, β are roots of $ax^2 + bx + c = 0 \Rightarrow \frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = \frac{\alpha^3 + \beta^3}{\frac{1}{\alpha^3} + \frac{1}{\beta^3}} = (\alpha^3 + \beta^3)\frac{(\alpha^3\beta^3)}{(\alpha^3 + \beta^3)} = (\alpha\beta)^3 =$

$\frac{c^3}{a^3}$

12. α, β are roots of $x^2 - px + q = 0$

$$\frac{\alpha^{-3} + \beta^3}{\alpha^3 + \beta^3} = \frac{\frac{1}{\alpha^3} + \frac{1}{\beta^3}}{\alpha^3 + \beta^3} = \frac{\alpha^3 + \beta^3}{\alpha^3\beta^3} \cdot \frac{1}{\alpha^3 + \beta^3} = \frac{1}{(\alpha\beta)^3} = \frac{1}{q^3}$$

13. $|\alpha - \beta| = 5 \Rightarrow (\alpha - \beta)^2 = 25 \Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = 25$

$\Rightarrow \frac{b^2}{a^2} - \frac{4c}{a} = 25 \Rightarrow b^2 - 4ac = 25a^2 \Rightarrow b^2 - 25a^2 = 4ac$

14. $x^2 + px + q = 0$ a = 1 b = p c = q

$b^3 + a^2c + ac^2 = 3abc$

$\Rightarrow p^3 + q + q^2 = 3pq\ \Rightarrow p^3 + q^2 = 3pq - q \Rightarrow p^3 + q^2 = q(3p - 1)$

15. $x^2 + px - 3 = 0 \Rightarrow \alpha^2 + \beta^2 = 10$

$(\alpha + \beta)^2 - 2\alpha\beta = 10 \Rightarrow p^2 - 2(-3) = 10 \Rightarrow p^2 = 4\ \therefore p = \pm 2$

16. $(x - 1)^2 - 5(x - 1) + 6 = 0 \Rightarrow x^2 - 2x + 1 - 5x + 5 + 6 = 0 \Rightarrow x^2 - 7x + 12 = 0$

17. $2(x+1)^2-3(x+1)+1=0$

$\Rightarrow 2(x^2+2x+1)-3x-3+1=0 \Rightarrow 2x^2+4x+2-3x-3+1=0 \Rightarrow 2x^2+x=0$

18. $2\left(\frac{x}{3}\right)^2-5\left(\frac{x}{3}\right)+6=0 \Rightarrow \frac{2x^2}{9}-\frac{5x}{3}+6=0 \Rightarrow 2x^2-15x+54=0$

19. α,β are roots of $ax^2+bx+c=0$

The Q.E whose roots is $\alpha+h,\beta+h$ is $a(x-h)^2+b(x-h)+c=0$

$\Rightarrow a(x^2-2hx+h^2)+bx-bh+c=0 \Rightarrow ax^2+(b-2ah)x+ah^2-bh+c=0$

Compared with $px^2+qx+r=0$

$\mathrm{a=p\ q=b-2ah\ c}=ah^2-bh+c=0$

$\Rightarrow \mathrm{q=b-2ph} \Rightarrow \mathrm{b-q=2ph} \Rightarrow h=\frac{1}{2}\left(\frac{b-q}{p}\right)$

$=\frac{1}{2}\left(\frac{b}{p}-\frac{q}{p}\right)=\frac{1}{2}\left(\frac{b}{a}-\frac{q}{p}\right)$

$(\because a=p)$

20. $2\left(\frac{1-x}{1+x}\right)^2+\frac{1-x}{1+x}+3=0$

$\Rightarrow 2(1-2x+x^2)+(1-x)(1+x)+3(1+2x+x^2)=0$

$\Rightarrow 4\mathrm{x}^2+2\mathrm{x}+6=0 \Rightarrow 2\mathrm{x}^2+\mathrm{x}+3=0$

21. roots of $x^2+x+1=0$ are ω,ω^2

$\alpha=\omega,\beta=\omega^2$ then $\alpha^{19}=\omega^{19}=\omega\,\beta^7=(\omega^2)^7=\omega^2$

$\therefore$ required equation is $x^2+x+1=0$

Level – III

22. $(x+a)(x+b)-8k=(k-2)^2$

$x^2+(a+b)x+ab-8k-(k-2)^2=0$

$\Delta=0$

$\Rightarrow (\mathrm{a+b})^2=4(\mathrm{ab})-32\mathrm{k}-4(\mathrm{k-2})^2=0 \Rightarrow (a+b)^2-4ab=-4[(k-2)^2+8k]$

$\Rightarrow (a-b)^2=-4(k+2)^2=0 \Rightarrow (a-b)^2+4(k+2^2)=0$

It is possible whess $a=b$ and $k=2$

23. $x^2-(-\sqrt{3}+\sqrt{3})x+(\sqrt{3})(-\sqrt{3})=0 \Rightarrow x^2-3=0$

Ans:1; both the statements are true

24. $x-2=\sqrt{x} \Rightarrow x^2-4x+4=x \Rightarrow x^2-5x+4=0 \Rightarrow (x-1)(x-4)=0$

$\mathrm{x=1,4}$

25. $x^2=6+x \Rightarrow x^2-x-6=0 \Rightarrow (x-3)(x+2)=0$

$\Rightarrow x = 3\ x = -2$ (not possible)

26. $x^2 = 3 - x$

$x^2 + x - 3 = 0$

$x = \frac{-1\pm\sqrt{1+12}}{2} = \frac{-1\pm\sqrt{13}}{2} \therefore x = \frac{-1+\sqrt{13}}{2}$ (Since x is positive)

27. $x^2 + x + 2 = 0, \alpha, \beta$ are roots

a) $\alpha + \beta = -1$

b) $\alpha\beta = 2$

c) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = (-1)^2 - 2(2) = 1 - 4 = -3$

d) $\alpha^2 + \beta^2 + \alpha\beta = -3 + 2 = -1$

28. $5x^2 - 4x + 38 + k(-4x^2 - 2x - 1) = 0 \Rightarrow x^2(5-4k) - x(4+2k) + (38-k) = 0$

$\alpha\beta = -5 \Rightarrow \frac{c}{a} = -5 \Rightarrow \frac{38-k}{5-4k} = -5$

$38 - k = -25 + 20k \Rightarrow 38 + 25 = 21k \Rightarrow 21k = 63 \Rightarrow k = 3$

29. Numericalls equal but opposite is sign then $f(-x) = 0$

$\Rightarrow 3(-x)^2 - 5(-x) - 7 = 0 \Rightarrow 3x^2 + 5x - 7 = 0$

$ax^2 + bx + c = 0$

$a = 3, b = 5, c = -7$

30. α, β are roots of $ax^2 + bx + c = 0 \ \therefore \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

a) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta} = \left(\frac{b^2}{a^2} - \frac{2c}{a}\right)\cdot\frac{a}{c} = \frac{b^2-2ac}{ac}$

b) $\frac{\alpha^2+\beta^2}{\alpha^{-2}+\beta^{-2}} = \frac{(\alpha^2+\beta^2)}{\left(\frac{1}{\alpha^2}+\frac{1}{\beta^2}\right)} = (\alpha^2+\beta^2)\left(\frac{\alpha^2\beta^2}{\alpha^2+\beta^2}\right) = (\alpha\beta)^2 = \frac{c^2}{a^2}$

c) $\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = \frac{-b^3}{a^3} - \frac{3c}{a}\left(\frac{-b}{a}\right) = \frac{3bc}{a^2} - \frac{b^3}{a^3} = \frac{3abc-b^3}{a^3}$

d) $\alpha^5\beta^8 + \alpha^8\beta^5 = \alpha^5\beta^5(\alpha^3+\beta^3) = \frac{c^5}{a^5}\left(\frac{3abc-b^3}{a^3}\right) = \frac{c^5(3abc-b^3)}{a^8}$

31. $\alpha, \beta \Rightarrow ax^2 + bx + c = 0$ then $\alpha + \beta = \frac{-b}{a}; \alpha\beta = \frac{c}{a}$

$$\frac{1}{a\alpha+b} + \frac{1}{a\beta+b} = \frac{a\beta+b+a\alpha+b}{a^2\alpha\beta+ab(\alpha+\beta)+b^2} = \frac{a(\alpha+\beta)+2b}{a^2\alpha\beta+ab(\alpha+\beta)+b^2}$$

$$= \frac{a\left(\frac{-b}{a}\right)+2b}{a^2\left(\frac{c}{a}\right)+ab\left(-\frac{b}{a}\right)+b^2} = \frac{-b+2b}{ac-b^2+b^2} = \frac{b}{ac}$$

32. $\alpha, \beta \to x^2 + 5x + 16 = 0 \Rightarrow \alpha + \beta = -5 \alpha\beta = 16$

$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 25 - 2(16) = 25 - 32 = -7 \Rightarrow \frac{\alpha\beta}{2} = \frac{16}{2} = 8$

Q.E is $x^2 - (\alpha^2+\beta^2+\alpha\beta)x + (\alpha^2+\beta^2)\alpha\beta = 0$

$\Rightarrow x^2 - (-7+8)x + (-7)(8) = 0 \Rightarrow x^2 - x - 56 = 0$

compared with $x^2 - px + q = 0 \ \therefore p = -1q = -56$

33. $px^2 + qx + 1 = 0$ having real roots $\Rightarrow q^2 - 4p \geq 0 \Rightarrow q^2 \geq 4p$

but $p, q \in \{1,2,3,4\}$

$2^2 \geq 4(1)$ (1,2) $3^2 \geq 4(2)$ (2,3) $4^2 \geq 4(3)$ $4^2 \geq 4(4)$

$3^2 \geq 4(1)$ (1,3) $4^2 \geq 4(2)$ (2,4) (3,4) (4,4)

$4^2 \geq 4(1)$ (1,4)

No. of possible equations are seven

WORKSHEET – 2

Level – I

1. A	2. A, C	3. B	4. B	5. B
6. B	7. D	8. B	9. 3	10. D

Level – II

1. C	2. A	3. D	4. B	5. A
6. B	7. C	8. 35	9. A	10. B
11. A	12. A, D	13. A	14. A	15. C
16. D	17. A	18. B	19. A	20. B
21. C	22. A	23. 2		

Level – III

24. C	25. C	26. A	27. B	28. A
29. a-t; b-p; c-q; d-r		30. 2	31. A, D	
32. a-t; b-p; c-q; d-r		33. A, C		

SOLUTIONS

Level – I

1. $(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$.

2. Coomon root $= \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$ or $\frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}$.

3. $(6 \times 2 - k \times 1)^2 = (-10 + 10)(-5k + 60) \Rightarrow (12 - k)^2 = 0(-5k + 60) \Rightarrow 12 - k = 0 \Rightarrow k = 12$

4. $b^2 - 4ac < 0$

5. $4 < x < 8 \Rightarrow (x - 4)(x - 8) < 0 \Rightarrow x^2 - 12x + 32 < 0 \Rightarrow 12x - x^2 - 32 > 0$.

6. $x^2 - 6x + 10$

$a = 1 > 0 \ \Delta = 6^2 - 4(10) = 36 - 40 = -4 < 0$

$\therefore \forall x \in R, \ x^2 - 6x + 10 > 0$.

7. $10x - 5x^2 - 1 = -5x^2 + 10x - 1$

$a = -5 < 0, b = 10, c = -1$

Maximum value $= \frac{4ac-b^2}{4a} = \frac{4(-5)(-1)-(10)}{4(-5)} = \frac{20-100}{-20} = \frac{-80}{-20} = 4.$

8. $x^2 - 5x + 6$

$a = 1 > 0, b = -5, c = 6$; Minimum value $= \frac{4(1)(6)-(-5)^2}{4(1)} = \frac{24-25}{4} = \frac{-1}{4}.$

9. $x^2 - 8x + 17$

$a = 1 > 0, b = -8, c = 17$; Minimum value $= \frac{4(1)(17)-(-8)^2}{4(1)} = \frac{68-64}{4} = \frac{4}{4} = 1.$

10. $\left(x - \frac{5}{3}\right)^2 + \frac{7}{2}$ Minimum value $= \frac{7}{2}.$

Level – II

1. $x^2 - 11x + a = 0 \Rightarrow x^2 - 14x + 2a = 0 \Rightarrow (a - 2a)^2 = (-14 + 11)(-22a + 14a)$

$a^2 = (-3)(-8a) \Rightarrow a^2 - 24a = 0 \Rightarrow a(a - 24) = 0 \Rightarrow a = 0$ or $a = 24.$

2. $x^2 + bx + c = 0 \Rightarrow x^2 + cx + b = 0$

let the common root be " k "

$\Rightarrow f(k) = g(x) \Rightarrow k^2 + bk + c = k^2 + ck + b \Rightarrow (b - c)k = (b - c)$

$\Rightarrow k = 1 \Rightarrow (1)^2 + b(1) + c = 0 \Rightarrow b + c = -1.$

3. $x^2 + ax + bc = 0 \Rightarrow x^2 + bx + ca = 0$

let the common root be " k " $\Rightarrow k^2 + ak + bc = k^2 + bk + ca$

$\Rightarrow (a - b)k = c(a - b) \Rightarrow k = c \Rightarrow c^2 + ac + bc = 0$

$\Rightarrow c(c + a + b) = 0 \Rightarrow a + b + c = 0.$

4. $x - 1$ is a common factor to $x^2 + ax + b \cdot x^2 + cx + d$

$\Rightarrow (1)^2 + a(1) + b = (1)^2 + c(1) + d \Rightarrow a + b = c + d \Rightarrow \frac{a + b}{d + c} = 1$

5. $(x - a)(b - x) = bx - x^2 - ab + ax = -x^2 + (b + a)x - ab$

Since co-efficient x^2 is $-1 < 0$

$\therefore$ maximum value $= \frac{4(-1)(-ab)-(b+a)^2}{4(-1)} = \frac{4ab-(a+b)^2}{-4}$

$= \frac{(a+b)^2-4ab}{4} = \frac{(a-b)^2}{4}.$

6. $3x - 5x^2 + 12$

$a = -5 < 0$, maximum value

having at $x = \frac{-b}{2a}. = \frac{-3}{2(-5)} = \frac{-3}{-10} = \frac{3}{10}$

7. $x^2 + ax + b, x^2 + cx + d$ have common factor $x - 1$

$\Rightarrow (1)^2 + a(1) + b = (1)^2 + c(1) + d \Rightarrow a + b - c - d = 0.$

8. $x^2 - 6x + 5 = 0, x^2 - 12x + p = 0$ have a common root

$5 \quad 1 \quad -6 \quad 5$
$p \quad 1 \quad -12 \quad p$

$(5 - p)^2 = (-12 + 6)(-6p + 60)$

$(p - 5)^2 = (-6)(p - 10)(-6)$

$p^2 - 10p + 25 = 36p - 360$

$p^2 - 46p + 385 = 0$

$(p - 11)(p - 35) = 0 \Rightarrow p = 11$ or $p = 35.$

9. $ax^2 + 2cx + b = 0$

$ax^2 + 2bx + c = 0$ have a common root.

$$\Rightarrow \begin{matrix} b & a & 2c & b \\ c & a & 2b & c \end{matrix}$$

$(ab - ac)^2 = (2ab - 2ac)(2c^2 - 2b^2)$
$a^2(b - c)^2 = 2a(b - c)2(c - b)(c + b)$
$a = -4(b + c)$
$\Rightarrow b + c = \frac{-a}{4}.$

10. $c + 2bx - x^2 = -x^2 + 2bx + c$

co-efficient of x^2 is $-1 < 0$

$\therefore$ maximum value $= \frac{4(-1)(c)-(2b)^2}{4(-1)} = \frac{-4c-4b^2}{-4} = \frac{4(c+b^2)}{-4} = c + b^2.$

11. $x + \frac{1}{x}$ $AM \geq GM \Rightarrow \frac{x+\frac{1}{x}}{2} \geq \sqrt{x \cdot \frac{1}{x}} \geq 1 \Rightarrow x + \frac{1}{x} \geq 2.$

12. $\frac{\cos^2 x + \sec^2 x}{2} \geq \sqrt{\cos^2 x \cdot \sec^2 x} \Rightarrow \cos^2 x + \sec^2 x \geq 2.$

13. $2x^2 + 4x + 7$

minimum value $= \frac{4ac-b^2}{4a} = \frac{4(2)(7)-4^2}{4(2)} = \frac{56-16}{8} = \frac{40}{8} = 5$

minimum value has at $\frac{-b}{2a} = \frac{-4}{2(2)} = \frac{-4}{4} = -1$

$\therefore$ m $= 5$ $\alpha = -1.$

14. $x < 2$ or $x > 3 \Rightarrow (x - 2)(x - 3) > 0 \Rightarrow x^2 - 5x + 6 > 0.$

15. $-\frac{1}{2} < x < 3 \Rightarrow \left(x + \frac{1}{2}\right)(x - 3) < 0$

$\Rightarrow x^2 - 3x + \frac{x}{2} - \frac{3}{2} < 0 \Rightarrow 2x^2 - 6x + x - 3 < 0 \Rightarrow 2x^2 - 5x - 3 < 0.$

16. $x^2 + px + q = 0$ and $x^2 - ax + b = 0$ have common root

$$\Rightarrow \begin{matrix} q & 1 & p & r \\ b & 1 & -a & b \end{matrix}$$

$\Rightarrow (q-b)^2 = (-a-p)(bp+aq) \Rightarrow q^2+b^2-2bq = -abp-a^2q-bp^2-apq$

$x^2-ax+b=0$ has equal roots

so $a^2 = 4b$.

$$q^2+b^2-2bq = -apb-(4b)q-\left(\frac{a^2}{4}\right)p^2-abq$$

$$\Rightarrow q^2+b^2-2bq+apb+4bq+\frac{a^2p^2}{4}+abq=0$$

$$\Rightarrow q^2+b^2+2bq+abp+\frac{a^2p^2}{4}+abq=0 \Rightarrow \left(b+q+\frac{ap}{2}\right)^2=0 \Rightarrow b+q+\frac{ap}{2}=0$$

$\Rightarrow 2(b+q) = -ap$

17. $3^x + 3^{1-x} - 4 < 0$ let $3^x = a$

$$\Rightarrow 3^x+\frac{3}{3^x}-4<0$$

$$\Rightarrow a+\frac{3}{a}-4<0 \Rightarrow a^2+3-4a<0 \Rightarrow a^2-4a+3<0 \Rightarrow (a-1)(a-3)<0$$

$\Rightarrow 1 < a < 3 \Rightarrow 1 < 3^x < 3 \Rightarrow 3^0 < 3^x < 3^1 \Rightarrow x \in (0,1)$.

18. $x^2-6x+a=0$

$x^2-cx+6=0$ let the common root is α.

the other roots of 1 st and 2 nd equations are β, γ

given $\beta:\gamma = 4:3$

let $\beta = 4k\ \gamma = 3k$.

$\alpha+4k=6$ $\quad\quad$ $\alpha(4k)=a$

$\alpha+3k=c$ $\quad\quad$ $\alpha(3k)=6$

$$\Rightarrow \frac{\alpha(4k)}{\alpha(3k)}=\frac{a}{6}$$

$$\Rightarrow a=\frac{4}{3}\times 6 \Rightarrow a=8$$

$3\alpha k = 6 \Rightarrow \alpha k = 2$.

$\alpha+4k=6$ $\quad$ $\alpha\dfrac{(6-\alpha)}{4}=2$ $\quad$ $\alpha+4k=6$

$\Rightarrow 4k = 6-\alpha$ $\quad$ $6\alpha-\alpha^2=8$ $\quad$ $4k=6-\alpha=6-2=4$

$\Rightarrow k=\dfrac{6-\alpha}{4}$ $\quad$ $\alpha^2-6\alpha+8=0$ $\quad$ $k=1$

$(\alpha-2)(\alpha-4)=0$ $\quad$ Required $\alpha=2$

$\alpha = 2$ or $\alpha = 4$

19. $\alpha, \beta \rightarrow x^2+ax+bc=0$

$\beta, \gamma \rightarrow x^2+bx+ca=0$

$\gamma, \alpha \rightarrow x^2+cx+ab=0$

$\alpha+\beta=-a\ \beta+\gamma=-b\ \gamma+\alpha=-c \Rightarrow \alpha+\beta+\beta+\gamma+\gamma+\alpha=-a-b-c$

$\alpha+\beta+\beta+\gamma+\gamma+\alpha=-a-b-c \Rightarrow 2(\alpha+\beta+\gamma)=-(a+b+c)$

$\alpha+\beta+\gamma=-\frac{1}{2}(a+b+c)$.

20. $x^2-6x+10$

$\Delta=36-4(10)=-4$

$a>0, \Delta<0 \ \therefore \forall x \in R, x^2-6x+10>0$.

21. $\Delta=(a+b)^2-4(a^2-ab+b^2)$

$=a^2+2ab+b^2-4a^2+4ab-4b^2=-3a^2+6ab-3b^2$

$=-3(a^2-2ab+b^2)=-3(a-b)^2$

Since $a \neq b, \Delta<0$

and $a=1>0 \ \forall x \in R, x^2-(a+b)x+(a^2-ab+b^2)>0$

So answer is $x \notin R$.

22. Let $y=\frac{x^2-x+1}{x^2+x+1} \Rightarrow (y-1)x^2+(y+1)x+(y-1)=0$

x is real $\Rightarrow (y+1)^2-4(y-1)^2 \geq 0$

$\Rightarrow -3y^2+10y-3 \geq 0 \Rightarrow 3y^2-10y+3 \leq 0 \Rightarrow (3y-1)(y-3) \leq 0 \Rightarrow \frac{1}{3} \leq y \leq 3$

$\therefore$ minimum value $=\frac{1}{3}$.

23. $y=\frac{x}{x^2-5x+9} \Rightarrow y^2-(5y+1)x+ay=0$

$\Rightarrow (5y+1)^2-4y(9y) \geq 0$ ($\because$ x is real)

$\Rightarrow -11y^2+10y+1 \geq 0 \Rightarrow 11y^2-10y-1 \leq 0 \Rightarrow (y-1)(11y+1) \leq 0 \Rightarrow \frac{-1}{11} \leq y \leq 1$

$\therefore$ maximum value $=1$.

Level – III

24. $x^2-6x+5=0; x^2-12x+p=0$ have a common root.

5	1	-6	5
p	1	-12	p

$(5-p)^2=(-12+6)(-6p+60)$

$25+p^2-10p=-6(-6p+60)$

$p^2-10p+25=36p-360$.

$p^2-10p+25-36p+360=0$

$p^2-46p+385=0$

$(p-11)(p-35)=0$

$p=11$ or $p=35$

25. s – 1; $x^2-3x+2<0 \Rightarrow (x-1)(x-2)<0 \Rightarrow 1<x<2$

s – 1 is true. $\Rightarrow$ s – 2 is true.

26. $x-x^2+1=-x^2+x+1$

$a=-1<0; b=1; c=1$ maximum at $x=\frac{-b}{2a}=\frac{-1}{2(-1)}=\frac{1}{2}$.

So, $m=\frac{1}{2}$, then $2m=1$

27. $x - x^2 + 1$

maximum value $= \frac{4ac-b^2}{4a} = \frac{4(-1)(1)-1^2}{4(-1)} = \frac{-4-1}{-4} = \frac{-5}{-4} = \frac{5}{4}$.

28. $x^2 - 5x + 6$

maximum value $= \frac{4ac-b^2}{4a} = \frac{4(1)(6)-5^2}{4(1)} = \frac{24-25}{4} = \frac{-1}{4}$.

29. a) Maximum value of $x^2 + 1$ at $x = \frac{-b}{2a} = \frac{-0}{2(1)} = 0$.

b) Minimum value of $x^2 + 1$ is 1

c) $5x - x^2 + 6$ has maximum value at $x = \frac{-b}{2a} = \frac{-5}{2(-1)} = \frac{5}{2}$

d) $5x - x^2 + 6$ maximum value $= \frac{4(-1)(6)-5^2}{4(-1)} = \frac{-24-25}{-4} = \frac{49}{4}$.

30. $x - 2$ is common factor of $x^2 + ax + b$ and $x^2 + cx + d$

$\Rightarrow 2^2 + 2a + b = 2^2 + 2c + d \Rightarrow 2(a - c) = d - b \Rightarrow 2 = \frac{b-d}{c-a}$.

31. $ax^2 + bx + c = 0, bx^2 + cx + a = 0$

ca b c

ab c a

$(bc - a^2)^2 = (ac - b^2)(ab - c^2)$

$b^2c^2 + a^4 - 2a^2bc = a^2bc - ac^3 - ab^3 + b^2c^2$

$a^3 - 2abc = abc - c^3 - b^3$

$a^3 + b^3 + c^3 = 3abc$

$\Rightarrow a + b + c = 0$

32. a)

$x^2 - 4x + 3$ $a = 1 > 0$

minimum $= \frac{4(1)(3) - (-4^2)}{4(1)} = \frac{12 - 16}{4} = \frac{-4}{4} = -1$

b) $6x - x^2 + 2$

$a = -1 < 0$

maximum $= \frac{4(-1)(27)-6^2}{4(-1)} = \frac{-108-36}{-4} = \frac{144}{4} = 36$.

c) $3x - x^2 + 4$

$a = -1 < 0$

maximum $= \frac{4(-1)(4)-3^2}{4(-1)} = \frac{-16-9}{-4} = \frac{25}{4}$.

d) $x^2 - 3x + 2$ $a = 1 > 0$

minimum $= \frac{4(1)(2)-(-3)^2}{4(1)} = \frac{8-9}{4} = \frac{-1}{4}$.

33. $(2a-5)x^2-4x-15=0$

$(3a-8)x^2-5x-21=0$ have a common root

$\Rightarrow -15 \quad 2a-5 \quad -4 \quad -15$
$\quad -21 \quad 3a-8 \quad -5 \quad -21$

$(-45a+120+42a-105)^2=[(-10a+25)+(12a-32)](84-75)$

$(3a+15)^2=(2a-7)(9)\Rightarrow 9(a-5)^2=9(2a-7)\Rightarrow a^2-12a+32=0$

$(a-4)(a-8)=0\Rightarrow a=4$ or $a=8$

WORKSHEET – 3

Level – I

1. A	2. B	3. A	4. A	5. D
6. B	7. B	8. B	9. A	10. A

Level - II

1. C	2. B	3. A	4. D	5. B
6. D	7. B	8. B	9. A	10. B
11. A	12. C	13. A	14. B	15. D
16. A	17. B	18. C	19. B	

Level – III

20. C, D	21. A	22. C	23. B	24. C
25. a-q; b-r; c-s; d-p		26. 5	27. D	28. B
29. B	30. A, B, C	31. a-s; b-p; c-q; d-r		

SOLUTIONS

Level – I

1. $x^2-3x+2<0$

$\Delta=(-3)^2-4(1)(2)=9-8=1>0$

$\Rightarrow (x-1)(x-2)<0\Rightarrow 1<x<2.$

2. $x^2-5x+6>0$

$\Delta=(-5)^2-4(1)(6)=25-24=1>0$

$\Rightarrow (x-2)(x-3)>0\Rightarrow x<2$ or $x>3.$

3. $x^2-x-6\leq 0$

$\Delta=(-1)^2-4(1)(-6)=1+24=25>0$

$\Rightarrow (x-3)(x+2)\leq 0\Rightarrow -2\leq x\leq 3$

$\therefore x\in[-2,3].$

4. $x^2 + x - 6 \geq 0\ \Delta = 1^2 - 4(1)(-6) = 1 + 24 = 25 > 0$

$\Rightarrow (x + 3)(x - 2) \geq 0 \Rightarrow x \leq -3$ or $x \geq 2\ \therefore x \in (-\infty, -3] \cup [2, \infty)$.

5. $a + (k) < 0 \Rightarrow a^2k^2 + abk + ac < 0$.

6. (b) Open upward parabola above the x-axis.

7. $33 - x(2 + 3x) > 0 \Rightarrow 33 - 2x - 3x^2 > 0$

$\Rightarrow 3x^2 + 2x - 33 < 0 \Rightarrow 3x^2 + 11x - 9x - 33 < 0$

$\Rightarrow x(3x + 11) - 3(3x + 11) < 0 \Rightarrow (x - 3)(3x + 11) < 0 \Rightarrow \frac{-11}{3} < x < 3$

least integral value $= -3$.

8. $4x - 5x^2 + 1 \geq 0$

$\Rightarrow 5x^2 - 4x - 1 \leq 0 \Rightarrow 5x^2 - 5x + 1x - 1 \leq 0$

$\Rightarrow 5x(x - 1) + 1(x - 1) \leq 0 \Rightarrow (x - 1)(5x + 1) \leq 0 \Rightarrow \frac{-1}{5} \leq x \leq 1$

highest value $= 1$.

9. $x^2 - 5x - 6 < 0$

$\Rightarrow (x - 6)(x + 1) < 0 \Rightarrow -1 < x < 6\ \therefore x \in (-1,6)$.

10. $-x^2 + 3x + 4 > 0 \Rightarrow x^2 - 3x - 4 < 0$

$\Rightarrow (x - 4)(x + 1) < 0 \Rightarrow -1 < x < 4\ \therefore x \in (-1,4)$.

Level – II

1. $\frac{x-3}{x-2} \geq 0 \Rightarrow (x - 3)(x - 2) \geq 0 \Rightarrow x \leq 2$ or $x \geq 3$

2. Each one of the roots of $x^2 - 4ax + 2a^2 - 3a + 5 = 0$ is greater than 2

Hence, 1) $D \geq 0 \Rightarrow 2a^2 + 3a - 5 \geq 0 \Rightarrow a \leq \frac{-5}{2}$ or $a \geq 1$

2. af $(k) > 0 \Rightarrow 4 - 8a + 2a^2 - 3a + 5 > 0\ \Rightarrow 2a^2 - 11a + 9 > 0$

$\Rightarrow (a - 1)(2a - a) > 0 \Rightarrow a < 1$ or $a > \frac{9}{2}$

3. $\frac{-b}{2a} < k \Rightarrow \frac{4a}{2} < 2 \Rightarrow a < 1$

From the equations (1),(2) and (3), we get the solution set as

$a > \frac{9}{2}$ or $a \in \left(\frac{9}{2}, \infty\right)$

3. Roots of $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3 .

Hence 1) $D \geq 0 \Rightarrow 4a^2 - 4(a^2 + a - 3) \geq 0$

$\Rightarrow a - 3 \leq 0 \Rightarrow a \leq 3$

2. $af(k) > 0 \Rightarrow 9 - 6a + a^2 + a - 3 > 0 \Rightarrow a^2 - 5a + 6 > 0$

$\Rightarrow (a - 2)(a - 3) > 0 \Rightarrow a < 2$ or $a > 3$

3. $\frac{-b}{2a} < k$

$\Rightarrow \frac{2a}{2} < 3 \Rightarrow a < 3$

From the above three equations we get the solution set as $a < 2$ or $a \in (-\infty, 2)$.

4. $x^2 + a^2 = 8x + 6a \Rightarrow x^2 - 8x + a^2 - 6a = 0$

Roots are real $\Delta \geq 0$

$\Rightarrow 64 - 4(1)(a^2 - 6a) \geq 0 \Rightarrow a^2 - 6a - 16 \leq 0 \Rightarrow a^2 - 8a + 2a - 16 \leq 0$

$\Rightarrow a(a-8) + 2(a-8) \leq 0 \Rightarrow (a+2)(a-8) \leq 0 \Rightarrow -2 \leq a \leq 8.$

5. Let $y = \frac{x^2+14x+9}{x^2+2x+3} \Rightarrow x^2(y-1) + x(2y-14) + 3y - 9 = 0$

As $x \in R, 4(y-7)^2 - 12(y-1)(y-3) \geq 0 \Rightarrow y^2 + y - 20 \leq 0 \Rightarrow y \in [-5,4]$

6. $2x^2 - 2(2a+1)x + a(a+1) = 0$

One root less than a and other root greater than a clearly $\Delta > 0$ and af $(k) < 0$

$\Rightarrow [-2(2a+1)]^2 - 4 \times 2 \times a(a+1) > 0$ and $2.f(a) < 0$

$\Rightarrow (2a+1)^2 - 2a(a+1) > 0$ and $2a^2 - 2(2a+1)a + a(a+1) < 0$

$\Rightarrow a > -\frac{1}{2} \Rightarrow 2a^2 - 4a^2 - 2a + a^2 + a < 0$

$\Rightarrow -a^2 - a < 0 \Rightarrow a(a+1) > 0 \Rightarrow a < -1$ or $a > 0$.

7. $(1-a^2)x^2 + 2ax - 1 = 0$ has roots belonging to (0,1)

Clearly $\Delta \geq 0$ and $f(0)f(1) > 0$

$(-1)(1 - a^2 + 2a - 1) > 0$

$(2a)^2 - 4(1-a^2)(-1) \geq 0$

$a^2 - 2a > 0$

True for all $a \in R$

$a(a-2) > 0$

$a < 0$ or $a > 2$.

8. $x^2 - 4x + 5 > 0$

$\Delta = (-4)^2 - 4(1)(5) = 16 - 20 = -4 < 0$

Since $a = 1 > 0 \; \forall x \in R, x^2 - 4x + 5 > 0.$

9. $4x - 5x^2 + 1 > 0 \Rightarrow 5x^2 - 4x - 1 < 0$

$\Delta = (-4)^2 - 4(5)(-1) = 16 + 20 = 36 > 0$

$\Rightarrow 5x^2 - 5x + 1x - 1 < 0 \Rightarrow 5x(x-1) + 1(x-1) < 0$

$\Rightarrow (x-1)(5x+1) < 0 \Rightarrow x \in \left(\frac{-1}{5}, 1\right).$

10. $[-2(4k-1)]^2 - 4(1)(15k^2 - 2k - 7) < 0$

$\Rightarrow (4k+1)^2 - (15k^2 - 2k - 7) < 0\ 2 < k < 4$

$\Rightarrow 16k^2 - 8k + 1 - 15k^2 + 2k + 7 < 0\ k = 3$

$\Rightarrow k^2 - 6k + 8 < 0\ (k-2)(k-4) < 0.$

11. $x^2 + ax + 1 = 0$ difference between the roots less than $\sqrt{5}$

$\Rightarrow \sqrt{a^2-4} < \sqrt{5} \Rightarrow a^2 - 4 < 5 \Rightarrow a^2 < 9 \Rightarrow a \in (-3,3).$

12. $\alpha + \beta = a - 2\ \alpha\beta = -(a+1)$

$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = a^2 - 4a + 4 + 2a + 2 = a^2 - 2a + 6$

least value $= \frac{4(1)(6)-(-2)^2}{4(1)} = \frac{24-4}{4} = \frac{20}{4} = 5.$

13. $x - \frac{1}{x^2-4} = 2 - \frac{1}{x^2-4}$

$x^2 - 4 \neq 0\ x \neq \pm 2$

by solving given equation $x - \frac{1}{x^2-4} = 2 - \frac{1}{x^2-4} \Rightarrow x = 2$

From the above $x \neq 2$

Given equation has no solutions

Number of solutions $= 0.$

14. $(x-a)^2 = 0\ -3 < a < 4$

$x = a\ a \in (-2,-1,0,1,2,3)$

sum of a values $= 3.$

15. $x^2 + \frac{1}{x^2} = A$, and $x - \frac{1}{x} = B$

$$\frac{A}{B} = \frac{x^2 + \frac{1}{x^2}}{x - \frac{1}{x}} = \frac{\left(x - \frac{1}{x}\right)^2 + 2}{\left(x - \frac{1}{x}\right)} = \left(\frac{x-1}{1}\right) + \frac{2}{\left(\frac{x-1}{x}\right)}$$

$$\left(x - \frac{1}{x}\right) + \frac{2}{\left(x - \frac{1}{x}\right)} \geq 2\sqrt{2}(AM \geq GM)$$

16. Both the roots are greater than 2 , so

$f(2) > 0, D \geq 0, \frac{-1}{2a} > 2 \Rightarrow m < 10$ and $m \in (-\infty, 1] \cup [9, \infty)$

and $m - 3 > 4 \Rightarrow m \in [9,10].$

17. $x^2 - 2(a+1)x + a(a-1) = 0\ 4(a+1)^2 - 4a(a-1)$

$\Rightarrow x^2 - 2ax - 2x + a^2 - a = 0\ 4(8a+1).$

18. $(2x+3)(2x+5)(x-1)(x-2) = 30 \Rightarrow [(2x+3)(x-1)][(2x+5)(x-2)] = 30$

$\Rightarrow (2x^2 + x - 3)(2x^2 + x - 10) = 30$

$\Rightarrow (y-3)(y-10) = 30\ (\because 2x^2 + x = y)$

$\Rightarrow y^2 - 13y + 30 - 30 = 0 \Rightarrow y^2 - 13y = 0$

$\Rightarrow y(y-13)=0$

$\therefore y=0$ or $y=13$

$y=0 \Rightarrow 2x^2+x=0$ $y=13 \Rightarrow 2x^2+x-13=0$

$\Rightarrow x(2x+1)=0$ $x=\dfrac{-1\pm\sqrt{1+104}}{4}$

$\Rightarrow x=0$ or $x=\frac{-1}{2}$ $=\frac{-1\pm\sqrt{105}}{4}$

No. of rational roots = 2.

19. $x^2-2mx-1=0$ both the roots lies between $-2,4$

$f(-2)f(4)>0$

$\dfrac{-3}{4}<m<\dfrac{15}{8}$

$-2<-\dfrac{b}{2a}<4$

$-2<\dfrac{2m}{2}<4 \Rightarrow -2<m<4.$

$\Rightarrow (4+4m-1)(16-8m-1)>0$

$\Rightarrow (4m+3)(8m-15)<0$

Level – III

20. $35x^2+13x-12<0 \Rightarrow 35x^2+28x-15x-12<0$

$\Rightarrow 7x(5x+4)-3(5x+4)<0 \Rightarrow (7x-3)(5x+4)<0 \Rightarrow \frac{-4}{5}<x<\frac{3}{7}.$

21. $x^2+2(a-3)x+9>0$

$\alpha<6<\beta \Rightarrow f(6)<0 \Rightarrow 36+12(a-3)+9<0 \Rightarrow 12a+9<0 \Rightarrow 12a<-9 \Rightarrow a<\dfrac{-3}{4}$

So both the statements are true.

22. $x^2-12x+32<0 \Rightarrow (x-4)(x-8)<0 \Rightarrow 4<x<8$ $x\in(4,8)$.

23. $x^2-9x+18>0 \Rightarrow (x-3)(x-6)>0 \Rightarrow x<3$ or $x>6$

x does not lies between 3 and 6 .

24. $x^2-2x+5\leq 0$

$a=1$ $\Delta=(-2)^2-4(1)(5)$

$=4-20=-16<0$

$\therefore \forall x\in R, x^2-2x+5>0$

$\therefore x^2-2x+5\leq 0$ has no solutions.

25. a) $x^2+3x-4\leq 0$

$(x+4)(x-1)\leq 0 \Rightarrow -4\leq x\leq 1$ $x\in[-4,1]$.

b) $x^2-x-6\geq 0 \Rightarrow (x-3)(x+2)\geq 0 \Rightarrow x\leq -2$ or $x\geq 3 \Rightarrow x\in(-\infty,-2]\cup[3,\infty)$.

c) $x^2 - x - 6 < 0 \Rightarrow -2 < x < 3 \Rightarrow x \in (-2,3)$

d) $x^2 - 8x + 15 > 0 \Rightarrow (x-3)(x-5) > 0 \Rightarrow x < 3$ or $x > 5 \Rightarrow x \in (-\infty, 3) \cup (5, \infty)$.

26. $x^2 - 3x - 10 \leq 0 \Rightarrow (x-5)(x+2) \leq 0 \Rightarrow -2 \leq x \leq 5$

maximum possible integral value = 5.

27. 3 lies between the roots of $x^2 - ax + 8 = 0$

$3^2 - 3a + 8 < 0 \quad \Delta > 0$
$\Rightarrow 17 - 3a < 0 \quad a^2 - 32 > 0$
$\Rightarrow 17 < 3a \quad a^2 - (4\sqrt{2})^2 > 0$

$\Rightarrow \frac{17}{3} < \text{a}$

$(a - 4\sqrt{2})(a + 4\sqrt{2}) > 0$

$\Rightarrow \text{a} > \frac{17}{3}$

$a < -4\sqrt{2}$ or $a > 4\sqrt{2}$.

28. $2^2 - \text{a}(2) + 4 < 0 \Rightarrow 8 - 2\text{a} < 0 \Rightarrow 8 < 2\text{a} \Rightarrow 2\text{a} > 8 \Rightarrow \text{a} > 4$.

29. $\text{k}^2 - 3\text{k} + 2 < 0 \Rightarrow (\text{k}-1)(\text{k}-2) < 0 \Rightarrow 1 < \text{k} < 2 \therefore \text{k} \in (1,2)$

30. $(x+3)(3x-2)^5(7-x)^3(5x+8)^2 \geq 0$

$\Rightarrow -(x+3)\left(x - \frac{2}{3}\right)^5 (x-7)^5(5x+8)^2 \geq 0$

$\Rightarrow (x+3)\left(x - \frac{2}{3}\right)(x-7)^3\left(x + \frac{8}{5}\right)^2 \leq 0$

$\therefore$ Critical points are $-3, \frac{-8}{5}, \frac{2}{3}, 7$ $x \in (-\alpha, -3) \cup \left[\frac{2}{3}, 7\right] \cup \left[\frac{-8}{5}\right]$

31. a) $\text{k}^2 - 256\text{a} \geq 0 \Rightarrow \text{a} \leq \frac{\text{k}^2}{256}$

$64 - 4ak \leq 0; 64 \geq 4ak; a \leq \frac{16}{15} \;\; \therefore \frac{k^2}{256} = \frac{16}{15} \;\; \Rightarrow k^3 = 16^3 \Rightarrow k = 16$

b) $\tan x_1 + \tan x_2 = \frac{4}{1+k} \Rightarrow \tan x_1 + \tan x_2 = \frac{k-1}{k+1}$

$\tan x_1 + \tan x_2 \geq 2\sqrt{\tan x_1 - \tan x_2}$

$$\frac{4}{1+k} \geq 2\sqrt{\frac{k-1}{k+1}} \Rightarrow \frac{4}{(k+1)^2} \geq \frac{k-1}{(k+1)}$$
$$\Rightarrow 4 \geq k^2 - 1 \;\; \Rightarrow k^2 - 1 \leq 4 \;\; \Rightarrow k^2 \leq 5$$

$\therefore$ maximum value of k^2 is 5

c) $x^2 - 8x + a^2 - 6a = 0 \Rightarrow 64 - 4(a^2 - 6a) \geq 0 \Rightarrow 16 - a^2 + 6a \geq 0$

$\Rightarrow a^2 - 6a - 16 \leq 0 \Rightarrow a^2 - 8a + 2a - 16 \leq 0 \Rightarrow (a-8)(a+2) \leq 0 \Rightarrow -2 \leq a \leq 8$

maximum value of a is 8

d) $D \geq 0 \Rightarrow 16 + 4(\log_3 4) \geq 4 \Rightarrow \log_3 a \geq 4 \Rightarrow a \geq \frac{1}{81}$

2. THEORY OF EQUATIONS

WORKSHEET – 1

LEVEL – I

1. C	2. A	3. C	4. D	5. D
6. B	7. C	8. D	9. A	10. A

LEVEL – II

1) B	2) C	3) D	4) C	5) D	6) A
7) B	8) B	9) D	10) C		11) B
12) C	13) A	14) D	15) B	16) B	17) C
18) B	19) D	20) D	21) D	22) B	

LEVEL – III

23. A, B, C	24. A	25. D	26. B	
27. $a-t, b-r, c-q, d-p$		28. 8	29. ABCD	30. D
31. C	32. B	33. 4	34. ABCD	
35. a-s, b-r, c-q, d-p				

SOLUTIONS

LEVEL – II

2. Number of changes of signs in $f(x)$ is 2 ⇒ Number +positive roots is almost two.

3. $f(x) = x^7 + 5x^6 - x^3 + 7x + 2 \Rightarrow x^7 + 5x^6 + x^3 - 7x + 2$

   Number of changes of signs in f(x) is two ⇒ $f(x) = 0$ has at most three negative roots.

   $\therefore f(x) = 0$ has at most five real roots ⇒ $f(x) = 0$ has at least two imaginary roots.

4. Required equation is $(x^2 - 6x + 7)(x^2 - 4x + 13) = 0 \Rightarrow x^4 - 10x^3 + 44x^2 - 106x + 91 = 0$

5. Required equation is $(x^2 - 4x + 1)(x^2 + 2x + 5) = 0 \Rightarrow x^4 - 6x^3 + 14x^2 - 22x + 5 = 0$

6. The quadratic equation having roots $1 + i$ is $x^2 - 2x + 2 = 0$

   The quadratic equation having roots $1 - \sqrt{2}$ is $x^2 - 2x + 1 = 0$

The quadratic equation is $(x^2-2x+2)(x^2-2x-1)=0 \Rightarrow x^4-4x^3+5x^2-2x-2=0$

7. Dividing by ' 3 ' standard result

8. $1+1+\alpha=6 \Rightarrow \alpha=4$

9. $\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)=\frac{1}{4}-1=-3/4$

10. $\alpha=\beta+\gamma$

11. $S_3=\frac{-d}{a}=\frac{4}{7}$

12. $\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}=\frac{\alpha+\beta+\gamma}{\alpha\beta\gamma}=5$

13. $\alpha+\beta+\gamma=0, \Rightarrow \alpha+\beta=-\gamma; \frac{-S_2}{S_3}$

14. $\alpha\beta\gamma=-3 \Rightarrow (-1)\gamma=-3; \gamma=3$ then $f(3)=0$

15. $\alpha+\beta=0 \Rightarrow \alpha=-\beta$

$\alpha+\beta+\gamma=-p \Rightarrow \gamma=-p \Rightarrow \alpha\beta+\beta\gamma+\gamma\alpha=q \Rightarrow \alpha\beta+\gamma(0)=q \Rightarrow \alpha\beta=q$

$\Rightarrow \alpha\beta\gamma=-r \Rightarrow q\cdot(-p)=-r \Rightarrow pq=r$

16. let the roots be $\alpha,-\alpha,\beta$: sum of roots $\beta=3$ replace x by 3 in given equation.

17. Let the roots be $\alpha,\beta,\gamma: \alpha\beta=8, \sum\alpha\beta=-10$ then we get $\gamma=3$ product of roots $=-\mathrm{k}=24$

18. $\alpha+\beta+\gamma=-p, \alpha+\beta=-p-\gamma$

$\therefore \alpha+\beta-3\gamma=-(p+4\gamma)$

$-[(p+4\alpha)(p+4\beta)(p+4\gamma)]=3p^3-16pq+64r$

19. $s_1=-p, \alpha+\beta+\gamma=-p, \gamma=-p$

Substitute $f(\gamma)=0$

20. $\sum\alpha^2\beta^2=s_2^2-2s_1s_3$

21. The requated polynomial equation whose roots $2\pm\sqrt{3}, 1\pm 2i$ is

$\left[x^2-(2+\sqrt{3}+2-\sqrt{3})x+(2+\sqrt{3})\right][x^2-(1+2i+1-2i)x+(1+2i)(1-2i)]=0$

$\Rightarrow (x^2-4x+1)[x^2-3x+5]=0 \Rightarrow x^4-6x^3+14x^2+5=0$

22. $S_1=0 \Rightarrow S_2+0S_1-7(2)=0 \Rightarrow S_2=14 \Rightarrow S_3+0S_2-7S_1+4(3)=0 \Rightarrow S_3=-12$

$S_4+0.S_3-7S_2+4S_1-3(4)=0 \Rightarrow S_4-98-12=0 \Rightarrow S_4=110$

$S_5+OS_4-7S_3+4S_2-3S_1=0 \Rightarrow S_5+84+56=0 \Rightarrow S_5=-140$

LEVEL – III

23. $\sqrt{a}+\sqrt{b}$ is a root then the remaining roots are $\sqrt{a}-\sqrt{b}, -\sqrt{a}-\sqrt{b}, -\sqrt{a}+\sqrt{b}$.

24. (1) $\alpha+\beta=-(p+\gamma)$, similarly $\beta+\gamma=-(p+\alpha)$ and $\gamma+\alpha=-(p+\beta)$

$\therefore(\alpha+\beta)(\beta+\gamma)(\gamma+\alpha)=-(p+\gamma)(p+\alpha)(p+\beta)=-p^3+p^3-pq+r=r-pq$

25.

(4) $\sum\frac{1}{\alpha}=\frac{\beta\gamma+\alpha\gamma+\alpha\beta}{\alpha\beta\gamma}=\frac{-q}{r}$

26. (2) $\sum\alpha^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)=p^2-2q$

27. Conceptual

28. $s_3=\frac{-d}{a}=\frac{-(-28)}{1}=28$

29. $f(x)=x^3+x^2+x+1=0$

$$\alpha^3+\beta^3+\gamma^3=(\alpha+\beta+\gamma)^3-3s_1s_2+3s_3=\frac{3abc-b^3-3a^2d}{a^3}=\frac{3-1-3}{1}=1$$

$f(x)=x^4-16x^3+86x^2-176x+105=0$

$s_1=16,\ s_2=86,\ s_3=176,\ s_4=105,$

30. $s_1+s_2=16+86=102,$

31. $s_1+s_2-s_3=16+86-176=-74,$

32. $s_1+s_2+s_3+s_4=16+86+176+105=383$

33. $1+\alpha+\beta=2\ \Rightarrow\alpha+\beta=1$

34. $(-4)^3+3(-4)^2+3(-4)+28=0$

$\Rightarrow-4+48-12+28=76-76=0$

$\Rightarrow-4$ is a root of $x^3+3x^2+28=0\Rightarrow\alpha=-4$

$x^3+3x^2+3x+28=0\Rightarrow(x+4)(x^2-x+7)=0$

$x^3+3x^2+3x+28=0\Rightarrow(x+4)(x^2-x+7)=0$

$$x^2-x+7=0\Rightarrow x=\frac{1\pm\sqrt{1-28}}{2}=\frac{1\pm\sqrt{27}i}{2}\Rightarrow\beta=\frac{1+\sqrt{27}i}{2}$$

$$r=\frac{1-\sqrt{27}i}{2}$$

$$2\alpha+3\beta+3r=2\alpha+3(\beta+r)=2(-4)+3\left[\frac{1+\sqrt{27}i}{2}+\frac{1-\sqrt{27}i}{2}\right]$$

$=8+3=5$

35. $a-s;b-r;c-q;d-p$

WORKSHEET – 2

LEVEL – I

1. C	2. C	3. A	4. B	5. A
6. D	7. A	8. B		

LEVEL – II

1) B	2) A	3) B	4) D	5) A
6) C	7) B	8) D	9) A	10) C
11) A	12) C	13) C	14) C	15) A
16) B				

LEVEL – III

17) A, B, D	18) A	19) C	20) A	21) D
22) a-s,b-r,c-p,d-q		23) 1	24) C, D	25) B
26) D	27) C	28) C	29) a-s,b-p,c-t,d-q	
30) 98	31) A, B, C	32) A	33) A	34) B

SOLUTIONS

LEVEL – I

1. Required equation is $6\left(\frac{1}{x}\right)^6 - 25\left(\frac{1}{x}\right)^5 + 31\left(\frac{1}{x}\right)^4 - 31\left(\frac{1}{x}\right)^2 + 25\left(\frac{1}{x}\right) - 6 = 0$

   $\Rightarrow 6x^6 - 25x^5 + 31x^4 - 31x^2 + 25x - 6 = 0$

2. Let α, β, γ are it roots of $ax^3 + bx^2 + cx + d = 0$

   $dx^3 + \left(x^2 + bx + a = 0 \Rightarrow \frac{a}{x^3} + \frac{b}{x^2} + \frac{c}{x} + d = 0 \Rightarrow \frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}\right.$ are the roots of this equation.

   $\therefore$ these are in H.P., since α, β, γ are in A.P

3. Let α, β, γ be the roots of $ax^3 + bx^4 + cx + d = 0$

   $dx^3 - cx^2 + bx - a = 0 \Rightarrow -\frac{a}{x^3} + \frac{b}{x^2} - \frac{c}{x} + d = 0$

   $\Rightarrow \mathrm{a}\left(\frac{-1}{\mathrm{x}}\right)^3 + \mathrm{b}\left(\frac{-1}{\mathrm{x}}\right)^2 + \mathrm{c}\left(\frac{1}{\mathrm{x}}\right) + \mathrm{d} = 0 = \frac{-1}{\alpha}, \frac{1}{\beta}, \frac{-1}{\gamma}$ an in A.P

4. Let α, β, γ are roots of $ax^3 + bx^2 + cx + d = 0$

   $ax^3 - bx^2 + cx - d = 0 \Rightarrow -ax^3 + bx^4 - cx + d = 0$

   $a(-x)^3 + b(-x)^4 + c(-x) + d = 0 \Rightarrow -\alpha, -\beta, -\gamma$ are the roots

   α, β, γ are in Gp $\Rightarrow -\alpha, -\beta - \gamma$ are inG.P

5. Required equation is $f(x+1) = 0 \Rightarrow (x+1)^3 + 2(x+1)^2 + 3(x+1) + 4 = 0$

   $\Rightarrow x^3 + 3x^2 + 3x + 1 + 2x^2 + 4x + 2 + 3x + 4 + 3 = 0 \Rightarrow x^3 + 5x^2 + 10x + 10 = 0$

6. $a = 2, b = -3, c = k, d = 6$ roots are in A.P

   $\Rightarrow 2b^3 + 27a^2d = 9abc \Rightarrow -54 + 27(24) = 9(2)(-3)k \Rightarrow -54k = 594 \Rightarrow k = -11$

7. $\alpha - d + \alpha + \alpha + d = \frac{12}{4} \Rightarrow 3\alpha = 3 \Rightarrow \alpha = 1$

   $4 - 126 + 11 + \mathrm{k} = 0 \Rightarrow \mathrm{k} = -3$

8. $\alpha^2+\beta^2+\gamma^2=(\alpha+\alpha+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma d)=10^2-2(6)=100-12=88$

LEVEL – II

1. $f(x)=2x^3-5x^2-7x+8=0$, Req equation
   $f(-x)=2(-x)^3-5(-x)^2-7(-x)+8=0$
   $=-2x^3-5x^4+7x+8=0=2x^3+5x^2-7x-8=0$
2. Substitute $f(x/2)=0$
3. Required answer $f(2x)=0$
4. let roots be $a-d,a,a+d$
   $s_1=3a=9,\Rightarrow a=3: s_3=(a^2-d^2)a=15\Rightarrow 3(9-d^2)=15,9-d^2=5\Rightarrow d^2=4$
   $\Rightarrow d=\pm 2$
5. Let the roots be $a-d,a,a+d\Rightarrow s_1=3\Rightarrow 3a=3\Rightarrow a=1,f(a)=0$
6. $x^3+3px^2+3qx+\gamma=0$ compare the above equation with $ax^3+bx^2+cx+d=0$ using the forumula $ac^3=b^3d$
7. Let α,β,γ be roots of the given equation an in H.P
   then $\beta=\frac{-3\gamma}{q}$, Replace x by $\frac{-3r}{q}$ in the given equation.
8. Let roots $bca-d,a,a+d s_1=3a\Rightarrow 12=3a\Rightarrow a=4$
   $s_3=(a^2-d^2)a=48,4(16-d^2)=48,16-d^2=12,d^2=4\Rightarrow d=\pm 2$
9. $-1+2+\alpha=\frac{-1}{2}\Rightarrow\alpha=-3/2$
10. sum of the roots $=a=0$. since coefficient of x^8 is $=0$
11. By verification
12. α,β,γ be roots of given equation are in H.P $\beta=\frac{3\alpha\beta\gamma}{\sum\alpha\beta}=\frac{r}{q}$
13. Required equation is f(x + 3) = 0

	1	−5	−20	3	17
	–	3	−6	−78	−225
$x=3$	1	−2	−26	−75	−208
	–	3	3	−69	−1
$x=3$	1	1	−23	−1	−144
	–	3	12	1	
$x=3$	1	4	−23		−11
	–	3	12		
$x=3$	1		7		

14. roots of $ax^3 + bx^2 + cx + d = 0$ are $2, -3, 5$

roots of $a(x-1)^3 + b(x-1)^2 + c(x-1) + d = 0$ are $2+1, -3+1, 5+1 = 3, -2, 6$

15. Using it Newtons formula $s_1 = 6, s_2 = 14, s_3 = 36$

$$p_1 = -s_1 = -6, p_2 = \frac{s_1^2 - s_2}{2} = 11, p_3 = \frac{s_1(s_2 - p_2) - s_3}{3} = -6$$

$$s_4 + p_1\, s_3 + p_2\, s_2 + p_3\, s_1 = 0 \Rightarrow s_4 = -p_1\, s_3 - p_2\, s_2 - p_3\, s_1 - 4p_4 = 98$$

16. $s_1 = -1,\ s_2 - 1(s_1) + 1(2) = 0 \Rightarrow s_2 = -1 \Rightarrow$

$s_3 + 1\, s_2 + 1\, s_2 + 1(3) = 0\ s_3 = -1$

17.

$x=1$	1	−7	14	−8
	0	1	−6	8
$x=2$	1	−6	8	0
	0	2	−8	
$x=4$	1	−4	0	
	1	4		
	1	0		

18. Both the statements are true.

19. $2(2)^3 + 27(1)^2(c) = 18 \Rightarrow c = \frac{2}{27}$

20. $k(3)^3 = (2)^3(1) \Rightarrow k = \frac{8}{27}$

21. $2c^3 + 27ad^2 = 9bcd$

$2(8) + 27(2)(9) = 9(k)(2)(3)$

$$251 = 27k \Rightarrow k = \frac{251}{27}$$

22. $a - s, b - rc - p, d - q$

23. $2b^3 + 27a^v d = 9bcd$

$2(-27) + 27(1)(3) = 9(-3)\mathrm{k}(5)$

$-2 + 3 = -1k$

$(-k) = 1$

24. If roots an in GP $ac^3 = b^3 d$

$1(56) \times 56 \times 56 = (-14)(-14)(-14)\mathrm{k}$

$-k = 64 \Rightarrow k = -64$

25. Use $2b^3 + 27a^2d = 9abc$.

26. Let roots be $a - d, a, a + d$.

$s_1 = 3a = 9 \Rightarrow a = 3 s_3 = (a^2 - d^2)a = 15$

$3(9 - d^2) = 15, 9 - d^2 = 5 \Rightarrow d^2 = 4 \Rightarrow d = \pm 2$

27. Let α, β, γ the roots of the given equation in H.P then $\beta = \frac{3r}{q}$

28. $x^3 + 3px^2 + 3qx + r = 0$ compare the equation with $ax^3 + bx^2 + cx + d = 0$ using the formula $ac^3 = b^3d$.

29. a) use the formula $2b^3 + 27a^2d = 9abc$

b) use the condition $ac^3 = b^3d$

c) use the condition $ac^3 = b^3d$

d) use the condition $2c^3 + 27ad^2 = 9bcd$

30. $f(x) = x^3 + 98x + 7 = 0\ s_2 = \frac{c}{a} = 98$

31. Let common ratio be r, then

$\alpha(1 + \gamma) = \alpha + \beta = 4\alpha^2(1 + \gamma) = 36$

$\gamma^2 = 9 \Rightarrow \gamma = 3 \therefore \text{a} = 1 \ \therefore \text{A} = \alpha\beta = \alpha^2\gamma = 3 \Rightarrow \text{B} = \gamma\text{S} = \alpha^2\gamma^5 = 243$

32. $\text{y} = (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = \gamma^2 - \frac{4\alpha\beta\gamma}{\gamma} = \frac{r^3 + 4\gamma}{\gamma} = \frac{x^3 + r + 3r}{x} = -\frac{qx + 3r}{x}$

$y + q = \frac{3r}{x} \Rightarrow x = \frac{3r}{y + q}$

33. put $x = -1/x$ in the given equation.

34. Given $\sum\alpha = 0 \sum\alpha\beta = q\alpha\beta = -r$

let $\gamma = \beta^2 + \beta\gamma + \gamma = (\beta + \gamma)^2 - 2\beta\gamma + \beta\gamma = (-\alpha)^2 - \beta\gamma = \alpha^2 - \frac{2\beta\gamma}{\alpha}$

$\text{Asx}^3 + \text{qx} + \gamma = 0$

1), 2, given $x(y + q) = 0 \Rightarrow y + yrq = 0$

The transformed equation is $(y + q)^3 = 0$

3. TRIGONOMETRY

WORKSHEET – 1

LEVEL – I

1. C	2. A	3. B	4. B	5. C
6. C	7. D	8. A	9. B	10. C

LEVEL – II

1. B	2. D	3. A	4. C	5. 1
6. D	7. A	8. C	9. B	10. A
11. B	12. B	13. C	14. C	15. B
16. B	17. A	18. A	19. B	20. A

LEVEL – III

21.B, C	22. B, C	23. C, D	24. C, D	25. B, C
26. A	27. C	28. A	29. B	
30.a-q; b-p, r; c-p, r; d-q		31. a-q; b-r; c-s; d-p		
32. B, C, D		33. A	34. B	35. B

SOLUTIONS

LEVEL – I

3. $\tan\theta \cdot \cot\theta = 1 \Rightarrow \frac{1}{\cot\theta} = \tan\theta \Rightarrow \frac{1}{\cot\theta} = \frac{\sin\theta}{\cos\theta}$

6. $\frac{D}{90} = \frac{2C}{\pi} \Rightarrow \frac{D}{90} = \frac{2.\frac{\pi}{6}}{\pi} \Rightarrow \frac{D}{90} = \frac{1}{3} \Rightarrow D = 30°$

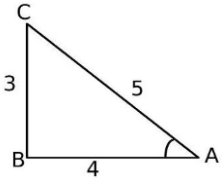
9. $\frac{G}{100} = \frac{2C}{\pi} \Rightarrow \frac{500}{100} = \frac{2C}{\pi} \Rightarrow C = \frac{5\pi^C}{2}$

$5\sin\theta = 3$

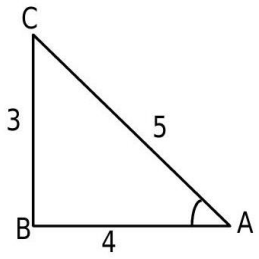
$\sin\theta = \frac{3}{5} = \frac{opp}{hyp}$

$\therefore adj = 4$

$\cos\theta = \frac{\text{Adj}}{Hyp} = \frac{4}{5}$



10. In $\triangle ABC, \underline{B} = 90$



$\text{Sin}\, A = \frac{3}{5}$

LEVEL – II

1. $\frac{D}{90} = \frac{G}{100} \Rightarrow \frac{330^\circ}{90} = \frac{G}{100}$

2. $\frac{D}{90} = \frac{2C}{\pi} \Rightarrow \frac{D}{90} = \frac{2(3\pi)}{\pi}$

$\Rightarrow \frac{11}{3} = \frac{G}{100}$

$\Rightarrow G = \frac{1100^g}{3}$

$\Rightarrow \frac{D}{90} = 6$

$\Rightarrow D = 540^\circ$

3. $\frac{D}{90} = \frac{G}{100} = \frac{2C}{\pi}$

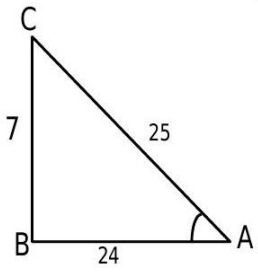
$\frac{D}{90} = \frac{200}{100}$ $\quad$ $\frac{200}{100} = \frac{2C}{\pi}$

$\Rightarrow \frac{D}{90} = \frac{2}{1}$ $\quad$ $\Rightarrow \frac{2}{1} = \frac{2C}{\pi}$

$\Rightarrow D = 180^\circ$ $\quad$ $\Rightarrow C = \frac{2\pi}{2}$

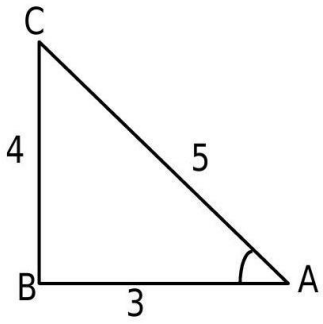
$\therefore 200^g = 180^\circ$ $\quad$ $\Rightarrow 200^g = \pi^c$

4. In $\triangle ABC$



$\therefore \text{Cosec}\, A = \frac{hyp}{opp} = \frac{25}{7}$

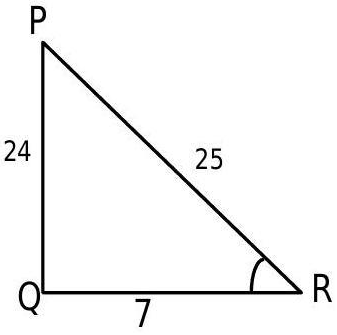
5.



$$\cos A = \frac{\text{adj}}{\text{hyp}} = \frac{3}{5}$$

$\sec A = \frac{hyp}{adj} = \frac{5}{3}$; $\therefore \cos A \cdot \sec A = \frac{3}{5} \cdot \frac{5}{3} = 1$

6.



$3\sin R + 4\cos R = 3\left(\frac{24}{25}\right) + 4\left(\frac{7}{25}\right) = \frac{72}{25} + \frac{28}{25} = \frac{100}{25} = 4$

7. $\sin^2\theta \cdot \text{cosec}^2\theta = (\sin\theta \cdot \text{cosec}\,\theta)^2 = 1^2 = 1$

8. $\sin\theta + \cos\theta = \frac{4}{5} + \frac{3}{5} = \frac{7}{5}$

9. $1^g = 100^1$

10. $\frac{D}{90} = \frac{G}{100} \Rightarrow \frac{D}{90} = \frac{800}{100}$: D=720

11. $\frac{D}{90} = \frac{2C}{\pi} \Rightarrow \frac{D}{90} = \frac{2\cdot\left(\frac{5\pi}{3}\right)}{\pi} \Rightarrow D = 90 \times \frac{10}{3} \Rightarrow D = 300°$

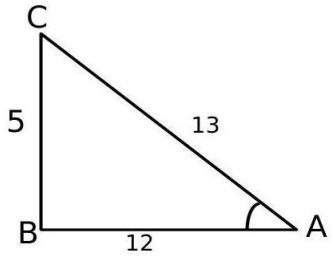
12. $Concpet$

13. $\frac{D}{90} = \frac{2C}{\pi} \Rightarrow \frac{40}{90} = \frac{2C}{\pi} \Rightarrow \frac{2\pi}{9} = C$

14. $\sin A + \cos A = \frac{12}{13} + \frac{5}{13} = \frac{17}{13}$

15. $\sin^4\theta - \sin^2\theta = \left(\frac{4}{5}\right)^4 - \left(\frac{4}{5}\right)^2 = \frac{256}{625} - \frac{16}{25} = \frac{256-400}{625} = \frac{-144}{625}$

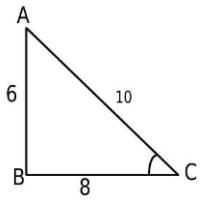
16.



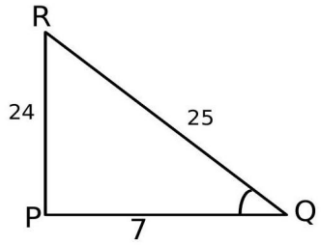
$\sin A\cos A + \sin C \cdot \cos C\ \left(\frac{5}{13}\right)\left(\frac{12}{13}\right) + \left(\frac{12}{13}\right)\left(\frac{5}{13}\right) = \frac{120}{169}$

17. $\cos^2\theta - \sin^2\theta = \left(\frac{21}{29}\right)^2 - \left(\frac{20}{29}\right)^2 = \frac{441}{841} - \frac{400}{841} = \frac{441-400}{841} = \frac{41}{841}$

18. $\frac{D}{90}=\frac{2C}{\pi}\Rightarrow\frac{D}{90}=\frac{2\left(\frac{-13\pi^C}{6}\right)}{\pi}\Rightarrow D=90\left(\frac{-13}{3}\right)\Rightarrow D=-390$

19. 

$\text{Tan}\,\theta=\frac{3}{8}k\Rightarrow\frac{6}{8}=\frac{3}{8}k\Rightarrow 6=3k\Rightarrow k=2$

20. 

$\tan Q-\tan R=\frac{24}{7}-\frac{7}{24}=\frac{576-49}{168}=\frac{527}{168}$

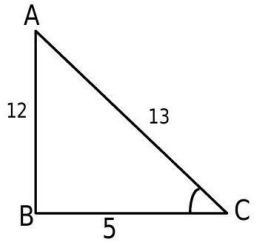
LEVEL – III

21. $\frac{D}{90}=\frac{G}{100}=\frac{2C}{\pi}\Rightarrow\frac{135}{90}=\frac{G}{100}=\frac{2C}{\pi}\Rightarrow\frac{3}{2}=\frac{G}{100}=\frac{2C}{\pi}\Rightarrow G=150, C=\frac{3\pi}{4}.$

22. $\frac{D}{90}=\frac{G}{100}=\frac{2C}{\pi}\Rightarrow\frac{D}{90}=\frac{G}{100}=\frac{2\left(\frac{5\pi}{4}\right)}{\pi}\Rightarrow\frac{D}{90}=\frac{G}{100}=\frac{5}{2}\Rightarrow D=225^\circ, \text{G}=250^g.$

24. $\frac{D}{90}=\frac{G}{100}=\frac{2c}{\pi}$ 25. $\cos\theta\cdot\sec\theta=1$

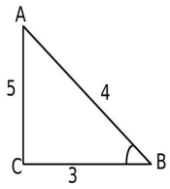
26. ST- 1: $\sin A+\cos A=\frac{12}{13}+\frac{5}{13}=\frac{17}{13}$

ST-2 

27. $\cos\theta+\sin\theta=\frac{5}{13}+\frac{12}{13}=\frac{17}{13}$

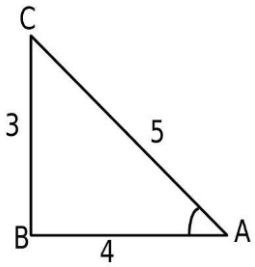
28. $\cos\theta-\sin\theta=\frac{5}{13}-\frac{12}{13}=\frac{-7}{13}$

29. $\sec\theta\cdot\text{cosec}\,\theta=\frac{13}{5}\cdot\frac{13}{12}=\frac{169}{60}$

31. 

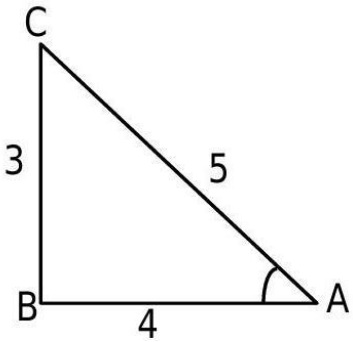
32. Conceptual

33.



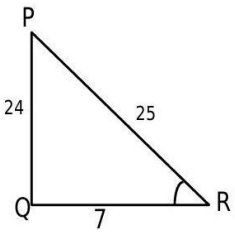
$$\sec^2 A - \tan^2 A = \left(\frac{5}{4}\right)^2 - \left(\frac{3}{4}\right)^2 = \frac{25}{16} - \frac{9}{16} = 1$$

34.



$$\frac{1+\cot A}{1-\cot A} = \frac{1+\frac{4}{3}}{1-\frac{4}{3}} = \frac{\frac{7}{3}}{\frac{-1}{3}} = -7$$

35.



$$1-\text{Cos}^2\theta = 1-\left(\frac{7}{25}\right)^2 = 1-\frac{49}{625} = \frac{576}{625}$$

WORKSHEET – 2

LEVEL – I

1) A	2) C	3) C	4) B	5) A
6) A	7) D	8) B		

LEVEL – II

1) A	2) B	3) B	4) D	5) 0
6) D	7) B	8) B	9) C	10) C
11) B	12) A	13) D	14)A	15) 1
16) 6	17) C	18) C	19) D	20) D
21) C	22) A	23) A	24) B	25) 3
26) 10				

LEVEL – III

27) B	28) A	29) C	30) C	31) B
32) a-r, b-s, c-p d-q		33) A, C, D		34) A
35) a-s, b-r, c-p d-q				

SOLUTIONS

LEVEL – II

1. $\sin 3A = \sin 3(30°) = \sin 90° = 1$

2. $\cos(63° + 27°) \cdot \cos(63° - 27°) = \cos 90° \cdot \cos 36° = 0$

3. $\cos 75° = \cos(45° + 30°) = \cos 45° \cdot \cos 30° - \sin 45° \sin 30° = \frac{\sqrt{3}-1}{2\sqrt{2}}$

4. $\cot 15° - \tan 15° = (2 + \sqrt{3}) - (2 - \sqrt{3}) = 2\sqrt{3}$

5. $\cos(240 + \theta) + \cos(240 - \theta) = 2\cos 240° \cdot \cos\theta$

   $\cos\theta + 2\cos 240° \cdot \cos\theta = \cos\theta + 2 \cdot \cos(180° + 60°) \cdot \cos\theta = \cos\theta + 2\left(\frac{-1}{2}\right)\cos\theta$

   $= 0$

6. $\cos(A - B) = \cos A \cdot \cos B + \sin A \cdot \sin B = \frac{3}{5} \cdot \frac{12}{13} + \frac{4}{5} \cdot \frac{5}{13} = \frac{56}{65}$

7. $40° + 80° = 120° \Rightarrow \tan(40° + 80°) = \tan 120° = -\sqrt{3}$

   $\therefore \tan(A + B) = \tan A + \tan B + \tan(A + B) \cdot \tan A \cdot \tan B$

   $= \tan 40 + \tan 80 - \sqrt{3} \cdot \tan 40 \cdot \tan 80°$

   $\therefore \tan(A + B) = -\sqrt{3}$

8. $\cot(A + B) \Rightarrow \cot(300°) = \frac{-1}{\sqrt{3}}$

9. $55^0 = 35^0 + 20^0$

   Take Tan on b.s

   $$\tan 55° = \tan(35° + 20°) \tan 55° = \frac{\tan 35° + \tan 20°}{1 - \tan 35° \cdot \tan 20°}$$

   $$\Rightarrow \tan 55° - \tan 35° \tan 20° \tan 55° = \tan 35° + \tan 20°$$

   Add tan 20 on both sides

   We get $\Rightarrow \tan 55° - \tan 35° \cdot \tan 20° \cdot \tan 55° + \tan 20° = \tan 35° + 2\tan 20°$

$\Rightarrow \tan 55^\circ = \tan 35^\circ + 2\tan 20^\circ$

$\therefore x = 55^0$

10. $\frac{\tan 225^\circ - \cot 81^\circ \cdot \cot 69^\circ}{\cot 261^\circ + \tan 21^\circ} = \frac{1 - \tan 9^\circ \cdot \tan 21^\circ}{\tan 9^\circ + \tan 21^\circ} = \cot 30^\circ = \sqrt{3}$

11. $A + B = 135^\circ$

Take tan on b.s & expand.

$\Rightarrow \tan A + \tan B = -1 + \tan A \tan B$

on solving $(1 - \tan A)(1 - \tan B) = 2$

12. $\therefore (1 + \tan A)(1 + \tan B) = 2$ If $A + B = 45^\circ$

13. $\sin^2 45^\circ - \sin^2 15^\circ = \frac{1}{2} - \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2 = \frac{1}{2} - \frac{4-2\sqrt{3}}{8} = \frac{1}{2} - \frac{1}{2} + \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{4}$

14. $\tan(\pi \cos\theta) = \tan\left(\frac{\pi}{2} - \pi \sin\theta\right) \Rightarrow \pi \cos\theta = \frac{\pi}{2} - \pi \sin\theta$

$\Rightarrow \pi(\cos\theta + \sin\theta) = \frac{\pi}{2} = \frac{1}{2}$

$\therefore \cos\left(\theta - \frac{\pi}{4}\right) = \cos\theta \cdot \cos\frac{\pi}{4} + \sin\theta \cdot \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}(\cos\theta + \sin\theta) = \frac{1}{\sqrt{2}}\left(\frac{1}{2}\right) = \frac{1}{2\sqrt{2}}$

15. $\therefore A + B + C = 360^\circ$

$\frac{A}{4} + \frac{B}{4} + \frac{C}{4} = \frac{360^\circ}{4} = 90^\circ \Rightarrow \frac{A}{4} + \frac{B}{4} = 90 - \frac{C}{4}$

Apply cot on both sides then $\sum \cot\frac{A}{4} = \pi \cot\frac{A}{4} \therefore k = 1$

16. $\frac{1}{\sin 10^\circ} + \frac{1}{\sin(60^\circ - 10^\circ)} - \frac{1}{\sin(60^\circ + 10^\circ)}$

$$= \frac{\sin^2 60^\circ - \sin^2 1^\circ 0 + \sin 10^\circ\left(\sin(60^\circ + 10^\circ) - \sin(60^\circ - 10^\circ)\right)}{\sin 10^\circ \cdot \sin(60^\circ - 10^\circ) \cdot \sin(60^\circ + 10^\circ)}$$

$$= \frac{\frac{3}{4} - \sin^2 10^\circ + \sin 10^\circ(2\cos 60^\circ \cdot \sin 10^\circ)}{\frac{1}{4}\sin 30^\circ} = \frac{\frac{3}{4} - \sin^2 10 + \sin^2 10}{\frac{1}{4} \times \frac{1}{2}} = 6$$

17. $2\cos^2\left(\frac{\pi}{4} + \alpha\right)$

$\sin^2\beta = \sin\alpha \cdot \cos\alpha$

$\cos 2\beta = 1 - 2\sin^2\beta$

$= 1 - \sin 2\alpha$

$[\therefore \sin 2\alpha = 2\sin\alpha\cos\alpha]$

$= 1 + \cos\left[\frac{\pi}{2} + 2\alpha\right]$

$\left[\therefore 1 + \cos\theta = 2\cos^2\frac{\theta}{2}\right]$

$= 2\cos^2\left(\frac{\pi}{4} + \alpha\right)$

18. $S = \sin\frac{\pi}{14}\cdot\sin\frac{3\pi}{14}\cdot\sin\frac{5\pi}{14}\cdot\sin\frac{9\pi}{14}\cdot\sin\frac{11\pi}{14}\cdot\sin\frac{13\pi}{14}$

$\cot c = \cos\frac{\pi}{14}\cdot\cos\frac{3\pi}{14}\cdot\cos\frac{5\pi}{14}\cdot\cos\frac{9\pi}{14}\cdot\cos\frac{11\pi}{14}\cdot\cos\frac{13\pi}{14}$

$$c.s = \frac{1}{64}\left[\begin{matrix}2\sin\frac{\pi}{14}\cos\frac{\pi}{14}\times 2\sin\frac{3\pi}{14}\cdot\cos\frac{3\pi}{14}\times 2\sin\frac{5\pi}{14}\cdot\cos\frac{5\pi}{14}.\\ 2\sin\frac{9\pi}{14}\cdot\cos\frac{9\pi}{14}\cdot 2\sin\frac{11\pi}{14}\cdot\cos\frac{11\pi}{14}\cdot 2\sin\frac{13\pi}{14}\cdot\cos\frac{13\pi}{14}\end{matrix}\right]$$

$c.s = \frac{1}{64}\sin\frac{2\pi}{14}\cdot\sin\frac{6\pi}{14}\cdot\sin\frac{10\pi}{14}\cdot\sin\frac{18\pi}{14}\cdot\sin\frac{22\pi}{14}\cdot\sin\frac{26\pi}{14}$

$\{\sin 2\ A = 2\sin A\cos A\}$

$c.s = \frac{1}{64}\cdot\cos\frac{\pi}{14}\cdot\cos\frac{3\pi}{14}\cdot\cos\frac{5\pi}{14}\cdot\cos\frac{9\pi}{14}\cdot\cos\frac{11\pi}{14}\cdot\cos\frac{13\pi}{14}$

$c.s = \frac{1}{64}\times c\ s = \frac{1}{64}$

19. $\frac{A+B}{2} = \frac{180-C}{2} \Rightarrow \tan\left(\frac{A}{2}+\frac{B}{2}\right) = \tan\left(90-\frac{C}{2}\right) = \cot\frac{C}{2}$

20. $90^\circ < \alpha < 180^\circ, 180^\circ < \beta < 270^\circ \Rightarrow 270^\circ < \alpha+\beta < 450^\circ \Rightarrow \alpha+\beta \in$ I or IV

$$\therefore \tan(\alpha+\beta) = \frac{\frac{-5}{12}+\frac{7}{24}}{1-\left(\frac{-5}{12}\right)\left(\frac{7}{24}\right)} < 0$$

$\therefore \alpha+\beta \in \mathbb{IV}$ Quadrant

21. $\therefore \tan\theta(1+\sec 2\theta)(1+\sec 2^2\theta)\ldots\ldots\ldots(1+\sec 2^n\theta) = \tan 2^n\theta$

22. Sol: $\operatorname{cosec}\theta + \cot\theta = \cot\theta/2$

$\cot\frac{\pi}{4}+\operatorname{cosec}\frac{\pi}{4} = \cot\frac{\pi}{8}$ $\cot\frac{\pi}{8}+\operatorname{cosec}\frac{\pi}{8} = \cot\frac{\pi}{16}$

$\cot\frac{\pi}{16}+\operatorname{cosec}\frac{\pi}{16} = \cot\frac{\pi}{32}$ $\therefore 1+\operatorname{cosec}\frac{\pi}{4}+\operatorname{cosec}\frac{\pi}{8}+\operatorname{cosec}\frac{\pi}{16} = \cot\frac{\pi}{32}$

23. $\frac{\cos 13^\circ-\sin 13^\circ}{\cos 13^\circ+\sin 13^\circ}+\frac{1}{\cot 148^\circ} = \frac{1-\tan 13^\circ}{1+\tan 13^\circ}+\tan 148^\circ$

$= \tan(45-13^\circ)+\tan(180-32) = \tan 32-\tan 32 = 0$

24. $\cos 170\cos 150\cos 130\cos 110$

$\cos(180-10)\cos(180-30)\cos(180-50)\cos(180-70)$

$\cos 10^\circ\cos 30^\circ\cos 50^\circ\cos 70^\circ$

$\cos 10^\circ\cos(60-10)\cos(60+10)\cos 30^\circ$

$\cos(60-\theta)\cos(60+\theta) = \frac{1}{4}\cos 3\theta$

$\frac{1}{4}\cos 3\theta\times\cos 30^\circ = \frac{1}{4}\cos 30^\circ\times\cos 30^\circ = \frac{1}{4}\times\frac{\sqrt{3}}{2}\times\frac{\sqrt{3}}{2} = \frac{3}{16}$

25. If $\frac{\tan x}{2} = \frac{\tan y}{3} = \frac{\tan z}{5} = k, x+y+z = \pi$

$\therefore \tan x = 2k, \tan y = 3k, \tan z = 5k$ $\therefore \tan(x+y) = -\tan z \Rightarrow k^2 = \frac{1}{3}$

$4k^2 + 9k^2 + 25k^2 = \frac{38}{k} \Rightarrow k^3 = 1 \Rightarrow k^2 \cdot k = 1 \Rightarrow k = 3$

26. $\cot\left(\frac{5\pi}{4}\right) = \cot\left(\pi + \frac{\pi}{4}\right) = \cot\frac{\pi}{4} = 1$ put $x = 1$ we get 10

LEVEL – III

27. $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \cdot \tan B} = \frac{\frac{1}{35} - \frac{18}{17}}{1 + \frac{1}{35}\cdot\frac{18}{17}} = \frac{-613}{613} = -1$

28. Conceptual

29. $(1 - \tan A)(1 - \tan B) = 1 - \tan A - \tan B + \tan A \tan B = 1 + 1 = 2$

30. $(1 + \cot A)(1 + \cot B) = 1 + \cot A + \cot B + \cot A \cot B = 1 + 1 = 2$

31. $\tan(A + B) = \tan 135 = \tan(90 + 45) = -\cot 45 = -1$

32. $a - r, b - s, c - pd - q$

Use $\cos 15° = \frac{\sqrt{3}+1}{2\sqrt{2}}, \sin 15° = \frac{\sqrt{3}-1}{2\sqrt{2}}$

$\tan 15° = 2 - \sqrt{3}, \cot 15° = 2 + \sqrt{3}$

33. $\cos^2(\alpha - \beta) + 2ab\sin(\alpha - \beta)$

$= \cos^2(\alpha - \beta) + 2\cos(\theta - \alpha)\sin(\theta - \beta)\sin(\alpha - \beta)$

$= \cos^2(\alpha - \beta) + [\sin\{2\theta - (\alpha + \beta)\} + \sin(\alpha - \beta)]\sin(\alpha - \beta)$

$= 1 + \sin\{2\theta - (\alpha + \beta)\}\sin(\alpha - \beta)$

$= 1 + \sin\{(\theta - \alpha + (\theta - \beta))\} \cdot \sin\{(\theta - \beta) - (\theta - \alpha)\}$

$= 1 + \sin^2(\theta - \beta) - \sin^2(\theta - \alpha) = \cos^2(\theta - \alpha) + \sin^2(\theta - \beta) = a^2 + b^2$

34. $\tan A = x\tan B$

$\frac{\tan A}{\tan B} = \frac{x}{1} \Rightarrow \frac{\tan A + \tan B}{\tan A - \tan B} = \frac{x + 1}{x - 1} \Rightarrow \frac{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}{\frac{\sin A}{\cos A} - \frac{\sin B}{\cos B}} = \frac{x + 1}{x - 1} \Rightarrow \frac{\sin(A + B)}{\sin(A - B)} = \frac{x + 1}{x - 1}$

35. $\cos 5\theta = a_0 + a_1\cos\theta + a_2\cos^2\theta + a_3\cos^3\theta + a_4\cos^4\theta + a_5\cos^5\theta$

$\cos 5\theta = \cos(3\theta + 2\theta) = \cos 2\theta \cdot \cos 3\theta - \sin 2\theta \cdot \sin 3\theta$

$= (2\cos^2\theta - 1)(4\cos^3\theta - 3\cos\theta) - 2\sin\theta\cos\theta(3\sin\theta - 4\sin^3\theta)$

On solving we get

$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta + 0$

Comparing the coefficients we get $a_5 = 16, a_3 = -20, a_1 = 5, a_0 = 0$.

WORKSHEET – 3

LEVEL – I

1. C	2. B	3. C	4. B	5. C
6. A	7. C	8. C	9. C	10. A

LEVEL – II

1. D	2. A	3. A	4. A	5. C
6. 0	7. C	8. B	9. A	10. B
11. C	12. D	13. A	14. A	15. A
16. A	17. C	18. B	19. B	20. B
21. D	22. B			

LEVEL – III

23. A, B, C	24. A	25. A	26. D	27. A
28.a-q; b-p; c- q; d-s		29. A, C	30. B	31. A
32. A	33. A, C	34. a-r; b-p; c- q; d-s		35.1

SOLUTIONS

LEVEL – II

1. $\cos^2 B - \cos^2 A = \sin(A+B)\cdot\sin(A-B) = \sin 54 \cdot \sin 30 = \frac{\sqrt{5}+1}{4}\cdot\frac{1}{2} = \frac{\sqrt{5}+1}{8}$

2. given $\tan\alpha = \frac{b}{a}$ $\therefore a\cos 2\alpha + b\sin 2\alpha = a\cdot\frac{1-\tan^2\alpha}{1+\tan^2\alpha} + b\cdot\frac{2\tan\alpha}{1+\tan^2\alpha} = \frac{a(a^2-b^2)}{a^2+b^2} + \frac{b\cdot 2ab}{a^2+b^2} = a$

3. $GE = 4(2\sin\theta\cdot\cos\theta)\cos 2\theta\cdot\cos 4\theta = 2(2\sin 2\theta\cdot\cos 2\theta)\cdot\cos 4\theta = 2\sin 4\theta\cdot\cos 4\theta = \sin 8\theta$

4. $\cos\theta = -\frac{4}{\sqrt{16+9}} = \frac{-4}{5}, 90^\circ < \frac{\theta}{2} < 135^\circ$ $\therefore \sin\frac{\theta}{2} = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{-4}{5}\right)}{2}} = \frac{3}{\sqrt{10}}$

5. $= \frac{\sin\theta+2\sin\theta\cos\theta}{1+\cos\theta+2\cos^2\theta-1} = \frac{\sin\theta(1+2\cos\theta)}{\cos\theta(1+2\cos\theta)} = \tan\theta$

6. $\sin 3a\cdot\sin a + \sin^2 a + \cos 3a\cdot\cos a - \cos^2 a = \cos(3a-a) = \cos 2a = 0$

7. $\cos\frac{2\pi}{15}\cdot\cos\frac{4\pi}{15}\cdot\cos\frac{8\pi}{15}\cdot\cos\frac{14\pi}{15} = \frac{2\sin\frac{2\pi}{15}}{2\sin\frac{2\pi}{15}}\left[\cos\frac{2\pi}{15}\cdot\cos\frac{4\pi}{15}\cdot\cos\frac{5\pi}{15}\cdot\cos\frac{14\pi}{15}\right] =$

$\frac{1}{2\sin\left(\frac{2\pi}{15}\right)}\left[2\sin\frac{2\pi}{15},\cos\frac{2\pi}{15}\cdot\cos\frac{4\pi}{15}\cdot\cos\frac{8\pi}{15}\cdot\cos\frac{14\pi}{15}\right] = \frac{1}{2^2\sin\left(\frac{2\pi}{15}\right)}\left[2\sin\frac{4\pi}{15}\cdot\cos\frac{4\pi}{15}\cdot\cos\frac{8\pi}{15}\cdot\cos\frac{14\pi}{15}\right]$

$= \frac{1}{2^3\sin\frac{2\pi}{15}}\left[2\sin\frac{8\pi}{15}\cdot\cos\frac{8\pi}{15}\cdot\cos\frac{14\pi}{15}\right] = \frac{1}{2^3\sin\left(\frac{2\pi}{15}\right)}\left[2\sin\frac{16\pi}{15}\cdot\cos\frac{14\pi}{15}\right] = \frac{1}{2^3\sin\left(\frac{2\pi}{15}\right)}\left[2\sin\left(\pi+\frac{\pi}{15}\right)\left(-\cos\frac{\pi}{15}\right)\right]$

$= \frac{1}{8\cdot\sin\left(\frac{2x}{15}\right)}\times\frac{2}{2}\left[\sin\frac{x}{15}\cdot\cos\frac{\pi}{15}\right] = \frac{1}{16\cdot\sin\frac{2\pi}{15}}\times\sin\frac{2\pi}{15} = \frac{1}{16}$

8. $\frac{\sin 3\theta}{1+2\cos 2\theta} = \frac{3\sin\theta-4\sin^2\theta}{1+2(1-2\sin^2\theta)} = \sin\theta$

9. $(\tan 9^0 + \cot 9^\circ) - (\tan 27^\circ + \cot 27^\circ) = \left(\frac{\sin 9^0}{\cos 9^0} + \frac{\cos 9^0}{\sin 9^0}\right) - \left(\frac{\sin 27^0}{\cos 27^0} + \frac{\cos 27^0}{\sin 27^0}\right)$

$$= \frac{1}{\sin 9^0 \cos 9^0} - \frac{1}{\sin 27^0 \cdot \cos 27^0} = \frac{2}{\sin 18^0} - \frac{2}{\sin 54^0}$$

$$= 2\left[\frac{4}{\sqrt{5}-1} - \frac{4}{\sqrt{5}+1}\right] = 8\left[\frac{\sqrt{5}+1-\sqrt{5}+1}{5-1}\right] = \frac{8(2)}{4} = 4$$

10. $\cos^2 \frac{3\pi}{5} + \cos^2 \frac{4\pi}{5} = \cos^2 108° + \cos^2 144° = \sin^2 18° + \cos^2 36°$

$$= \left(\frac{\sqrt{5}-1}{4}\right)^2 + \left(\frac{\sqrt{5}+1}{4}\right)^2 = \frac{2(5+1)}{16} = \frac{3}{4}$$

11. $\frac{3\cos\theta + 4\cos^3\theta - 3\cos\theta}{3\sin\theta - (3\sin\theta - 4\sin^3\theta)} = \frac{4\cos^3\theta}{4\sin^3\theta} = \cot^3\theta$

12. $GE = \cos^2 76° + 1 - \sin^2 16° - \left(\frac{1}{2}\right)[2\cos 76° \cdot \cos 16°]$

$$= 1 + \cos 92° \times \cos 60° - \frac{1}{2}[\cos 92° + \cos 60]$$

$$= 1 + \cos 92°\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)\cos 92° - \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 1 - \frac{1}{4} = \frac{3}{4}$$

13. $\frac{(\cos A + \sin A)^2 - (\cos A - \sin A)^2}{(\cos A - \sin A)(\cos A + \sin A)} = \frac{4\cos A \cdot \sin A}{\cos^2 A - \sin^2 A} = \frac{2 \cdot \sin 2A}{\cos^2 2A} = 2 \cdot \tan 2A$

14. $\frac{\cos 10° - \sqrt{3}\sin 10°}{\sin 10° \cdot \cos 10°} = \frac{2\left(\frac{1}{2}\cos 10° - \frac{\sqrt{3}}{2}\sin 10\right)}{\frac{1}{2}(2\sin 10° \cdot \cos 10°)} = \frac{2(\sin(30° - 10°))}{\frac{1}{2} \cdot \sin 20°} = 4$

15. Sum of roots $= \sin^2 18^0 + \cos^2 36^0 = \left(\frac{\sqrt{5}-1}{4}\right)^2 + \left(\frac{\sqrt{5}+1}{4}\right)^2 = \frac{3}{4}$

Product of roots $= \sin^2 18^0 \cdot \sin^2 36° = \frac{1}{16}$

$\therefore$ required Q.E is $x^2 - \left(\frac{3}{4}\right)x + \frac{1}{16} = 0 \Rightarrow 16x^2 - 12x + 1 = 0$

16. $\frac{(\cos A + \cos 5A) + (\cos 3A + \cos 7A)}{(\sin A + \sin 5A) + (\sin 3A + \sin 7A)} = \frac{2\cos 3A \cdot \cos 2A + 2\cos 5A \cdot \cos 2A}{2\sin 3A \cdot \cos 2A + 2\sin 5A \cdot \cos 2A}$

$$= \frac{2\cos 4A \cdot \cos A}{2\sin 4A \cdot \cos A} = \cot 4A \;\; \therefore x = 4A$$

17. $\tan(145° - 125°) = \tan 20° = \cot 70° = \frac{1}{\tan 70°} = \frac{1}{\tan 2(35°)} = \frac{1 - \tan^2 35°}{2\tan 35°} = \frac{1-k^2}{2k}$

18. If, a,b,c are in H.P then $b = \frac{2ac}{a+c}$

$$\therefore \cos x = \frac{2\cos(x-y) \cdot \cos(x+y)}{\cos(x-y) + \cos(x+y)} = \frac{2(\cos^2 x - \sin^2 y)}{2\cos x \cdot \cos y} \Rightarrow (1 - \cos y)\cos^2 x = \sin^2 y$$

$$\Rightarrow 2\sin^2 \frac{y}{2} \cdot \cos^2 \frac{y}{2} \Rightarrow \cos^2 x \cdot \sec^2 \frac{y}{2} = 2 \Rightarrow \cos x \cdot \sec \frac{y}{2} = \pm\sqrt{2}$$

19. $\frac{1}{\cos\theta} - \cos\theta = 1$

20. $\cos^3\theta + \cos^3\left(\frac{2\pi}{3} + \theta\right) + \cos^3\left(\frac{4\pi}{3} + \theta\right) = 3\cos\theta \cdot \cos\left(\frac{2\pi}{3} + \theta\right) \cdot \cos\left(\frac{2\pi}{3} + \theta\right)$

$$\therefore a\cos 3\theta = 3\cos\theta \cdot \cos\left(\frac{2\pi}{3} + \theta\right) \cdot \cos\left(\frac{2\pi}{3} - \theta\right) = 3\cos\theta\left(\cos^2\theta - \sin^2\frac{2\pi}{3}\right)$$

$= 3 \cdot \cos\theta\left(\cos^2\theta - \frac{3}{4}\right) = \frac{3}{4}(\cos^3\theta - 3\cos\theta) = \frac{3}{4}\cos 3\theta \ \therefore a = \frac{3}{4}$

21. $\tan\frac{(\alpha+\beta)}{2} = \frac{26/8}{1-\frac{15/8}{8}} = -\frac{26}{7}$ $\tan\left(\frac{\alpha}{2}+\frac{\beta}{2}\right) = \frac{\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}}{1-\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}}$

$$\Rightarrow \cos(\alpha+\beta) = \frac{1-\left(\frac{-26}{7}\right)^2}{1+\left(\frac{-26}{7}\right)^2} = \frac{-627}{725}$$

22. $2\sec 2\theta = \tan\phi + \cot\phi \Rightarrow \frac{2}{\cos 2\theta} = \frac{\sin^2\phi+\cos^2\phi}{\sin\phi\cdot\cos\phi} = \frac{2}{\cos 2\theta} = \frac{1}{\sin\phi\cdot\cos\phi}$

$\Rightarrow \cos 2\theta = \sin 2\phi \Rightarrow 2\theta = 90^\circ - 2\phi \Rightarrow \theta + \phi = \frac{\pi}{4}$

LEVEL – III

25. $\frac{2\tan A}{1-\tan^2 A} = \frac{-24}{7}$

26. $\tan 3A = \frac{3\tan A-\tan^3 A}{1-3\tan^2 A} = -\frac{44}{117}$

27. $\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A} = \frac{1-\frac{9}{16}}{1+\frac{9}{16}} = \frac{7}{25}$

28. a) $- = \tan\frac{\theta}{2}$ b) $\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sin^2\frac{\theta}{2}} = \cot\frac{\theta}{2}$

c) $\frac{(1-\cos\theta)\sin\theta}{(1+\cos\theta)+\sin\theta} = \frac{2\sin^2\frac{\theta}{2}+2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}+2\sin\frac{\theta}{2}\text{coa}\frac{\theta}{2}} = \frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}} = \tan\frac{\theta}{2}$

d) $\frac{(1+\cos 2\theta)\sin\theta}{(1-\cos 2\theta)+\sin\theta} = \frac{2\cos^2\theta+2\sin\theta\cos\theta}{2\sin^2\theta+2\sin\theta+\cos\theta} = \cot\theta$

29. $x = \tan\frac{\theta}{2}$, then $\frac{2x}{1-x^2} = \tan\theta$

30. $A = 45^\circ$, in $\sin\frac{A}{2} = \sqrt{\frac{1-\cos A}{2}}$

31. Put $A = 45^\circ$, in $\cos\frac{A}{2} = \sqrt{\frac{1-\cos A}{2}}$

32. $\tan\frac{A}{2} = \frac{\sin\frac{A}{2}}{\cos\frac{A}{2}}$

33. By verfication.

34. a) $\sin^2\frac{3\pi}{5} + \sin^2\frac{4\pi}{5} = \sin^2 108^\circ + \sin^2 144^\circ$f

$= \cos^2 18^0 + \sin^2 36^0$

$= \frac{10+2\sqrt{5}}{16} + \frac{10-2\sqrt{5}}{16} \Rightarrow \frac{20}{16} = \frac{5}{4}$

b) $\cos^2\frac{3\pi}{5} + \cos^2\frac{4\pi}{5} = \cos^2 108^\circ + \cos^2 144^0$

$= \sin^2 18^0 + \cos^2 36^0 \Rightarrow \frac{(\sqrt{5}-1)^2}{16} + \frac{(\sqrt{5}+1)^2}{16} = \frac{3}{4}$

c) $\cos^2\frac{\pi}{5} - \cos^2\frac{2\pi}{5} = \frac{10+2\sqrt{5}}{16} - \frac{(\sqrt{5}+1)^2}{16} \Rightarrow \frac{10-2\sqrt{5}-6-2\sqrt{5}}{16} = \frac{4}{16} \Rightarrow \frac{1}{4}$

d) $\cos^2\frac{\pi}{12} + \cos^2\frac{3\pi}{12} + \cos^2\frac{5\pi}{12} = \left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)^2 + \frac{1}{2} + \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2$

$= \frac{4+2\sqrt{3}+4+4-2\sqrt{3}}{8} \Rightarrow \frac{12}{8} = \frac{3}{2}.$

35. $\frac{\cos^4 x}{\cos^2 y} = \frac{\sin^4 x}{\sin^2 y} = 1$

$\frac{(1-\sin^2 x)^2}{1-\sin^2 y} + \frac{\sin^4 y}{\sin^2 y} = 1 \Rightarrow \frac{1-\sin^4 x - 2\sin^2 x}{1-\sin^2 y} + \frac{\sin^4 x}{\sin^2 y} = 1$

$\frac{(1-\sin^4 x - 2\sin^2 x)\sin^2 y + \sin^4 y(1-\sin^2 y)}{\sin^2 y(1-\sin^2 y)} = 1$

$1 = \frac{\sin^2 y + \sin^4 x \cdot \sin^2 y - 2\sin^2 x \sin^2 y + \sin^4 x - \sin^4 x \cdot \sin^4 y}{\sin^2 y(1-\sin^2 y)}$

$\frac{\sin^2 y - 2\sin^2 x \cdot \sin^2 y + \sin^4 x}{\sin^2 y(1-\sin^2 y)} = 1$

$\sin^2 y - 2\sin^2 x \sin^2 y + \sin^4 x = \sin^2 y - \sin^4 y;\ (\sin^2 y - \sin^2 x)^2 = 0$

$\sin^2 x = \sin^2 y \Rightarrow \sin x = \sin y \Rightarrow x = y$

$\Rightarrow \cos^2 x = \cos^2 y = \frac{\cos^2 y}{\cos^2 x} + \frac{\sin^4 y}{\sin^2 x} = \frac{\cos^2 y}{\cos^2 y} + \frac{\sin^4 y}{\sin^2 y} = 1$

WORKSHEET – 4

LEVEL – I

1. A	2. A	3. B	4. B	5. C
6. B	7. D	8. A	9. A	10. B

LEVEL – II

1. A	2. C	3. C	4. C	5. B
6. 0	7. C	8. A	9. A	10. A
11. C	12. C	13. B	14. A	15. B
16. D	17. B	18. C	19. B	20. D
21. C				

LEVEL – III

22. B, C, D	23. B	24. A	25. C
26.a-r; b-s; c- p; d-q		27. A, C	28. A	29. A
30. B	31. a-s; b-r; c- p; d-q		32. C	33. C
34. a-r; b-s; c- p; d-q		35. 0		

SOLUTIONS
LEVEL – II

1. From option (i) verify using transformations

2. $\cos 70^\circ - \cos 10^\circ = -2\sin\frac{70^\circ+10^\circ}{2}\cdot\sin\frac{70^\circ-10^\circ}{2} = -2\sin 40^\circ\cdot\sin 30^\circ = -\sin 40^\circ$

   $32(1-2\sin^2\theta)\cdot\cos 4\theta = 2(\cos 2\theta)\cdot\cos 4\theta$

   $= \cos(4\theta+2\theta)+\cos(4\theta-2\theta) = \cos 6\theta + \cos 2\theta$

4. $G\cdot E = 2\cos 48^\circ\cdot\cos 36^\circ + 2\cos 144^\circ\cdot\cos 12^\circ = 2\cos 48^\circ\cdot\cos 36^\circ - 2\cos 36^\circ\cdot\cos 12^\circ$

   $[\cos 144^\circ - \cos(180^\circ-36^\circ)]$

   $= 2\cos 36^\circ(\cos 48^\circ - \cos 12^\circ) = 2\cos 36^\circ\cdot 2\sin 30^\circ\cdot\sin(-18^\circ)$

   $= 2\frac{\sqrt{5}+1}{4}2\cdot\frac{1}{2}\left[-\frac{\sqrt{5}-1}{4}\right] = \frac{-1}{2}$

5. $\frac{\cos 20^\circ+\cos 40^\circ}{\cos 70^\circ+\cos 50^\circ} = \frac{2\cos 30^\circ\cdot\cos 10^\circ}{2\cos 60^\circ\cdot\cos 10^\circ}\begin{bmatrix}\sin 70^\circ = \cos(90^\circ-20^\circ)\\ \sin 40^\circ = \cos(90^\circ-50^\circ)\end{bmatrix} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$

6. $2\sin 30^\circ\cdot\cos 48^\circ + \cos(180^\circ-48^\circ) = \cos 48^\circ - \cos 48^\circ = 0$

7. If $A+B+C = 180^\circ$ then $\tan\left(\frac{A+B}{2}\right) =$

   $\tan\frac{A+B}{2} = \tan\left(90^\circ-\frac{C}{2}\right) = \cot\frac{C}{2}$

8. $\frac{x}{y} = \frac{\sin\alpha+\sin\beta}{\cos\alpha+\cos\beta} = \frac{2\sin\frac{\alpha+\beta}{2}\cdot\cos\frac{\alpha-\beta}{2}}{2\cos\frac{\alpha+\beta}{2}\cdot\cos\frac{\alpha-\beta}{2}} = \tan\frac{\alpha+\beta}{2}$

9. $2\sin(A+B)\cdot\cos(A-B)+2\sin C\cdot\cos C = 2\sin C[\cos(A-B)-\cos(A+B)]$

   $= 4\sin A\cdot\sin B\cdot\sin C$

10. $\tan 5x\cdot\tan 3x\cdot\tan 2x =$

    take $5x = 3x+2x$, take tan on bothe sides

11. By using transformations. $\frac{\sin\alpha+\sin\beta}{\cos\alpha+\cos\beta} = \frac{a}{b} \Rightarrow \frac{2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}}{2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}} = \frac{a}{b} \Rightarrow \tan\frac{\alpha+\beta}{2} = \frac{a}{b}$

$$\Rightarrow \cos(\alpha+\beta) = \frac{1-\tan^2\left(\frac{\alpha+\beta}{2}\right)}{1+\tan^2\left(\frac{\alpha+\beta}{2}\right)} = \frac{1-\frac{a^2}{b^2}}{1+\frac{a^2}{b^2}} = \frac{b^2-a^2}{b^2+a^2}$$

12. Squaring & adding given equations

    we get $\cos^2\frac{\alpha-\beta}{2} = \frac{9}{65} \Rightarrow \cos\frac{\alpha-\beta}{2} = \frac{-3}{\sqrt{130}}$

13. Put $A = B = C = 90^\circ$ Substitute 14. Put $A = B = C = 60^\circ$& verify option

14. Put $A = B = C = 60^\circ$& verify option

15. Put $A = 60^\circ, B = 45^\circ, C = 30^\circ$

    (or) $\frac{\cos C-\cos A}{\sin A-\sin C} = \frac{2\sin\frac{C+A}{2}\cdot\sin\frac{A-C}{2}}{2\cos\frac{A+C}{2}\cdot\sin\frac{A-C}{2}} = \tan\frac{A+C}{2} = \tan\frac{2B}{2} = \tan B$

17. If use $\cot(A-B)=\frac{\cot A\cdot\cot B+1}{\cot B-\cot A}$ 18. Put $A=B=30^\circ$ and verify

18. Put $\sin C+\sin D$ formula $\Rightarrow \frac{2\sin 5A\cos 4A+\sin 5A}{2\cos 5A\cos 4A+\cos 5A}=\frac{\sin 5A(2\cos 4A+1)}{\cos 5A(2\cos 4A+1)}=\tan 5A$ (or)

19. By using componendo & dividendo method

$$\frac{\sin x+\sin y}{\sin x-\sin y}=\frac{1+2}{1-2}\Rightarrow\frac{2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)}{2\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)}=-3\Rightarrow\tan\frac{x+y}{2}\cot\frac{x-y}{2}=-3$$

similarly, $\frac{\cos x+\cos y}{\cos x-\cos y}=\frac{3+2}{3-2}\Rightarrow\frac{2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)}{-2\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)}=5\Rightarrow-\cot\frac{x+y}{2}\cot\frac{x-y}{2}=5$

$$\Rightarrow-\tan^2\frac{x+y}{2}=-\frac{3}{5}\Rightarrow\tan^2\frac{x+y}{2}=\frac{3}{5}$$

21. Put x = $\tan\alpha$ y = $\tan\beta$ z = $\tan\gamma$ and simplify

22. Squaring on both sides

LEVEL – III

23. Put A = B = C = 90°

24. Put A = B = C = 90° (or) use $\sin C+\sin D$ formula

25. Use $\sin^2\theta=\frac{1-\cos 2\theta}{2}$

26. a) $\sin 48^\circ\sin 12^\circ=\frac{1}{2}[\cos 36^\circ-\cos 60^\circ]=\frac{\sqrt{5}-1}{8}$

b) $\cos 48^\circ\cos 12^\circ=\frac{1}{2}[\cos 60^\circ+\cos 36^\circ]=\frac{\sqrt{5}+3}{8}$ c) $2\sin 45^\circ\cos 30^\circ=\frac{\sqrt{3}}{\sqrt{2}}$

d) $\sin 24^\circ+\sin 84^\circ=2\sin 54^\circ\cdot\cos 30^\circ=\frac{\sqrt{15}+\sqrt{3}}{4}$

27. In G.P $S_\infty=\frac{a}{1-r}$

28. $3A=2A+A\Rightarrow\tan 3A=\tan(2A+A)$

29. $5x=3x+2x\Rightarrow\tan 5x=\tan(3x+2x)$

30. $\tan(A+C)=\tan B$

31. Use $\cos^2=\frac{1+\cos^2 A}{2}\Rightarrow\sin^2 A=\frac{1-\cos^2 A}{2}$

32. $\cos^2(\alpha+\beta)+\cos^2(\alpha-\beta)-\cos 2\alpha\cdot\cos 2\beta$

$$=\frac{1+\cos(2\alpha+2\beta)}{2}+\frac{1+\cos(2\alpha-2\beta)}{2}-\cos 2\alpha\cdot\cos 2\beta$$

$$=1+\frac{1}{2}[2\cos 2\alpha\cdot\cos 2\beta]-\cos 2\alpha\cdot\cos 2\beta=1$$

33. $\sin(2\alpha+\beta)=3\sin\beta\Rightarrow\frac{\sin(2\alpha+\beta)}{\sin\beta}=\frac{3}{1}$ [By C&D method]

$$\Rightarrow\frac{\sin(2\alpha+\beta)+\sin\beta}{\sin(2\alpha+\beta)-\sin\beta}=\frac{3+1}{3-1}\Rightarrow\frac{\tan(\alpha+\beta)}{\tan\alpha}=2\Rightarrow\tan(\alpha+\beta)=2\tan\alpha$$

34. $\frac{a}{b}=\frac{\cos x+\cos y}{\sin x+\sin y} \Rightarrow \frac{a}{b}=\frac{2\cos\left(\frac{x+y}{2}\right)\cdot\cos\left(\frac{x-y}{2}\right)}{2\sin\frac{x+y}{2}\cdot\cos\frac{x-y}{2}}$

$\Rightarrow \frac{a}{b}=\cot\frac{(x+y)}{2} \Rightarrow \frac{b}{a}=\tan\frac{(x+y)}{2} \quad \therefore \sin(x+y)=\sin 2\frac{(x+y)}{2}=\frac{2\tan\frac{(x+y)}{2}}{1+\tan^2\left(\frac{x+y}{2}\right)}$

35. $1+\cos 2x+\cos 4x+\cos 6x-4\cos x\cdot\cos 2x\cdot\cos 3x$

$\Rightarrow 2\cos^2 x+2\cos 5x\cdot\cos x-4\cos x\cdot\cos 2x\cdot\cos 3x$

$\Rightarrow 2\cos x[\cos x+\cos 5x]-4\cos x\cdot\cos 2x\cdot\cos 3x$

$\Rightarrow 2\cos x[2\cos 3x\cdot\cos 2x]-4\cos x\cdot\cos 2x\cdot\cos 3x=0$

WORKSHEET – 5

LEVEL – I

1. D	2. C	3. C	4. C	5. B
6. B	7. C	8. B	9. B	10. C

LEVEL – II

1. B	2. 24	3. C	4. A	5. D
6. A	7. B	8. A	9. A	10. D
11. B	12. D	13. D	14. C	15. B
16. B	17. A	18. A	19. B	20. A
21. A	22. A	23. D	24. A	25. B
26. B				

LEVEL – III

27. B, D	28. A	29. A	30. D
31.a-t; b-r; c-q,s; d-p		32. A, C	33.a-s; b-p; c-q; d-r;
34.2			

SOLUTIONS

LEVEL – II

1. period of $\sin^n x$, if n is odd is 2π

2. $\sin\frac{\pi}{4}x$ period $=\frac{2\pi}{\pi/4}=8$ $\cos\frac{\pi}{6}$ xperiod $=\frac{2\pi}{\pi/6}=12$ then lcm of 8,12 is 24

3. $\sin x\cdot\sin(120+x)\sin(120-x)=\frac{1}{4}\sin 3x \quad \therefore$ period is $\frac{2\pi}{3}$

4. $\sin x\left[\cos\frac{\pi}{4}\cos x+\sin\frac{\pi}{4}\sin x\right]=\frac{1}{\sqrt{2}}\cdot\sin x(\cos x+\sin x) \Rightarrow \frac{1}{\sqrt{2}}\left[\frac{1}{2}\sin 2x+\sin^2 x\right]$

$\therefore$ period is, LCM of $\frac{2\pi}{2}$ and π is π

5. LCM of $\frac{\pi}{2}$ and $\frac{2\pi}{4}$ is $\frac{\pi}{2}$

6. $\sin 2x - \sqrt{3}\cos 2x$ campare with a $a\sin\theta + b\cos\theta + c$

$\therefore$ Min $= c - \sqrt{a^2 + b^2} = 0 - \sqrt{(1)^2 + (-\sqrt{3})} = -2$

7. $\cos x\cos(60 + x) \cdot \cos(60 - x) = \frac{1}{4}\cos 3x$ $\therefore$ Max value of $\frac{1}{4}\cos 3x$ is $\frac{1}{4}$.

8. $-1 \le \cos(4x - 5) \le 1 \Rightarrow -3 \le 3\cos(4x - 5) \le 3 \Rightarrow 1 \le 4 + 3\cos(4x - 5) \le 7$

9. $\sec(x + 8x + 27x) = \sec 36x =$ period is $\frac{2\pi}{36} = \pi/18$

10. It is not periodic function. so does not satisfy

11. $f(x + t) = f(x) \Rightarrow \sqrt{\cos(x + t)} = \sqrt{\cos x} \Rightarrow t = 2\pi, 4\pi$

But t is the least positive value, period is 2π

12. $f(x) = \sin^2\left(\frac{\pi}{8} + \frac{x}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{x}{2}\right) = \sin\left(\frac{\pi}{8} + \frac{x}{2} + \frac{\pi}{8} - \frac{x}{2}\right)$.

$\sin\left(\frac{\pi}{8} + \frac{x}{2} - \frac{\pi}{8} + \frac{x}{2}\right) = \sin\frac{\pi}{4} \cdot \sin x$ period is 2π

13. period of $\sin\frac{\pi x}{2} = \frac{2\pi}{\pi/2} = 4$

period of $\cos\frac{\pi x}{3} = \frac{2\pi}{\pi/2} = 6$ LCM of 4,6,4, is 12

period of $\tan\frac{\pi x}{4} = \frac{\pi}{\pi/4} = 4$

14. LCM of $\left(\frac{2\pi}{n}, 2n\pi\right) \Rightarrow 2n\pi = 4\pi \Rightarrow n = 2$

15. $\sqrt{7}\sin x + \sqrt{29}\cos x + 7$, min $= 7 - \sqrt{-1 + 29} = 7 - 6 = 1$

max $= 7 + \sqrt{7 + 29} = 7 + 6 = 13$ max value is ' 1 '

16. $\tan(x(1 + 2 + 3 \cdot\cdot n)) = \tan\frac{n(n+1)}{2} \cdot x$

period is $\frac{\pi}{\frac{n(n+1)}{2}} = \frac{2\pi}{n(n+1)}$

17. $4\cos x^2 \cdot \cos\left(\frac{\pi}{3} + x^2\right)\cos\left(\frac{\pi}{3} - x^2\right) = \cos(3x^2) \Rightarrow$ extreamvalues $-1,1$

18. $\sin^2\left(\frac{\pi}{3} + x\right) + \sin^2\left(\frac{\pi}{3} - x\right) = 1 - \cos^2\left(\frac{\pi}{3} + x\right) + \sin^2\left(\frac{\pi}{3} - x\right)$

$= 1 - \cos\left(\frac{\pi}{3} + x + \frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x - \frac{\pi}{3} + x\right) = 1 - \cos\frac{2\pi}{3} \cdot \cos 2x = 1 + \frac{1}{2}\cos 2x$

$\therefore$ min value $= 1 - \frac{1}{2} = \frac{1}{2}$

19. $x - [x] = \{x\}$. Its period is 1

20. $\sin^6 x + \cos^6 x = 1 - 3\sin^2 x \cdot \cos^2 x = 1 - \frac{3}{4}\sin^2 2x = 1 - \frac{3}{4}\left(\frac{1-\cos 4x}{2}\right) = \frac{5}{8} + \frac{3}{8}\cos 4x$

$\therefore -1 \le \cos 4x \le 1 \Rightarrow \frac{5}{8} - \frac{3}{8} \le f(x) \le \frac{5}{8} + \frac{3}{8} \Rightarrow \frac{1}{4} \le f(x) \le 1$

21. $\left(\sin x-\frac{3}{2}\right)^2-\frac{1}{4}$ we know $-1\le\sin x\le 1\Rightarrow -1-\frac{3}{2}\le\left(\sin x-\frac{3}{2}\right)\le 1-\frac{3}{2}$

$\Rightarrow\frac{-5}{2}\le\sin x-\frac{3}{2}\le\frac{-1}{2}\Rightarrow\frac{1}{4}\le\left(\sin x-\frac{3}{2}\right)^2\le\frac{25}{4}$

$\frac{-1}{4}+\frac{1}{4}\le\left(\sin x-\frac{3}{2}\right)^2-\frac{1}{4}\le\frac{25}{4}-\frac{1}{4}\Rightarrow 0\le\left(\sin x-\frac{3}{2}\right)^2-\frac{1}{4}\le 6$

22. $f(x)=\sqrt{4-|\sec x|}$.

$|\sec x|\ge 1\Rightarrow 4-|\sec x|\le 3, 0\le\sqrt{4-|\sec x|}\le\sqrt{3}$

$\therefore 0\le\sqrt{4-|\sec x|}\cdot\le\sqrt{3}$

23. $\sec[x(1^2+2^2+3^2+\cdots n^2)]=\sec\left[\frac{n(n+1)(2n+1)}{6}\right]x$

period is $\frac{2\pi}{\frac{n(n+1)(2n+1)}{6}}=\frac{12\pi}{n(n+1)(2n+1)}$

24. $3\sin 2x+4\cos 2nx$

Range $=\left[c-\sqrt{a^2+b^2},c+\sqrt{a^2+b^2}\right]$

25. sq on both sidesthen $u^2=a^2+b^2+$

$2\sqrt{(a^4+b^4)\sin^2\theta\cdot\cos^2\theta+a^2b^2(\cos^4\theta+\sin^4\theta)}$

$a^2+b^2+2\sqrt{(a^4-b^4)\sin^2\theta\cdot\cos^2\theta+a^2b^2}$

$\therefore 4^2$ has max when $\theta=\pi/4$, min when $\theta=0$

then $u^2=a^2+b^2+2\sqrt{(a^4+b^4)\sin^2\theta\cdot\cos^2\theta+a^2b^2(\cos^4\theta+\sin^4\theta)}$

$=2a^2+2b^2-a^2-b^2-2ab=(a-b)^2$

26. $A.M\ge GM$ 27. Conceptual.

27. Maximum value $=\sqrt{49+576}=25$; Minimum value $=-\sqrt{49+576}=-25$

28. $5\cos\theta+3\cos\left(\theta+\frac{\pi}{3}\right)+3$

$=5\cos\theta+3\left(\frac{\cos\theta}{2}-\frac{\sqrt{3}}{2}\sin\theta\right)+3=\frac{13}{2}\cos\theta-\frac{3\sqrt{3}}{2}\sin\theta+3$

minimum value $=3-\sqrt{\frac{169}{4}+\frac{27}{4}}\Rightarrow 3-7=-4$

maximum value $=3+7=10$

30. $3\cos\theta+5\sin\left(\theta+\frac{\pi}{6}\right)=3\cos\theta+5\left(\frac{\sqrt{3}}{2}\sin\theta+\frac{1}{2}\cos\theta\right)=\frac{11}{2}\cos\theta+\frac{5\sqrt{3}}{2}\sin\theta$ minimum

value $=-\sqrt{\frac{121}{4}+\frac{75}{4}}=-7$; maximum value $=7$.

31. a-t; b-r; c-q,s; d-p

a) maximum value of $\sqrt{3}\cos x+\sin x=2$

This will be obtained at $x=\frac{\pi}{6}$

b) $\sin\cos x = \frac{1}{2}\cdot \sin 2x$ $\therefore$ Least value $= -\frac{1}{2}$

c) Extreme value of $a\sin\theta - b\cos\theta$ are $\pm\sqrt{a^2+b^2}$

d) $\frac{\tan^2\theta+\cot^2\theta}{2} \geq \sqrt{\tan^2\theta \times \cot^2\theta} \Rightarrow \tan^2\theta + \cot^2\theta \geq 2$

32. $1 + 8\sin^2(x^2)\cdot\cos^2(x^2) = 1 + 2[2\sin(x^2)\cdot\cos(x^2)]^2$

$= 1 + 2\sin^2(2x^2) = 1 + (1 - \cos 4x^2) = 2 - \cos(4x^2)$

Extrema value are $= c \pm \sqrt{a^2+b^2} = 2 \pm 1 \geq$ range$[1,3]$

33. a-s; b-p;c-q;d-r

a) $2\cos^2 x + \sin^2 x = 1 + \cos^2 x \in [1,2]$

b) $3 - \cos\theta + \cos\left(\theta + \frac{\pi}{3}\right)$

$= 3 - \cos\theta + \left[\cos\theta \times \frac{\sqrt{1}}{2} - \sin\theta \times \frac{\sqrt{3}}{2}\right] = 3 - \frac{1}{2}\cos\theta + -\frac{\sqrt{3}}{2}\sin\theta$

Range $= [3-1, 3+1] = [2,4]$

c) $\cos^2 x + \sin^4 x = \cos^2 x + (1-\cos^2 x)^2 = \cos^4 x - \cos^2 x + 1$

Range $= \left[\frac{3}{4}, 1\right]$

d) $3\sin^2 x + 4\cos^2 x = 3 + \cos^2 x$

Range $= [3,4]$. 34. 2π

WORKSHEET – 6

LEVEL – I

1) C	2) B	3) D	4) A	5) A
6) B	7) B	8) A	9) B	10) D

LEVEL – II

1) B	2) A	3) D	4) B	5) 3
6) B	7) B	8) A	9) D	10) B
11) A	12) A	13) C	14) B	15) B
16) C	17) A	18) D	19) D	20) B
21) C	22) B	23) B	24) A	
25) a-q, b-r, c-p, d-s		26) A	27) C	28) A
29) B	30) 1-r, 2-p,r, 3-q, 4-s		31) A, C	32) B
33) B, C	34) 1			

SOLUTIONS

LEVEL – II

1. $\cos\theta = \cos\frac{5\pi}{4} = -\cos\frac{\pi}{4} = \cos\left(\pi - \frac{\pi}{4}\right) = \cos\frac{3\pi}{4} \Rightarrow \theta = 2n\pi \pm \frac{3\pi}{4}$ $n \in Z$

2. $\sin^2\theta = 3 \Rightarrow \sin^2\theta = \frac{3}{4} \Rightarrow \theta = n\pi \pm \frac{\pi}{3}, n \in Z$

3. $\cos 2\theta \Rightarrow \cos^2\theta \Rightarrow 2\cos^2\theta - 1 = \cos^2\theta \Rightarrow \cos^2\theta = 1 \Rightarrow \theta = n\pi, n \in Z$

4. $\tan\theta = -1 \Rightarrow \tan\theta = \tan\left(\frac{-\pi}{4}\right) \Rightarrow \theta = n\pi\frac{-\pi}{4}, n \in z$

   $\cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \cos\theta = \cos\frac{\pi}{4} \Rightarrow \theta = 2n\pi \pm \frac{\pi}{4}, n \in Z$

   common solution is $= 2n\pi - \frac{\pi}{4} = 2n\pi + 2\pi - \frac{\pi}{4} = 2n\pi + \frac{7\pi}{4}$

5. $\tan\theta + \sec\theta = \sqrt{3} \Rightarrow \sin\theta + 1 = \sqrt{3}\cos\theta$

   $\Rightarrow \frac{\sqrt{3}}{2}\cos\theta - \frac{1}{2}\sin\theta = \frac{1}{2} \Rightarrow \cos\theta \cdot \cos\frac{\pi}{6} - \sin\theta \cdot \sin\frac{\pi}{6} = \frac{1}{2} \Rightarrow \cos\left(\theta + \frac{\pi}{6}\right) = \frac{1}{2}$

   $\therefore$ p.s of $\theta + \frac{\pi}{6} = \pi/3$

6. On verification $\pi/4$ satisfies

7. $\tan m\theta = \cot n\theta = \tan\left(\frac{\pi}{2} - n\theta\right); m\theta = k\pi + \frac{\pi}{2} - n\theta \Rightarrow \theta = \frac{k\pi + \frac{\pi}{2}}{m+n} = \frac{(2k+1)\pi}{2(m+n)}$

8. $\frac{1}{2}\sin 3x = \frac{1}{8} \Rightarrow \sin 3x = \frac{1}{2} \Rightarrow 3x = n\pi + (-1)^n \cdot \frac{\pi}{6}$

9. $\sin x = \cos^2 x \Rightarrow \sin^2 x + \sin x - 1 = 0 \Rightarrow \sin x = \frac{-1 \pm \sqrt{1+4}}{2} \Rightarrow \sin x = \frac{\sqrt{5}-1}{2}$

10. If $3\cos 2\theta + 2 = 7\sin\theta$

    $\Rightarrow 3(1 - 2\sin^2\theta) + 2 = 7\sin\theta \Rightarrow 6\sin^2\theta + 7\sin\theta - 5 = 0 \Rightarrow \sin\theta = \frac{1}{2}$ or $\frac{5}{3}$

    $\Rightarrow \theta = n\pi + (-1)^n\frac{\pi}{6}$

11. $3\tan 3x = 0 \Rightarrow \tan 3x = n\pi \Rightarrow x = \frac{n\pi}{3}$

12. $\frac{2}{3}\cos 3\alpha = \frac{3\sqrt{3}}{4} \Rightarrow \cos 3\alpha = \sqrt{3}(> 1)$ There exist no solution.

13. $3x - 78 = 90 - (x + 28)$ or $90 + x + 28°$

    $\Rightarrow 4x = 90 - 28 + 78 = 140 \Rightarrow x = \frac{140}{4} = 35° \Rightarrow x = 8°$

14. $2 + \sqrt{3}\sec x - 4\cos x = 2\sqrt{3} \Rightarrow 4\cos^2 x - 2\cos x + 2\sqrt{3}\cos x - \sqrt{3} = 0$

    $\Rightarrow (2\cos x - 1)(2\cos x + \sqrt{3}) = 0 \Rightarrow \cos x = \frac{1}{2}$ or $\frac{-\sqrt{3}}{2} \Rightarrow x = 2n\pi \pm \frac{\pi}{3}$ or $2n\pi \pm \frac{5\pi}{6} n \in Z$

15. $\frac{4\cos^3\theta - 3\cos\theta}{2(2\cos^2\theta - 1) - 1} = \frac{1}{2} \Rightarrow \frac{(4\cos^2\theta - 3)\cos\theta}{4\cos^2\theta - 3} = \frac{1}{2} \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}$

16. $\cos 20° = k, \cos x = 2k^2 - 1 = 2\cos^2 20 - 1 = \cos 40°$

    on verification, $x = 40\&320$ satisfy

17. $\cos^2 x + \sin^3 x = 1 \Rightarrow \sin^3 x = \sin^2 x \Rightarrow \sin^2 x(\sin x - 1) = 0 \Rightarrow \sin x = 0$ or $\sin x = 1$

    which possible at $x = 0, \pi/2, \pi\&2\pi \forall x \in [0,2\pi]$

18. $\Rightarrow y^2 - 2y + 1 = 0$ when $y = 3^{\cos x} \Rightarrow (y-1)^2 = 0 \Rightarrow y = 1 \Rightarrow 3^{\cos x} = 3^\circ$

$\Rightarrow \cos x = 0 \Rightarrow x = (2n+1)\pi/2$

19. $\left(e^{\sin x}\right)^2 - 4e^{\sin x} - 1 = 0 \Rightarrow e^{\sin x} = \frac{4 \pm \sqrt{20}}{2} = 2 \pm \sqrt{5} \Rightarrow \sin x = \log 2 \pm \sqrt{5} \Rightarrow x$ has no real roots. As $\sin x > 1$ or $\sin x < -1$ is not possible.

20. $8^y = 64 \Rightarrow y = 2$

21. $2^{\sin x + \cos y} = 1 = 2^\circ \Rightarrow \sin x + \cos y = 0 \Rightarrow \cos y = -\sin x$

$16^{\sin^2 x + \cos^2 y} = 4 = 16^{1/2} \Rightarrow \sin^2 x + \cos^2 y = \frac{1}{2}$

$\Rightarrow \sin^2 x = \frac{1}{4} \Rightarrow \sin x = \pm\frac{1}{2} = x = n\pi + (-1)^n\left(\frac{\pm\pi}{6}\right)$

$\cos y = -\sin x = \mp\frac{1}{2} \Rightarrow y = 2n\pi \pm \frac{2\pi}{3}$ or $2n\pi \pm \pi/3$

22. $\sin 6\theta - \sin 4\theta + \sin 2\theta = 0 \Rightarrow \sin 4\theta(2\cos 2\theta - 1) = 0$

$\Rightarrow \sin 4\theta = 0, \cos 2\theta = \frac{1}{2} \Rightarrow 4\theta = n\pi, 2\theta = 2n\pi \pm \pi/3 \Rightarrow \theta = \frac{n\pi}{4}$ or $n\pi \pm \frac{\pi}{6}$

23. $b^2\sin^2 x = (c - a\cos x)^2$

$\Rightarrow b^2(1 - \cos^2 x) = c^2 + a^2\cos^2 - 2ac\cos x \Rightarrow (a^2 + b^2)\cos^2 x - 2ac\cos x + c^2 - b^2 = 0$

$\therefore \cos\alpha + \cos\beta = \frac{2ac}{a^2 + b^2}$

24. write as $(a^2 - b^2)\tan^2\theta - 2ac\tan\theta + c^2 - b^2 = 0$

Then roots are $\tan\alpha, \tan\beta$

$\tan\alpha + \tan\beta = \frac{2ac}{a^2 - b^2}$ $\tan\alpha \cdot \tan\beta = \frac{c^2 - b^2}{a^2 - b^2}$ $\therefore \tan(\alpha + \beta) = \frac{2ca}{a^2 - c^2}$

25. Conceptual 26. Both I and II are correct, as $|c| > \sqrt{a^2 + b^2}$

26. $\sin^{10} x - \cos^{10} x = 1 \Rightarrow \sin^{10} x = 1 + \cos^{10} x$

$\sin^{10} x \le 1 \Rightarrow 1 + \cos^{10} x \le 1 \Rightarrow \cos^{10} x = 0 \Rightarrow \cos x = 0 \Rightarrow x = (2n+1)\pi/2$

28. $\cos^{100} x - \sin^{100} x = 1 \Rightarrow \cos^{100} x = 1 + \sin^{100} x$

$\therefore \cos^{100} x \le 1 + \sin^{100} x \le 1 \Rightarrow \sin^{100} x = 0 \Rightarrow \sin x = 0 \Rightarrow x = n\pi$

29. $\therefore \tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right) = 4$

$\Rightarrow (1 + \tan\theta)^2 + (1 - \tan\theta)^2 = 4(1 - \tan\theta)(1 + \tan\theta)$

$\Rightarrow 2(1 + \tan^2\theta) = 4(1 - \tan^2\theta) \Rightarrow 3\tan^2\theta \Rightarrow \tan^2\theta = \frac{1}{3} \Rightarrow \theta = n\pi \pm \frac{\pi}{6}$

30. conceptual

31. (1,3)$\sin\theta = \cos\theta \Rightarrow \cos\left(\frac{\pi}{2} - \theta\right) = \cos\phi$

$\Rightarrow \frac{\pi}{2} - \theta = 2n\pi = \theta \Rightarrow \frac{1}{\pi}\left(\theta \pm \phi - \frac{\pi}{2}\right) = -2n \Rightarrow$ 0&2 are possible

32. conceptual 33. By verification

33. For $x = 1$ we get L.H.S = R.H.S

WORKSHEET – 7

LEVEL – I

1. C	2. A	3. B	4. A	5. A

LEVEL – II

1. B	2. A	3. 4m	4. A	5. C
6. D	7. B	8. C	9. A	10. A
11. B	12. C	13. B	14. C	15. B
16. B	17. C	18. B	19.12	

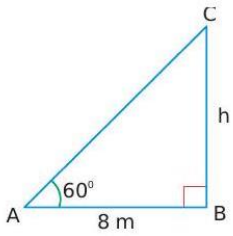
LEVEL – III

20. A, C	21. B	22. C	23. A	24. C
25.a-r;b-s;c-p;d-q;		26. C	27.B, C, D	28. B
29. C	30. B			

SOLUTIONS

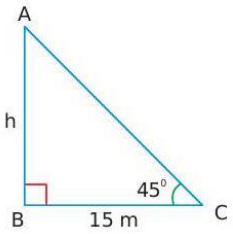
LEVEL – II

1.



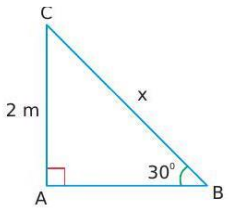
from figure, $\tan 60^\circ = \frac{h}{8} \Rightarrow h = 8 \times \tan 60^\circ = 8\sqrt{3}m$

2.



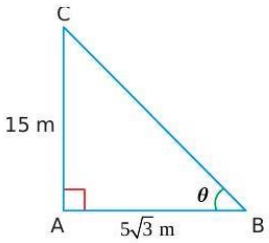
from figure, $\tan 45^0 = \frac{h}{15} \Rightarrow h = 15 \times \tan 45^0 = 15m$

3.



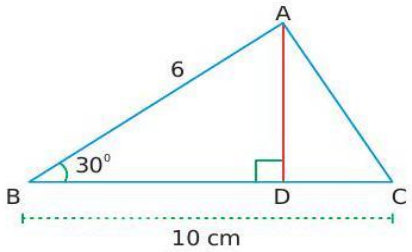
length of slide = x from figure, $\sin 30^0 = \frac{2}{x} \Rightarrow x = 4m$

4.



from figure $\tan\theta = \dfrac{15}{5\sqrt{3}}$; $\tan\theta = \sqrt{3} \Rightarrow \theta = 60^0$

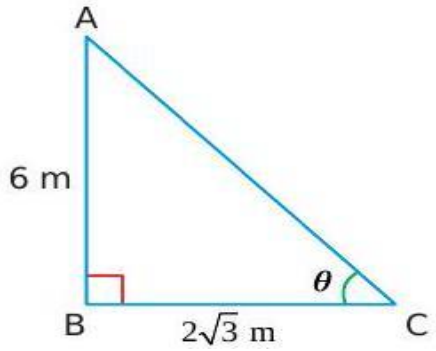
5.



From figure $\sin 30° = \frac{AP}{AB} \Rightarrow \frac{1}{2} = \frac{AD}{6} \Rightarrow AD = 3$ cm

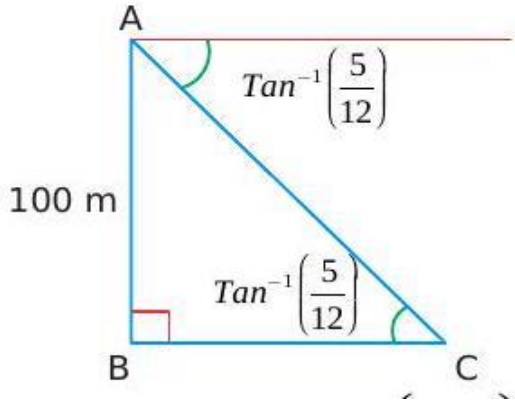
height = 3 cm

6.



From figure $\tan\theta = \frac{6}{2\sqrt{3}} = \frac{2\times3}{2\times\sqrt{3}} = \sqrt{3} \Rightarrow \theta = 60°$

7.



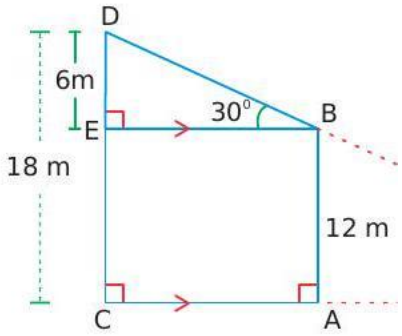
let $\theta = \tan^{-1}\left(\frac{5}{12}\right) \Rightarrow \tan\theta = \frac{5}{12} \Rightarrow \frac{100}{BC} = \frac{5}{12} \Rightarrow BC = 240$ m

Distance from ship = 240 m

8. From figure $\sin 30° = \frac{h}{AC} \Rightarrow \frac{1}{2} = \frac{h}{15} \Rightarrow h = \frac{15}{2}m$

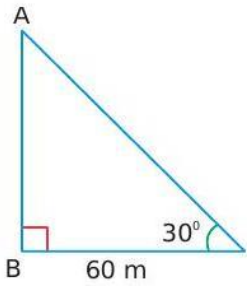
Height of kite above ground $= \frac{15}{2}m$

9.



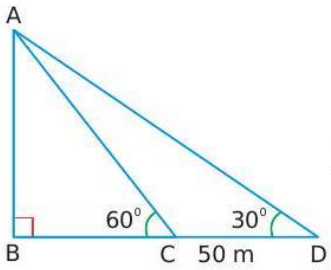
From figure $\sin 30° = \frac{DE}{DB} \Rightarrow \frac{1}{2} = \frac{6}{DB} \Rightarrow DB = 12$ m. Length of rope = 12 m

10.



From figure $\frac{AB}{BC} = \tan 30° \Rightarrow \frac{AB}{60} = \frac{1}{\sqrt{3}} \Rightarrow AB = \frac{60}{\sqrt{3}} = 20\sqrt{3}m$

11.

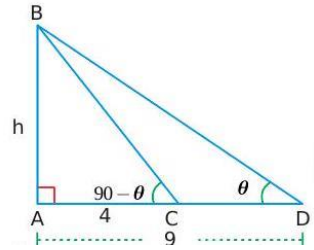


$$In\ \Delta ABC \tan 60^0 = \frac{h}{BC}$$

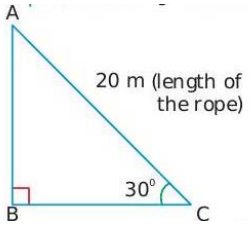
$\Rightarrow BC = \frac{h}{\sqrt{3}} \rightarrow (1)$

In $\triangle ABD \tan 30° = \frac{AB}{BD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{BC+50} \Rightarrow BC = \sqrt{3}h - 50 \rightarrow (2)$

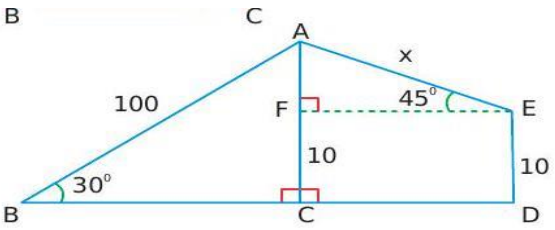
from (1), (2) $\frac{h}{\sqrt{3}} = \sqrt{3}h - 50 \Rightarrow h = 25\sqrt{3}$

12. 

$$h^2 = ab \Rightarrow h^2 = 4\times 9 \Rightarrow h^2 = 36 \Rightarrow h = 6m$$

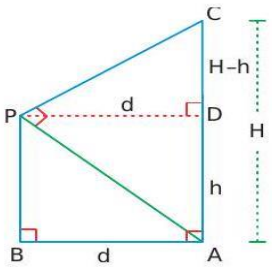
13. 

$$\text{from figure} \sin 30^0 = \frac{AB}{AC} \Rightarrow \frac{1}{2} = \frac{AB}{20} \Rightarrow AB = 10m$$

14. 

$$\text{In } \triangle ABC, \sin 30° = \frac{AC}{AB} \Rightarrow \frac{1}{2} = \frac{AC}{100} \Rightarrow AC = 50 \text{ m}$$

$$\text{In } \triangle AEF, \sin 45° = \frac{AF}{AE} \Rightarrow \frac{1}{\sqrt{2}} = \frac{AC-FC}{x} \Rightarrow \frac{1}{\sqrt{2}} = \frac{50-10}{x} \Rightarrow x = 40\sqrt{2}m$$

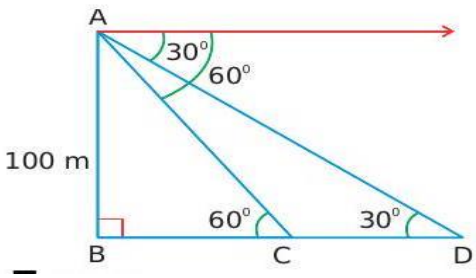
15. 

$$AP^2 = AB^2 + BP^2 = h^2 + d^2$$

$$PC^2 = (H-h)^2 + d^2$$

$$AC^2 = AP^2 + PC^2$$

$$\Rightarrow H^2 = h^2 + d^2 + (H-h)^2 + d^2 \Rightarrow H = \frac{h^2+d^2}{h}$$

16. 

From $\triangle ABD$,

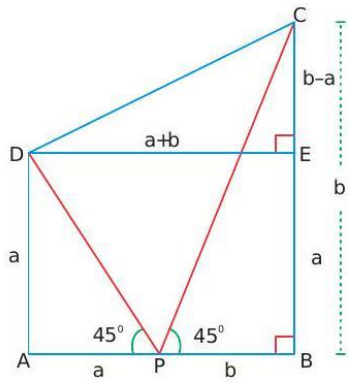
$\tan 30 = \frac{100}{BD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{100}{BD} \Rightarrow BD = 100\sqrt{3}$

from $\triangle ABC$

$\tan 60 = \frac{100}{BC} \Rightarrow \sqrt{3} = \frac{100}{BC} \Rightarrow BC = \frac{100}{\sqrt{3}}$ $\therefore$ distance travelled by a car

$= CD = BD - BC = 100\sqrt{3} - \frac{100}{\sqrt{3}} = \frac{200}{\sqrt{3}} m$

17.



In $\triangle BPC, \tan 45° = \frac{b}{PB} \Rightarrow PB = b$

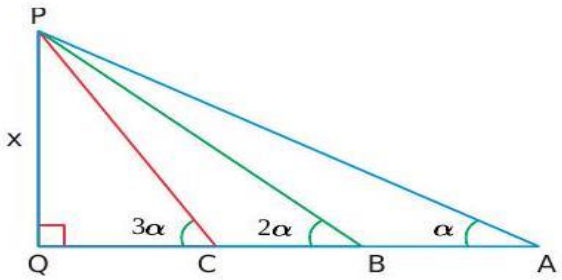
Similarly a = AP

$\therefore$ DE = a + b and CE = b − a

In $\triangle DEC, DC^2 + EC^2$

$\Rightarrow DC^2 = (a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$

18.



$\tan \alpha = \frac{x}{AQ}; \Rightarrow AQ = x\cot \alpha$:

$\tan 2\alpha = \frac{x}{BQ} \Rightarrow BQ = x\cot 2\alpha$:

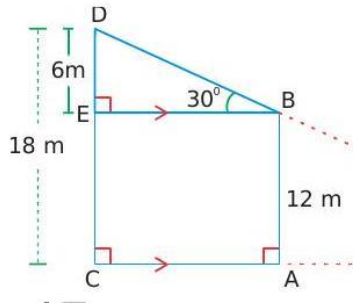
$\tan 3\alpha = \frac{x}{CQ} \Rightarrow CQ = x \cdot \cot 3\alpha$

$$\therefore \frac{AB}{BC} = \frac{AQ - BQ}{BQ - CQ} = \frac{x\cot\alpha - x\cot 2\alpha}{x\cot 2\alpha - x\cot 3\alpha} = \frac{\sin 3\alpha}{\sin\alpha} = \frac{3\sin\alpha - 4\sin^3\alpha}{\sin\alpha} = 3 - 4\sin^2\alpha$$

$$= 3 - 2(1 - \cos 2\alpha)$$

$$= 1 + 2\cos 2\alpha$$

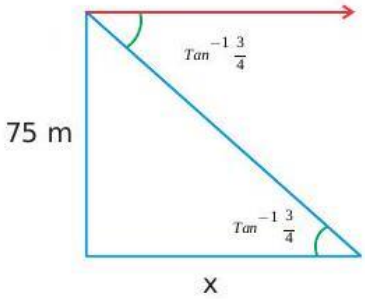
19.



from figure,

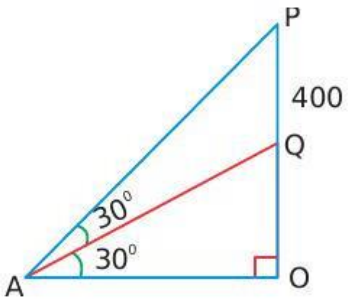
$$\frac{AE}{AC} = \sin 30° \Rightarrow AC = 6 \times 2 = 12m$$

20.



$$\tan[\tan^{-1} 3/4] = \frac{75}{x} \Rightarrow \frac{3}{4} = \frac{75}{x} \Rightarrow x = 100mts \Rightarrow x - 90 = 100 - 90 = 10m$$

21.



from $\triangle OAQ, \tan 30 = \frac{OQ}{OA} \Rightarrow OQ = OA \cdot \tan 30° = OA/\sqrt{3}$

$$\tan 60 = \frac{PQ + OQ}{OA} \Rightarrow \sqrt{3}(OA) = 400 + \frac{OA}{\sqrt{3}} = (\because \sqrt{3} = 1 \cdot 732)$$

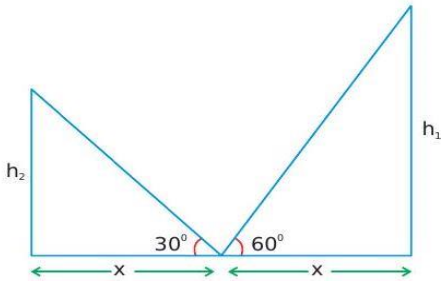
$$OA = 200\sqrt{3} = 346.4 \text{ cm} \Rightarrow \frac{\sqrt{3}}{2} = \frac{200\sqrt{3}}{AQ} \Rightarrow AQ = 400 \text{ cm}$$

22. $\sin 30° = \frac{OQ}{AQ} \Rightarrow \frac{1}{2} = \frac{OQ}{400} \Rightarrow OQ = 200$ cm and $OP = OQ + PQ = 200 + 400 = 600$ cm

23. AQ = 400 cm

24. Both statements I&II are true

25. a



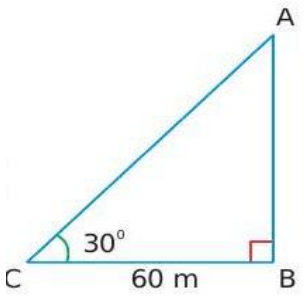
$\frac{h_2}{x} = \tan 30° \Rightarrow h_2 = \frac{x \cdot 1}{\sqrt{3}} = \frac{x}{\sqrt{3}} \Rightarrow d = 25\sqrt{6}$

$h_1 : h_2 = x\sqrt{3} : \frac{x}{\sqrt{3}} = 3:1 \Rightarrow h_1 = x\sqrt{3}$

Ratio of hight of tower and flag staff

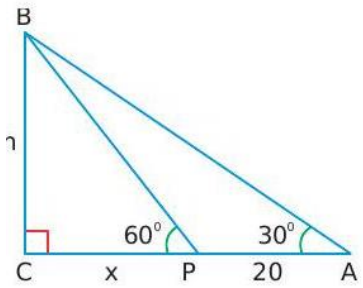
b) $\frac{\cos 75°}{\cos 45°} = \frac{\frac{\sqrt{3}-1}{2\sqrt{2}}}{\frac{1}{\sqrt{2}}} = \frac{\sqrt{3}-1}{2} = \sqrt{3} - 1 : 2$

c)



$\tan 30° = \frac{ab}{60} \Rightarrow ab = \frac{60}{\sqrt{3}} = 20\sqrt{3}m$

d) h

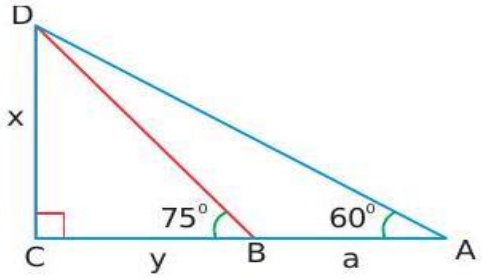


$\tan 60° = \frac{h}{x} \Rightarrow \sqrt{3} = \frac{h}{x} \Rightarrow h = \sqrt{3}x \rightarrow (1)$

$\tan 30° = \frac{h}{x+20} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x+20} \Rightarrow h = \frac{x+20}{\sqrt{3}} \rightarrow (2)$

height $= \sqrt{3}x = 10 \times \sqrt{3}m$

26.

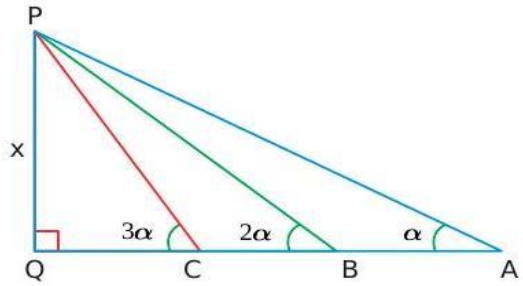


$\tan 60° = \frac{x}{y+a} \Rightarrow \sqrt{3} = \frac{x}{y+a} \Rightarrow x = \sqrt{3}(y+a) \rightarrow (1)$

$\tan 75° = \frac{x}{y} \Rightarrow x = y(2+\sqrt{3}) \rightarrow (2)$ from

from (1),(2) $y = \frac{\sqrt{3}}{2}a \therefore x = \sqrt{3}\left[a + \frac{\sqrt{3}}{2}a\right] = \frac{3a}{2[2\sqrt{3}-3]}$

27.



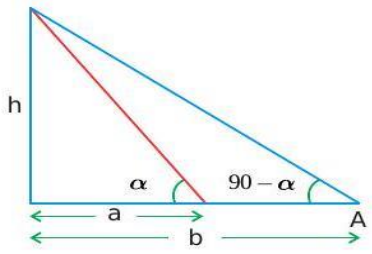
$\tan\alpha = \frac{x}{AQ} \Rightarrow AQ = x\cot\alpha$

similarly $BQ = x\cot 2\alpha$

$CQ = x \cdot \cot 3x$

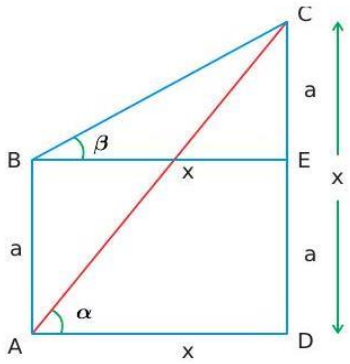
$\frac{AB}{BC} = \frac{AQ - BQ}{BQ - CQ} = 1 + \cos 2\alpha = 4\cos^2\alpha - 1 = 3 - 4\sin^2\alpha$

28.



$\tan\alpha = \frac{h}{a}; \tan(90-\alpha) = \frac{h}{b} \Rightarrow \tan\alpha \cdot \tan(90-\alpha) = \frac{h}{a} \cdot \frac{h}{b} \Rightarrow 1 = \frac{h^2}{ab} \Rightarrow h = \sqrt{ab}$

29.



$\tan\beta = \frac{a}{x} \Rightarrow x = a \cdot \cot\beta \rightarrow (1)$

$\tan\alpha = \frac{2a}{x} \Rightarrow \tan\alpha = \frac{2a}{a \cdot \cot\beta} \Rightarrow \tan\alpha \cdot \cot\beta = 2 \Rightarrow \cot\alpha \cdot \tan\beta = 1/2$

30. $\alpha = \cot^{-1}(3/5) \Rightarrow \cot\alpha = 3/5$ $\beta = \cot^{-1}(2/5) \Rightarrow \cot\beta = 2/5$

$h = \frac{32}{\frac{3}{5} - \frac{2}{5}} = 160m$

4. PROPERTIES OF TRIANGLES

WORKSHEET – 1

LEVEL – I

1) D	2) B	3) C	4) C	5) C	6) B
7) A	8) A	9) C	10) B		

LEVEL – II

1) A	2) C	3) A	4) B	5) A	6) C
7) 5cm	8) 2	9) B	10) C	11) A	12) A
13) D	14) C	15) B	16) C	17) D	18) 4
19) A	20) C	21) 0			

LEVEL – III

22) A, B, C	23) C	24) B	25) A	26) a-q,t; b-s; c-p,r, d-p,r
27) A, B		28) a-r,b-r,c-t,d-p	29) 2	30) 1
31) C	32) B, D			

SOLUTIONS

LEVEL – I

1. $\frac{b}{\sin B} = 2R \Rightarrow \frac{10}{\sin \mathbf{B}} = 2 \times 5 \Rightarrow b = 90^\circ$
2. $\frac{c-a}{c+a}$
3. $\underline{C} = 90^\circ$
4. $C = 180^\circ - (60^\circ + 30^\circ) = 90^\circ \Rightarrow a : b : c = \sin A : \sin B : \sin C = \frac{\sqrt{3}}{2} : \frac{1}{2} : 1 = \sqrt{3} : 1 : 2$
5. $a : b : c = 2 : \sqrt{6} : \sqrt{3} + 1 \Rightarrow a : b : c = \frac{2}{2\sqrt{2}} = \frac{\sqrt{6}}{2\sqrt{2}} = \frac{\sqrt{3}+1}{2\sqrt{2}} \Rightarrow \underline{C} = 75^\circ$
6. $\frac{a}{\sin A} = 2R \Rightarrow \frac{8}{\frac{\sqrt{3}}{2}} = 2R \Rightarrow R = 8/\sqrt{3}$.
7. Area $= \frac{1}{2}ab = \frac{1}{2} \times 3 \times 4 = 6$ sq.cm
9. $a\cos B = b\cos A \Rightarrow 2R\sin A\cos B = 2R\sin B \cdot \cos A \Rightarrow \sin A \cdot \cos B = \sin B \cdot \cos A$
   $\Rightarrow A - B = 0 \Rightarrow A = B \Rightarrow a = b$
10. Apply $c^2 = a^2 + b^2 - 2ab\cos c$

LEVEL – II

1. Let $A = B = C = 60^\circ$ and a = b = c
   $\therefore \Sigma a^2(\cos^2 B - \cos^2 C) = 0$
2. $a\cos B + b \cdot \cos A = c$

3. 5,12,13 form right angled triangle ⇒ hypotenuse = 13

   $\therefore$ circum radius $= \frac{\text{Hypotenuse}}{2} = \frac{13}{2} = 6.5$ cm

4. $\frac{a\cdot\cos B+b\cdot\cos A-b\cos A}{a\cdot\cos C+c\cdot\cos A-c\cdot\cos A} = \frac{\cos B}{\cos C}$

5. $\cos A = \frac{b^2+c^2-a^2}{2bc} \Rightarrow \frac{1}{2} = \frac{3^2+c^2-4^2}{2\times3\times c} \Rightarrow c^2-7 = 3c \Rightarrow c^2-3c-7 = 0$

6. Clearly if ratio is $3:4:5$, right angled triangle is formed

7. $\frac{a}{\sin A} = 2R \Rightarrow \frac{5}{\frac{1}{2}} = 2R \Rightarrow R = 5$ cm

8. $\lfloor A = 180^\circ - (30^\circ + 45^\circ) = 105^0$

   $\frac{a}{\sin A} = 2R \Rightarrow \frac{\sqrt{3}+1}{\frac{\sqrt{3}+1}{2\sqrt{2}}} = 2R \Rightarrow R = \sqrt{2}$ and $\frac{c}{\sin c} = 2R \Rightarrow \frac{c}{\frac{1}{\sqrt{2}}} = 2\sqrt{2} \Rightarrow c = 2$

9. $\frac{a-b\cos C}{b-a\cdot\cos C} = \frac{b\cos C+c\cdot\cos B-b\cos C}{a\cos C+c\cdot\cos A-a\cos C} = \frac{c\cdot\cos B}{c\cdot\cos A} = \frac{\cos B}{\cos A}$

10. Let $a = b = c = 1$ and $A = B = C = 60^\circ \Rightarrow\therefore \Sigma a^3\cos(B-C) = 3abc$

11. Given A, B, C are in $A.P \Rightarrow 2B = A + C \Rightarrow \underline{B} = \pi/3$

    Now $\cos B = \frac{a^2+c^2-b^2}{2ac} \Rightarrow a^2+c^2-b^2 = ac \Rightarrow b^2 = a^2+c^2-ac \rightarrow (1)$

    From (1)

    Now, $\frac{a+c}{\sqrt{a^2-ac+c^2}} = \frac{a+c}{b} = \frac{\sin A+\sin C}{\sin B} = 2\text{Cos}\frac{A-C}{2}$

12. $\text{Cot}\,A + \text{Cot}\,B + \text{Cot}\,C = \frac{\text{Cos}\,A}{\text{Sin}\,A} + \frac{\text{Cos}\,B}{\text{Sin}\,B} + \frac{\text{Cos}\,C}{\text{Sin}\,C}$

$$= \frac{b^2+c^2-a^2}{\frac{2bc}{\frac{a}{2R}}} + \frac{c^2+a^2-b^2}{\frac{2ca}{\frac{b}{2R}}} + \frac{a^2+b^2-c^2}{\frac{2ab}{\frac{c}{2R}}} = \frac{R}{abc}[a^2+b^2+c^2] = \frac{a^2+b^2+c^2}{4\Delta}$$

13. Given ratio of angles = $3:5:10$; Smallest angle = 30°; Greatest angle = 100°

    Ratio of smallest to greatest sides = $\sin 30^\circ : \sin 100^\circ$

$$= \frac{1}{2} : \sin(90+10^\circ) = \frac{1}{2} : \text{Cos}\,10^\circ = 1 : 2\cos 10^\circ$$

14. $(c^2)^2 + (a^2)^2 + (b^2)^2 - 2a^2c^2 - 2b^2c^2 + 2a^2b^2 = a^2b^2 \Rightarrow (a^2+b^2-c^2)^2 = a^2b^2$

    $\Rightarrow \left(\frac{a^2+b^2-c^2}{2\mathbf{a}b}\right) = \left(\frac{1}{2}\right)^2$

    $\cos C = \frac{1}{2} = \cos\frac{\pi}{3} = C = 60^\circ$

15. Let third side be c. Then , $1 + c > 2004 \Rightarrow$

    $c > 2003$ and $1 + 2004 > c \Rightarrow 2005 > c \Rightarrow 2005 > c > 2003$

    The only possible integral value of c is 2004.

16. $\sin A : \sin B : \sin C = \frac{1}{\sqrt{2}} : \frac{2}{\sqrt{5}} : \frac{3}{\sqrt{10}}$ then $\text{Sin}\,A : \text{Sin}\,B : \text{Sin}\,C = \sqrt{5} : 2\sqrt{2} : 3$

17. Given SinA,SinB are roots of $c^2x^2 - c(a+b)x + ab = 0$

$\Rightarrow c^2x^2 - cax - cbx + ab = 0 \Rightarrow cx(cx-a) - b(cx-a) = 0 \Rightarrow (cx-a)(cx-b) = 0$

or $cx - b = 0 \Rightarrow x = \frac{a}{c}$ or $x = \frac{b}{c}; \therefore \text{Sin}\, A = \frac{a}{c} : \text{Sin}\, B = \frac{b}{c} \Rightarrow \text{Sin}\, C = \frac{c}{c} = 1 \Rightarrow C = 90°$

18. $\frac{a}{\text{Sin}\, A} = 2R \Rightarrow \frac{4}{\frac{1}{2}} = 2R \Rightarrow R = 4$

19. Apply sine rule $\frac{\text{Sin}\lfloor BAD}{x} = \frac{\text{Sin}\, 60°}{AD}$ and $\frac{\text{Sin}\lfloor CAD}{3x} = \frac{\text{Sin}\, 45°}{AD}$

20. Solving we get $\frac{a^2+b^2+c^2}{2abc} = \frac{(a+b+c)^2-2(ab+bc+ca)}{2abc}$

21. $\sum \frac{\sin B\cos C - \cos B\sin C}{4R^2 \sin B \sin C} = \frac{1}{4R^2}\sum(\cot C - \cot B)$

$= \frac{1}{4R^2}(-\cot B + \cot A - \cot C + \cot B - \cot A + \cot C) = \frac{1}{4R^2} \times 0 = 0$

LEVEL – III

22. $\frac{a}{\text{Sin}\, A} = \frac{b}{\text{Sin}\, B} = \frac{c}{\text{Sin}\, C} = 2R$

23. $\frac{a}{\text{Sin}\, A} = 2R$

24. $\frac{b}{\text{Sin}\, B} = 2R \Rightarrow \frac{10}{\text{Sin}\, B} = 2 \times 10 \Rightarrow B = 30°; \therefore \lfloor A + \lfloor = 150°$

25. $\frac{c}{\text{Sin}\, C} = 2R$

26. Conceptual

27. Given $\frac{\text{Cos}\, A}{a} = \frac{\text{Cos}\, B}{b} = \frac{\text{Cos}\, C}{c} \Rightarrow \frac{\text{Cos}\, A}{2R\text{Sin}\, A} = \frac{\text{Cos}\, B}{2R\text{Sin}\, B} = \frac{\text{Cos}\, C}{2R\text{Sin}\, C} \Rightarrow A = B = C \therefore \triangle ABC$ is equilateral.

Area $= \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times 2^2 = \sqrt{3}$

28. a) $(a+c)^2 - b^2 = 3ac \Rightarrow a^2 + c^2 + 2ac - b^2 = 3ac \Rightarrow a^2 + c^2 - b^2 = ac$

$\text{Cos}\, B = \frac{a^2+c^2-b^2}{2ac} = \frac{ac}{2ac} = \frac{1}{2} \Rightarrow B = 60° = \frac{\pi}{3}$

b) $\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{\frac{1}{4}(6+2+2\sqrt{2})+2-3}{2\times\frac{1}{2}(\sqrt{6}+\sqrt{2})\times\sqrt{2}} = \frac{2+\sqrt{3}+2-3}{2\sqrt{3}+2}$

$\cos A = \frac{\sqrt{3}+1}{2(\sqrt{3}+1)} = \frac{1}{2} = \cos 60° = A = \frac{\pi}{3}$

c) $\frac{a}{\text{Sin}\, A} = 2R \Rightarrow R = 5$

d) $a + b + c = 6 \times \frac{1}{3}[\text{Sin}\, A + \text{Sin}\, B + \text{Sin}\, C] \Rightarrow a + b + c = 6 \times \frac{1}{3}\left[\frac{a+b+c}{2R}\right] \Rightarrow 6R = 6 \Rightarrow R = 1$

Given a = 1 $\Rightarrow \frac{a}{\text{Sin}\, A} = 2 \times 1 \Rightarrow \frac{1}{\text{Sin}\, A} = 2 \Rightarrow A = 30°$

$\therefore a \to r; b \to r; c - t; d \to p$

29. $A = 75°, B = 45° \Rightarrow C = 60° \Rightarrow 2R = \frac{a}{\sin 75} \Rightarrow 2R = \frac{a}{\cos 15°}$

$2R = \frac{a}{\frac{\sqrt{3}+1}{2\sqrt{2}}} \Rightarrow 2R = \frac{2\sqrt{2}a}{\sqrt{3}+1} \Rightarrow b + c\sqrt{2} = (2R\sin B + 2R\sin C\sqrt{2})$

$= 2R(\sin B + \sin C\sqrt{2}) = 2R\left(\sin 45° + \sin 60° \times \sqrt{2}\right) = \frac{2\sqrt{2}a}{\sqrt{3}+1}\left(\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}\sqrt{2}\right)$

$= \frac{2\sqrt{2}a}{\sqrt{3}+1}\left(\frac{\sqrt{3}+1}{\sqrt{2}}\right) = 2a.$

30. Ratio of angles $= 2:3:7$

I angle $= 2 \times 15 = 30°$; II angle $= 3 \times 15° = 45°$

III angle $= 7 \times 15 = 105°$

Ratio of sides $= \text{Sin}\,30°: \text{Sin}\,45°: \text{Sin}\,105°$

$= \frac{1}{2} : \frac{1}{\sqrt{2}} : \frac{\sqrt{3}+1}{2\sqrt{2}} = \sqrt{2}: 2: \sqrt{3}+1$

31. $\frac{b}{\text{Sin}\,B} = \frac{c}{\text{Sin}\,C} \Rightarrow \frac{\text{Sin}\,B}{\text{Sin}\,C} = \frac{b}{c} \Rightarrow \frac{\text{Sin}\,B}{\text{Sin}(90+B)} = \frac{b}{c} \Rightarrow \frac{\text{Sin}\,B}{\text{Cos}\,B} = \frac{b}{c} \Rightarrow \text{Tan}\,B = \frac{b}{c} = 3$

$[\therefore C - B = 90]$

32. Given $c^2x^2 - cax - cbx + ab = 0 \Rightarrow cx[cx - a] - b[(c - a)] = 0$

$\Rightarrow x = \frac{a}{c}, y = \frac{b}{c} \Rightarrow \text{Sin}\,A = \frac{a}{c}, \text{Sin}\,B = \frac{b}{c} \quad \therefore \text{Sin}\,C = \frac{c}{c} = 1 \Rightarrow C = 90°$

and $\text{Sin}\,A + \text{Sin}\,B = \frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$

WORKSHEET – 2

LEVEL – I

1) 1	2) 4	3) 1	4) 1	5) 2
6) 1	7) 2			

1. B	2. D	3. D	4. A	5. A
6. C	7. A	8. 3	9. B	10. B
11. C	12. A	13. 4	14. A	15. C
16. B	17. D	18. B	19. A	20. B
21. B	22. A			

LEVEL – III

23. A, B, C 24. D 25. A 26. D 27. C
28. C 29. D 30. A, B 31. C, D

32) $a - p; b - q; c - r; d - s$

SOLUTIONS
LEVEL – II

1. $s = 10$

$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$\Delta = \sqrt{10(10-4)(10-7)(10-9)}$

$= \sqrt{10 \times 6 \times 3 \times 1} = \sqrt{180}$

$\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$

$\text{Tan}\frac{A}{2} = \frac{\sqrt{180}}{10(6)} = \frac{\sqrt{180}}{10 \times 6} = \frac{3\sqrt{20}}{10 \times 6} = \frac{1}{\sqrt{20}}$

or

$= \sqrt{\frac{3 \times 1}{10 \times 6}} = \frac{1}{\sqrt{20}}$

2. $\text{Tan}\frac{A}{2} = \frac{\Delta}{s(s-a)} = \frac{30}{15 \times 10} = \frac{1}{5}$

3. $(a^2 + b^2 - 2ab)\cos^2\frac{C}{2} + (a^2 + b^2 + 2ab)\sin^2\frac{C}{2}$

$= (a^2 + b^2)\left(\cos^2\frac{C}{2} + \sin^2\frac{C}{2}\right) - 2ab\left(\cos^2\frac{C}{2} - \sin^2\frac{C}{2}\right) = a^2 + b^2 - 2ab\cos C$

4. $\cos^2\frac{C}{2} = \frac{s(s-c)}{ab}; \cos^2\frac{B}{2} = \frac{s(s-b)}{ac} \Rightarrow \frac{s(s-a)}{a} + \frac{s(s-b)}{a} = \frac{2\,s^2 - s(b+c)}{a}$

$= \frac{2\,S^2 - 2\,S^2 + Sa}{a} = \frac{sa}{a} = s$

5. $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

6. a, b, c are in A.P. s-a, s-b, s-c are in A.P ; so $\text{Tan}\frac{A}{2}, \text{Tan}\frac{B}{2}, \text{Tan}\frac{C}{2}$ are in H.P

7. $\cot\frac{A}{2}, \cot\frac{B}{2}, \cot\frac{C}{2}$ are in A.P $\Rightarrow$ a, b, c are in AP then $\tan\frac{A}{2}\tan\frac{B}{2}\tan\frac{C}{2}$ are in HP

$\cot\frac{A}{2}\cot\frac{B}{2}\cot\frac{C}{2}$ are in AP

8. $a \cdot \frac{s(s-c)}{ab} + c \cdot \frac{s(s-a)}{bc} = \frac{kb}{2} \Rightarrow \frac{s}{b}[s - c + s - a] = \frac{kb}{2} \Rightarrow \frac{s}{b}[b] = \frac{kb}{2} \Rightarrow a + b + c = kb$ Given

a, b, c are in AP $\Rightarrow$ 2 b = a + c $\Rightarrow b + 2b = kb \Rightarrow 3b = kb \Rightarrow k = 3$

9. $b + c = 3a \Rightarrow s = \frac{a+b+c}{2} = \frac{a+3a}{2} = 2a$ $\cot\frac{B}{2} \cdot \cot\frac{C}{2} = \frac{s(s-b)}{\Delta}\frac{s(s-c)}{\Delta} = \frac{s \times s(s-b)(s-c)}{s(s-a)(s-b)(s-c)} =$

$\frac{s}{s-a} = \frac{2a}{a} = 2$

10. $\frac{\sin\left(\frac{A}{2}+B\right)}{\sin\left(\frac{A}{2}\right)} = \frac{2 \cdot \cos\frac{A}{2} \cdot \sin\left(B+\frac{A}{2}\right)}{2 \cdot \sin\frac{A}{2} \cdot \cos\frac{A}{2}} = \frac{\sin(A+B)+\sin B}{\sin A} = \frac{\sin C + \sin B}{\sin A} = \frac{b+c}{a}$

11. $a=b=c=1; A=B=C=60^\circ$ or $abc\sqrt{\frac{s(s-a)}{bc}}\times\sqrt{\frac{s(s-b)}{ca}}\times\sqrt{\frac{s(s-c)}{ab}}=$

$abc\sqrt{\frac{s^2\times s(s-a)(s-b)(s-c)}{(abc)^2}}=abc\times s\frac{\Delta}{abc}=\Delta s$

12. $\frac{2s(2s-2a)}{bc}=3\Rightarrow 4\ s(\ s-a)=3bc\Rightarrow\frac{s(s-a)}{b-c}=\frac{3}{4}\Rightarrow\cos\frac{A}{2}=\frac{\sqrt{3}}{2}=\frac{A}{2}=30^\circ\Rightarrow A=60^\circ$

13. $\cos A+\sin A-\frac{2}{\cos B+\sin B}=0$

$\Rightarrow \cos A\cos B+\sin A\sin B+\cos A\sin B+\cos B\sin A=2\Rightarrow\cos(A-B)+\sin(A+B)=2$

$\Rightarrow \cos(A-B)=1,\ \sin(A+B)=1$

$\Rightarrow A-B=0\Rightarrow A=B$, So $a=b$ and $\sin 2A=1\Rightarrow A=45^\circ$ or 135°(Not possible)

$\therefore\frac{a+b}{c}=\frac{2a}{a\sqrt{2}}=\sqrt{2}\Rightarrow\left(\frac{a+b}{c}\right)^4=(\sqrt{2})^4=4$

14. Given $A=3C\Rightarrow A+B+C=180^\circ\Rightarrow 2C=90-\frac{B}{2}$

$\frac{a-c}{2c}=\frac{\sin A-\sin C}{2\cdot\sin C}=\frac{2\cos\frac{A+C}{2}\cdot\sin\frac{A-C}{2}}{2\cdot\sin C}$

$=\frac{2\cdot\cos 2C\cdot\sin C}{2\cdot\sin C}=\cos 2C=\cos\left(90-\frac{B}{2}\right)=\sin\frac{B}{2}$

15. $a+b+c=2\ s$

$\tan\frac{A}{2}=\frac{\Delta}{s(s-a)},\tan\frac{B}{2}=\frac{\Delta}{s(s-b)}$

$(a+b+c)\left(\tan\frac{A}{2}+\tan\frac{B}{2}\right)=2s\left[\frac{\Delta}{s(s-a)}+\frac{\Delta}{s(s-b)}\right]=\frac{2s\Delta}{s}\left[\frac{1}{s-a}+\frac{1}{s-b}\right]$

$=2\Delta\left[\frac{s-b+s-a}{(s-a)(s-b)}\right]=2\Delta\left[\frac{2s-(a+b)}{(s-a)(s-b)}\right]=\frac{2\Delta c}{(s-a)(s-b)}=2c\cot\left(\frac{C}{2}\right)$

$\therefore\Delta=\sqrt{s(s-a)(s-b)(s-c)}\Rightarrow\frac{\Delta}{(s-a)(s-b)}=\sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$

16. $a+b+c=2R[\sin A+\sin B+\sin C]=2R\left(4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}\right)=8R\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}.$

17. $\sqrt{\frac{(s-b)(s-c)}{bc}}\times\frac{s(s-b)}{ac}=\frac{a+c-b}{2c}\times k\Rightarrow\frac{(s-b)}{c}\sqrt{\frac{s(s-c)}{ab}}=\frac{s-b}{c}\times k\Rightarrow k=\sqrt{\frac{s(s-c)}{ab}}=\cos\frac{C}{2}$

18. $\frac{\Delta}{s(s-a)}+\frac{\Delta}{s(s-c)}=\frac{\Delta}{s}\left(\frac{b}{(s-a)(s-c)}\right)=\frac{\Delta}{s}\cot\frac{B}{2}$

19. $\Delta=(a+b-c)(a-b+c)$

$\Delta^2=[2(s-b)2(s-c)]^2\Rightarrow s(s-a)(s-b)(s-c)=16(s-b)^2(s-c)^2$

$\frac{(s-b)(s-c)}{s(s-a)}=\frac{1}{16}\Rightarrow\text{Tan}\left(\frac{A}{2}\right)=\frac{1}{4}\Rightarrow\text{Tan}^2A=\frac{2\tan\left(\frac{A}{2}\right)}{1-\tan^2\left(\frac{A}{2}\right)}=\frac{2\left(\frac{1}{4}\right)}{1-\left(\frac{1}{16}\right)}=\frac{8}{15}$

20. We know from geometry that the greatest altitude is perpendicular to the shortest side.

Let a = 17, b = 25 and c = 28.

Now $\Delta = \frac{1}{2} AD \times BC$ or $AD = \frac{2\Delta}{17}$

Where $\Delta^2 = s(s-a)(s-b)(s-c) = (210)^2 \Rightarrow AD = \frac{420}{17}$

21. Let us consider $\frac{b-c}{a}$

which is involved in three of the options

$$\frac{b-c}{a} = \frac{\sin B - \sin C}{\sin A} = \frac{2\cos\frac{B+C}{2} \cdot \sin\frac{B-C}{2}}{2\sin\left(\frac{A}{2}\right)\cos\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{A}{2}\right)\sin\left(\frac{B-C}{2}\right)}{\sin\left(\frac{A}{2}\right)\cos\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{B-C}{2}\right)}{\cos\left(\frac{A}{2}\right)}$$

$$\therefore (b-c)\cos\left(\frac{A}{2}\right) = a\sin\left(\frac{B-C}{2}\right)$$

22. $a^4 + b^4 + c^4 = b^2c^2 + 2c^2a^2 + 2a^2b^2$

$(a^2)^2 + (b^2)^2 + (c^2)^2 - 2a^2c^2 - 2a^2b^2 + 2b^2c^2 = 3b^2c^2 \Rightarrow (b^2 + c^2 - a^2)^2 = 3b^2c^2$

$(2bc\cos A)^2 = 3b^2c^2 \Rightarrow \cos^2 A = \left(\frac{\sqrt{3}}{2}\right)^2$

LEVEL – III

23. use $\text{Tan}\frac{A}{2}$ formula

24. $\sin\left(\frac{A}{2}\right) = \sqrt{\frac{(s-b)(s-c)}{bc}}$

25. $\cos\left(\frac{A}{2}\right) = \sqrt{\frac{s(s-a)}{bc}}$

26. $\text{Tan}\frac{A}{2} = \frac{\Delta}{s(s-a)}$

27. $\frac{bx}{c} + \frac{cy}{a} + \frac{az}{b} = b\sin B + c\sin C + a\sin A = \frac{b^2+c^2+a^2}{2R}$

k = 2R x = csin A, y = asin C, z = bsin A

= bsin C = csin A z = asin B

28. $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{a^2}{4\Delta^2} + \frac{b^2}{4\Delta^2} + \frac{c^2}{4\Delta^2} = \frac{a^2+b^2+c^2}{4\Delta^2}$

$$= \frac{R}{abc}(b^2 + c^2 + a^2) = \frac{R}{abc}\left(\frac{4\Delta^2}{x^2} + \frac{4\Delta^2}{y^2} + \frac{4\Delta^2}{z^2}\right)$$

$$= \frac{4\Delta^2 R}{abc}\left(\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}\right) = \frac{4\Delta R}{abc} \cdot \Delta\left(\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}\right) = \Delta\left(\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}\right)$$

29. $\sum \frac{c\sin B+b\sin C}{x}, \sum \frac{x+x}{x} = 6$

30. $\cos\left(\frac{A}{2}\right) = \sqrt{\frac{b+c}{2c}} \Rightarrow \sqrt{\frac{b+c}{2}} = \sqrt{\frac{s(s-a)}{bc}} \Rightarrow b^2 + bc = 2s(s-a)$

$\Rightarrow b^2 + bc = 2s(s-a) = (a+b+c)\frac{(b+c-a)}{2} \Rightarrow a^2 + b^2 = c^2 \Rightarrow \underline{C} = \frac{\Pi}{2}$

31. $AD = c\sin B[$ as $\angle ADP = \angle AEP = \frac{\Pi}{2}]$

using sine rule in triangle APQ, we get

$\frac{PQ}{\sin A} = 2R^1 = AD = c\sin B \Rightarrow PQ = c\sin B\sin A = c\frac{b}{2R}\sin A = \frac{2\Delta}{2R} = \frac{\Delta}{R}$

Also $PQ = c\sin B\sin A = 2R\sin C\sin B\sin A$

32. A) $(b+c)^2 - a^2 = \lambda bc$ (or) $b^2 + c^2 - a^2 = (\lambda - 2)bc$

$\frac{b^2+c^2-a^2}{2bc} = \frac{\lambda-2}{2}, \cos A = \frac{\lambda-2}{2}$ (or) $\lambda - 2 < 2;$

$$\lambda < 4 \text{ or } \lambda = 3$$

B) $\tan A + \tan B + \tan C = 9$ in any triangle $\tan A + \tan B + \tan C = \tan A\tan B\tan C$

$$\frac{\tan^2 A + \tan^2 B + \tan^2 C}{3} \geq (\tan A\tan B\tan C)^{\frac{2}{3}}$$

$k \geq 3(9)^{\frac{2}{3}};\ k \geq 9 \cdot 3^{\frac{1}{3}}$

C) Since the line joining the circum center to the incentre is parallel to BC

$r = R \cdot \cos A$

$$4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} = R\cos A$$

$$-1 + \cos A + \cos B + \cos C = \cos A \Rightarrow \cos B + \cos C = 1$$

D) a = 5, b = 4

$$\cos(A-B) = \frac{31}{32}\tan\frac{A-B}{2} = \frac{a-b}{a+b}\cot\frac{C}{2} = \frac{1}{9}\cot\frac{C}{2} \Rightarrow \cos(A-B) = \frac{1-\tan^2\frac{A-B}{2}}{1+\tan^2\frac{A-B}{2}}$$

$$\frac{31}{32} = \frac{1-\frac{1}{81}\cot^2\frac{C}{2}}{1+\frac{1}{81}\cot^2\frac{C}{2}} \Rightarrow 31 + \frac{31}{81}\cot^2\frac{C}{2} = 32 - \frac{32}{81}\cot^2\frac{C}{2} \Rightarrow \frac{7}{9}\cot^2\frac{c}{2} = 1 \Rightarrow \cot^2\frac{c}{2} = \frac{9}{7}$$

$$\cos C = \frac{1-\text{Tan}^2\frac{C}{2}}{1+\tan^2\frac{C}{2}} = \frac{1-\frac{7}{9}}{1+\frac{7}{9}} = \frac{2}{16} = \frac{1}{8}$$

$$\Rightarrow \cos C = \frac{a^2+b^2-c^2}{2ab} \Rightarrow \frac{25+16-c^2}{2\times 20} = \frac{1}{8}$$

$25 + 16 - c^2 = 5 \Rightarrow c^2 = 36 \Rightarrow c = 6$

WORKSHEET – 3

1) A	2) B	3) C	4) B	5) A
6) A	7) A	8) A	9) A	

LEVEL – II

1) A	2) B	3) C	4) A	5) B
6) C	7) D	8) B	9) 1	10) B
11) B	12) A	13) 2	14) C	15) A
16) 5	17) D	18) C	19) 3	20) A
21) C	22) 3			

LEVEL – III

23) A, B	24) B	25) A	26) C
27) a-q;b-s;c-r;d-p		28)2,3,4 29)4	30) A
31) C	32) a-r;b-p;c-s;d-q		33) A, C, D

SOLUTIONS

LEVEL – I

1. $\frac{s-a}{\Delta}+\frac{s-b}{\Delta}+\frac{s-c}{\Delta}=\frac{3s-2s}{\Delta}=\frac{s}{\Delta}=\frac{1}{r}$

2. $r\cdot r_1\cdot r_2\cdot r_3=\frac{\Delta}{s}\cdot\frac{\Delta}{s-a}\cdot\frac{\Delta}{s-b}\cdot\frac{\Delta}{s-c}=\frac{\Delta^4}{s(s-a)(s-b)(s-c)}=\frac{\Delta^4}{\Delta^2}=\Delta^2$

3. $\frac{\Delta}{s-a}=r_1$

4. $r_2=\frac{\Delta}{s-b}$

5. $4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}=r$

6. $a=4,b=5,c=3\Rightarrow a+b+c=25\Rightarrow 4+5+3=12=25\Rightarrow s=\frac{12}{2}=6$

7. $s\cdot\tan\frac{A}{2}=r_1$

8. $r=\frac{\Delta}{s}$

9. $\Delta=8,\ s=4\Rightarrow\therefore r=\frac{\Delta}{s}=\frac{8}{4}=2$

LEVEL – II

1. $a=4,b=5,c=3\Rightarrow s=\frac{a+b+c}{2}=\frac{4+5+3}{2}=6$

   $\Delta=\sqrt{s(s-a)(s-b)(s-c)}=\sqrt{6\times2\times1\times3}=\sqrt{36}=6\Rightarrow\therefore r=\frac{\Delta}{s}=\frac{6}{6}=1$

2. $a=5,b=12,c=13\Rightarrow s=\frac{5+12+13}{2}=\frac{30}{2}=15$

$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{15 \times 10 \times 3 \times 2} = \sqrt{900} = 30 \Rightarrow r_1 = \dfrac{\Delta}{s-a} = \dfrac{30}{10} = 3$

3. $s = 15, b = 7, \Delta = 24 \Rightarrow r_2 = \frac{\Delta}{s-b} = \frac{24}{15-7} = \frac{24}{8} = 3$

4. $\Delta = 48, s = 16, c = 4 \Rightarrow r_3 = \frac{\Delta}{s-c} = \frac{48}{16-4} = \frac{48}{12} = 4$

5. $r = 2, r_1 = 4, r_2 = 6, r_3 = 12$

$\Delta^2 = rr_1r_2r_3 = 2 \times 4 \times 6 \times 12 = 576 \Rightarrow \Delta = \sqrt{576} = 24$

6. $\Delta = 96, 2s = 16 \Rightarrow s = \frac{16}{2} = 8 \Rightarrow r = \frac{\Delta}{s} = \frac{96}{8} = 12$

7. $\Delta = 6,\ r = 2 \Rightarrow s = \frac{\Delta}{r} = \frac{6}{2} = 3 \Rightarrow$ perimeter $= 2s = 6$ units

8. $r = (s-b)\tan\dfrac{B}{2}$ 9. $\dfrac{1}{r_1} + \dfrac{1}{r_2} + \dfrac{1}{r_3} = \dfrac{1}{r} = \dfrac{k}{r} \Rightarrow k = 1$

$10(r_1 - r)(r_2 - r)(r_3 - r)$

$= 4R\sin^2\dfrac{A}{2} \times 4R\sin^2\dfrac{B}{2} 4R\sin^2\dfrac{C}{2} = 4R\left(4R\sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2}\right)^2 = 4Rr^2 = 4Rr^2$

11. $r + r_3 + r_1 - r_2 =$

$\Rightarrow 4R\sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2} + 4R\cos\left(\dfrac{A}{2}\right)\cos\left(\dfrac{B}{2}\right)\sin\left(\dfrac{C}{2}\right) + 4R\sin\left(\dfrac{A}{2}\right)\cos\left(\dfrac{B}{2}\right)\cos\left(\dfrac{C}{2}\right)$

$-4R\cos\left(\dfrac{A}{2}\right)\sin\left(\dfrac{B}{2}\right)\cos\left(\dfrac{C}{2}\right)$

$= 4R\sin\dfrac{A}{2}\left[\sin\dfrac{B}{2}\sin\dfrac{C}{2} + \cos\dfrac{B}{2}\cos\dfrac{C}{2}\right] - 4R\left[\cos\dfrac{A}{2}\right]\left[\sin\dfrac{B}{2}\cos\dfrac{C}{2} - \cos\dfrac{B}{2}\sin\dfrac{C}{2}\right]$

$= 4R\sin\dfrac{A}{2}\left[\cos\left(\dfrac{B-C}{2}\right)\right] - 4R\cos\dfrac{A}{2}\left(\sin\left(\dfrac{B-C}{2}\right)\right)$

$= 4R\sin\left[\dfrac{A-B+C}{2}\right] = 4R\sin(90 - B) = 4R\cos B$

12. $\cos A + \cos B + \cos C = 1 + 4\sin\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)\sin\left(\frac{C}{2}\right)$

$1 + \dfrac{\left[4R\sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2}\right]}{R} = 1 + \dfrac{r}{R}$

13. $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \Rightarrow \frac{1}{r} = \frac{1}{3} + \frac{1}{10} + \frac{1}{15} = \frac{10+3+2}{30} = \frac{1}{2} \Rightarrow r = 2$

14. Given $rr_3 = r_1r_2 \Rightarrow \frac{\Delta}{s}\frac{\Delta}{s-c} = \frac{\Delta}{s-a}\frac{\Delta}{s-b} \Rightarrow \frac{(s-a)(s-b)}{s(s-c)} = 1$

$\Rightarrow \tan^2\dfrac{C}{2} = 1 \Rightarrow \tan\dfrac{C}{2} = 1 = \tan\dfrac{\pi}{4} \Rightarrow \dfrac{C}{2} = \dfrac{\pi}{4} \Rightarrow C = \dfrac{\pi}{2} = 90^\circ$

$C = 90^\circ$

15. $a = 30, b = 24, c = 18 \Rightarrow s = \frac{30+24+18}{2} = \frac{72}{2} = 36,$

$\frac{1}{r^1} : \frac{1}{r^2} : \frac{1}{r^3} ; \frac{s-a}{\Delta} : \frac{s-b}{\Delta} : \frac{s-c}{\Delta} = s-a : s-b : s-c = (36-30) : (36-24) : (36-18)$

$= 6 : 12 : 18 = 1 : 2 : 3$

16. Distance between circumcentre and orthocentre is ' O '

$\therefore k = 0 \Rightarrow k + 5 = 0 + 5 = 5$

17. Given $r = \frac{R}{6}$ and $r_1 = 7r \Rightarrow r_1 = \frac{7R}{6}$

$r_1 - r = 4R\sin^2\frac{A}{2} \Rightarrow \frac{7R}{6} - \frac{R}{6} = 4R\sin^2\frac{A}{2}$

$\frac{6R}{6} = 4R\sin^2\frac{A}{2} \Rightarrow \frac{1}{4} = \sin^2\frac{A}{2} \Rightarrow \sin\frac{A}{2} = \frac{1}{2} = \sin\frac{\pi}{6} \Rightarrow \frac{A}{2} = \frac{\pi}{2} \Rightarrow A = \frac{\pi}{3}$

18. $r_1 = 4, r_2 = 4, r_3 = 6$

Given $r_1 = r_2 \ \frac{\Delta}{s-a} = \frac{\Delta}{s-b}$

$s - a = s - b \Rightarrow a = b$ triangle is isosceless

19. $r \cdot r_1 r_2 \cdot r_3 = \Delta^2 = \frac{\Delta^2}{k/3} \Rightarrow k = 3$

20. $\left(\frac{1}{r} - \frac{1}{r_1}\right)\left(\frac{1}{r} - \frac{1}{r_2}\right)\left(\frac{1}{r} - \frac{1}{r_3}\right)$

$\left(\frac{r_1 - r}{rr_1}\right)\left(\frac{r_2 - r}{rr_2}\right)\left(\frac{r_3 - r}{rr_3}\right) = \frac{4R\sin^2\frac{A}{2}\,4R\sin^2\frac{B}{2}\,4R\sin^2\frac{C}{2}}{r^3 r_1 r_2 r_3} = \frac{4Rr^2}{r^2\Delta^2} = \frac{abc/\Delta}{\Delta^2} = \frac{abc}{\Delta^3}$

21. Given $r_3 = r_1 + r_2 + r \Rightarrow r_3 - r = 4R\sin^2\frac{C}{2}, r_1 + r_2 = 4R\cos^2\frac{C}{2}$

$r_3 - r = r_1 + r_2$

$4R\sin^2\frac{C}{2} = 4R\cos^2\frac{C}{2} \Rightarrow \sin\frac{C}{2} = \cos\frac{C}{2} \Rightarrow \sin 45^\circ = \cos 45^\circ \Rightarrow c = 90^\circ$

$C = 90^\circ \Rightarrow A + B = 90^\circ \Rightarrow c = \sqrt{(r_3 - r)(r_1 + r_2)} = \sqrt{(15-2)(10+3)} = \sqrt{13 \times 13}$

$= 13$

22. Area of triangle $= \frac{1}{2} \times 2 \times 3\sin C = 3\sin C$

Maximum value of $\sin C = 1$

$\therefore$ Area maximum values is $3 \times 1 = 3$ $\therefore$ Area can not exceed 3

LEVEL – III

23. $r = \frac{\Delta}{s} \Rightarrow r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

24. Given $a = 6, b = 10, c = 8 \Rightarrow s = \frac{6+10+8}{2} = \frac{24}{2} = 12$

$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{12(6)(2)(4)} = \sqrt{24 \times 24} = 24 \Rightarrow r = \frac{\Delta}{s} = \frac{24}{12} = 2$

25. $r_1 = \frac{\Delta}{s-a} = \frac{24}{6} = 4$ 26. $r_3 = \frac{\Delta}{s-c} = \frac{26}{4} = 6$

27. (a) $\Delta = 400\sqrt{3}, s = 20(2+\sqrt{3}) \therefore r = \frac{\Delta}{s} = \frac{400\sqrt{3}}{20(2+\sqrt{3})} = \frac{20\sqrt{3}}{2+\sqrt{3}} = 20\sqrt{3}(2-\sqrt{3})$

(b) $s - a = 20\sqrt{3}: r_1 = \frac{\Delta}{s-a} = \frac{400\sqrt{3}}{20\sqrt{3}} = 20$

(c) $a + b - c = 40(2-\sqrt{3}) \Rightarrow 2s - 2c = 40(2-\sqrt{3}) \Rightarrow s - c = 20(2-\sqrt{3})$

$r_3 = \frac{400\sqrt{3}}{20(2-\sqrt{3})} = 20\sqrt{3}(2+\sqrt{3})$

(d) $abc = 64000\sqrt{3}; R = \frac{abc}{4\Delta} = \frac{64000\sqrt{3}}{4\times400\sqrt{3}} = 40$

28. Given $r_1 = 8 \Rightarrow r_2 = 12 \Rightarrow r_3 = 24$

$$a = \frac{r_1(r_2+r_3)}{\sqrt{r_1r_2 + r_2r_3 + r_3r_1}} = \frac{8\times36}{24} = 12$$
$$b = \frac{r_2(r_3+r_1)}{\sqrt{r_1r_2 + r_2r_3 + r_3r_1}} = \frac{12\times32}{24} = 16$$
$$c = \frac{r_3(r_1+r_2)}{\sqrt{r_1r_2 + r_2r_3 + r_3r_1}} = \frac{24\times20}{24} = 20$$

Also we know $r_1r_2r_3 = rs^2$

$r = \frac{r_1r_2r_3}{s^2}$ $\because s = \frac{a+b+c}{2}$

$r = \frac{8\times12\times24}{24\times24}$ $= \frac{12+16+20}{2}$

$r = 4$ $= \frac{48}{2} = 24$

29. $\frac{\Delta}{s-a} : \frac{\Delta}{s-b} : \frac{\Delta}{s-c} = 1:2:3$

Let $\frac{\Delta}{\frac{s-a}{1}} = \frac{\Delta}{\frac{s-b}{2}} = \frac{\Delta}{\frac{s-c}{3}} = \frac{\Delta}{6k}$

$\Rightarrow \frac{1}{s-a} = \frac{1}{6k}, \frac{1}{s-b} = \frac{1}{3k}, \frac{1}{s-c} = \frac{\Delta}{2k} \Rightarrow s-a = 6k, s-b = 3k, s-c = 2k$

$\Rightarrow s = 11k \Rightarrow \therefore a = 5k$, b = 8k, c = 9k

Hence, the ratio of sides is $5:8:9$

30. Area of the triangle $\Delta = k^2\sqrt{11\times6\times3\times2} = 6\sqrt{11}k^2$

So, $r = \frac{\Delta}{s} = \frac{6}{\sqrt{11}}k$

$[\because s = 11k]$

and $R = \frac{abc}{4\Delta} = \frac{5\times8\times9}{4\times6\times\sqrt{11}}k = \frac{15}{\sqrt{11}}k$

$\therefore R:r = 5:2$

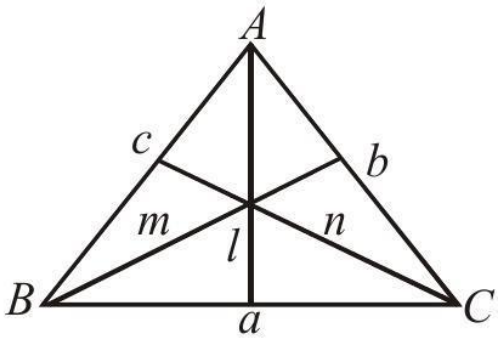
31. Since 'C' is the greatest side, ' C ' is the greatest angle.

$$\cos C = \frac{a^2+b^2-c^2}{2ab} = \frac{25+64-81}{2\times5\times8} = \frac{89-81}{80} = \frac{8}{80} = \frac{1}{10}$$

32. $a \to r; b \to p; c \to s; d \to q$

(a) $2(r+R) = 2\left((s-c)\tan\frac{C}{2}+\frac{c}{2}\right)\ 2\left(s-\frac{c}{2}\right) = a+b$

(b) $\sin C = \frac{l}{b}$



$\sin B = \frac{n}{a}$

$\sin A = \frac{m}{c}$

$2R\left(\frac{bl}{c}\right) + 2R\left(\frac{cm}{a}\right) + 2R\left(\frac{an}{b}\right)$

$\Rightarrow \frac{c}{\sin c}\frac{bl}{c} + \frac{a}{\sin A}\frac{cm}{a} + \frac{b}{\sin B}\frac{an}{b}$

From (i), (ii), (iii) ;

$2R\left(\frac{bl}{c}+\frac{cm}{a}+\frac{an}{b}\right) = a^2+b^2+c^2$

$\Rightarrow \sqrt{2R\left(\frac{bl}{c}+\frac{cm}{a}+\frac{an}{b}\right)+2ab+2bc+2ac} = a+b+c$

(c) $R = \frac{b}{2\sin B} = \frac{c}{2\sin C} \Rightarrow Rb^2\sin 2C + Rc^2\sin 2B \Rightarrow b^2 c\cos C + c^2 b\cos B$

$\Rightarrow bc(b\cos C + c\cos B) \Rightarrow abc$

(d) $2R(\cos A + \cos B) \Rightarrow 2R\left(-\frac{b}{\sqrt{a^2+b^2}}+\frac{a}{\sqrt{a^2+b^2}}\right)$

$\because\ R = \sqrt{a^2+b^2}/2;\ 2R\left(-\frac{b}{\sqrt{a^2+b^2}}+\frac{a}{\sqrt{a^2+b^2}}\right) = a-b$

33. Given that $s-a, s-b \& s-c$ are in A.P $\Rightarrow a, b, c$ are in A.P

$\Rightarrow \frac{\Delta}{s-a}, \frac{\Delta}{s-b}, \frac{\Delta}{s-c}$ are in H.P $\Rightarrow r_1, r_2, r_3$ are in H.P

Also $\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{b^2+c^2-(2b-c)^2}{2bc} = \frac{4c-3b}{2c}$

WORKSHEET – 4

LEVEL – I

1) C	2) B	3) A	4) B	5) A
6) A				

LEVEL – II

1) A	2) A	3) A	4) A	5) B
6) A	7) 2	8) C	9) B	10) 2
11) C	12) B	13) A	14) 2	15) B
16) C	17) 3	18) D	19) D	20) 8

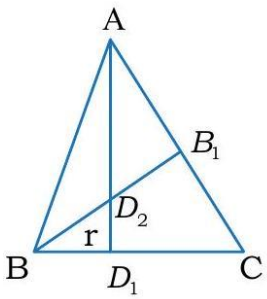
LEVEL – III

21) A, B, C	22) A, B, C	23) A	24) A	25) A
26) A - p ; B - q ; C - s ; D - t		27) B	28) D	29) C
30) a - q ; b - r ; c - p ; d – t				

SOLUTIONS

LEVEL – II

1. Acute angled
2. Equilateral
3. 1: 2
4. $R\sqrt{1-8\cos A\cdot\cos B\cdot\cos C}$
5. R/2
6. $\frac{2\Delta}{3a}$
7. ' G ' divides in the ratio $= 2:1 \Rightarrow \therefore x = 2$
8. In $\triangle ABD, \angle ABD_2 = \angle D_2BD_1 = \frac{\angle B}{2}$



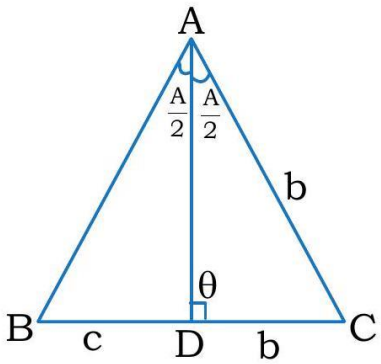
Hence triangle ABD_2 and D_1D_2B one congurent $\Rightarrow \frac{AD_1}{AD_2} = \frac{2}{1} = 2$

9. 2: 1

10. Circumradius = 2R ⇒ k = 2

11. The circumradius of an excentral $\Delta^{\ell e} = 2R$

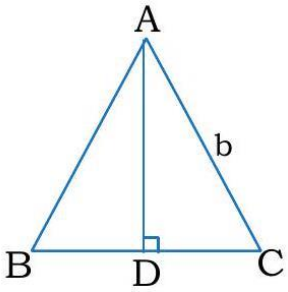
12. $r + R = \frac{a}{2}\cot\frac{\pi}{n} + \frac{a}{2}\operatorname{cosec}\frac{\pi}{n} = \frac{a}{2}\cot\frac{\pi}{2n}$



$$BD = \frac{ac}{b+c}, CD = \frac{ab}{b+c} \Rightarrow \sin\theta = \frac{\sin A/2}{CD} \Rightarrow \sin\theta = \frac{b+c}{2R\cos\frac{A}{2}} = \cos\left(\frac{B-C}{2}\right)$$

14. $\frac{R}{r} = 2$

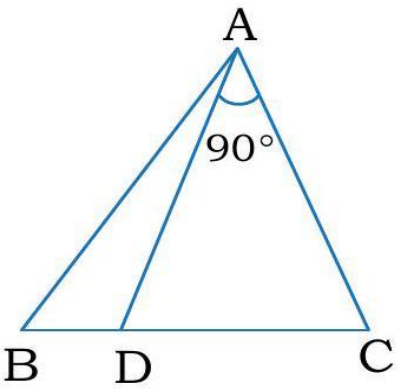
15. $\lfloor C = 23^\circ \Rightarrow \frac{AD}{\sin c} = b; \frac{abc}{b^2-c^2} = b\sin c$



$$\frac{\sin A\sin C}{\sin^2 B - \sin^2 C} = \sin C \Rightarrow \frac{\sin A\sin C}{\sin(B+C)\sin(B-C)} = \sin C$$

$$\frac{\sin A}{\sin A\sin(B-C)} = 1 \Rightarrow \sin(B-C) = 1 \Rightarrow B - C = 90^\circ \Rightarrow B = 113^\circ$$

16. $\lfloor BAD = \lfloor A - 90^\circ, BD = DC; \not\Vdash AD = 90 + C$



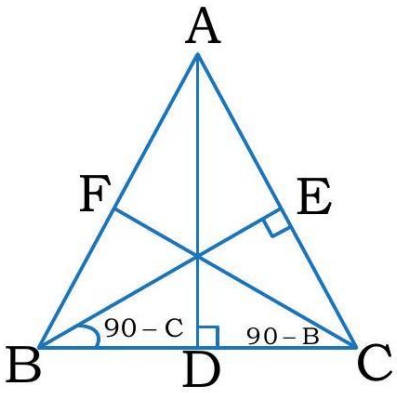
Apply sinc rule in $\triangle ABD \Rightarrow \frac{a}{2\sin(A-90^\circ)} = \frac{c}{\cos C} \Rightarrow \frac{a}{-2\cos A} = \frac{c}{\cos C}$

$$\Rightarrow \frac{2abc}{-2(b^2+c^2-a^2)} = \frac{2abc}{a^2+b^2-c^2} \Rightarrow -2b^2+2(a^2-c^2) = b^2+a^2-c^2$$

$$a^2-c^2 = 3b^2 \Rightarrow \frac{a^2-c^2}{b^2} = 3$$

17. $Ga = \frac{2\Delta}{3a} = \frac{2\Delta}{ka} \Rightarrow k = 3$

18. Circum diameter of $\triangle ABC = \frac{a}{\sin A} = \frac{a}{\sin A} = 2R = 8$



19. $h_a \geq a\ ;\ h_b \geq b$

$\therefore b\sin C \geq a; a\sin C \geq b$ $\quad \frac{c}{\sin c} = 2R$

$\therefore ab \leq ab\sin^2 C$ $\quad \frac{4}{1} = 2R$

$\sin^2 C = 1 \sin C = 1$ $\quad R = 2$

$\angle C = 90° \; a^2+b^2 = 16$ $\quad \Delta = \frac{abc}{4R}$

$\Rightarrow a = b \; 2a^2 = 16 \Rightarrow a = 2\sqrt{2}$

$$\Delta = \frac{(2\sqrt{2})(2\sqrt{2})\times 4}{4(2)} = 4$$

20. Distance $= R\sqrt{1-8\cos A\cos B\cos C} \Rightarrow p = 8$

LEVEL – III

21. Sides are $R\sin 2A, R\sin 2B, R\sin 2C$

22. Angles of pedal triangle are $\pi-2A, \pi-2B, \pi-2C$

23. $AD = 6$ cm $\Rightarrow AG = \frac{2}{3}\times AD \Rightarrow AG = \frac{2}{3}\times 6 = 4$ cm

24. $BG = 10$ cm $\Rightarrow BE = \frac{3}{2}\times BG = \frac{3}{2}\times 10 = 15$ cm

25. $AG: GD = 2:1 \Rightarrow AG: AD = 2:(2+1) = 2:3$

26. a) $BD = \frac{1}{2}\sqrt{2a^2+2c^2-b^2} = 7$

b) $DE = CD - CE = 2$ cm

c) $a = 7, b = 8, c = 9 \Rightarrow S = \frac{24}{2} = 12 \Rightarrow \Delta = \sqrt{12(5)(4)(3)} = 12\sqrt{5} \Rightarrow \Delta = \frac{1}{2}(AC)\cdot$

$\text{BE}\ 12\sqrt{5} = \frac{1}{2} \cdot 8 \cdot \text{BE}$

$3\sqrt{5} = \text{BE}$

d) $\text{CD} = 4$

27. Median passing through

$$c = \frac{1}{2}\sqrt{-c^2 + 2b^2 + 2a^2} = \frac{1}{2}\sqrt{-36 + 2(49) + 2(64)} = \frac{1}{2}\sqrt{-36 + 98 + 128} = \frac{1}{2}\sqrt{190}$$

$$= \sqrt{\frac{190}{4}} = \sqrt{\frac{95}{2}}$$

28. $\Delta = \frac{1}{2} \times h \times 6 = 3h$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}\,; s = \frac{a+b+c}{2} = \frac{7+8+6}{2} = \frac{21}{2}$$

$$= \sqrt{\frac{21}{2}\left(\frac{21}{2} - 8\right)\left(\frac{21}{2} - 7\right)\left(\frac{21}{2} - 6\right)} = \frac{1}{4}\sqrt{21 \times 5 \times 7 \times 9} = \frac{1}{4} \times 7 \times 3\sqrt{15}$$

$$\therefore 3h = \frac{1}{4} \times 21\sqrt{15} \Rightarrow h = \frac{21\sqrt{15}}{12} = \frac{7\sqrt{15}}{4}$$

29. $\frac{R}{r} = \frac{\frac{abc}{4\Delta}}{\frac{\Delta}{s}} = \frac{s(abc)}{4\Delta^2} = \frac{21}{2} \times \frac{8 \times 7 \times 6}{(21) \times 15} \times 4 = \frac{32}{15}$

30. $\text{c}^2 = \text{b}^2 + \text{a}^2 - 2\text{abcos}\,\text{c} \Rightarrow \text{c}^2 = 2^2 + (\sqrt{3}+1)^2 - 2(\sqrt{3}+1)\cos 60$

$$c^2 = 6 \Rightarrow c = \sqrt{6} \Rightarrow \frac{c}{\sin c} = 2R \Rightarrow \frac{\sqrt{6}}{\frac{\sqrt{3}}{2}} = 2R \Rightarrow \sqrt{2} = R\ R' = \frac{R}{2} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

a) $\frac{2}{\sin A} = 2R$ and $R' = \frac{R}{2}$

b) Area of triangle is of $\frac{1}{2}ab\sin C$

c) $r' = \frac{\Delta'}{s'} = \frac{1}{\sqrt{6}+\sqrt{2}} \times \frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}-2} = \frac{\sqrt{6}-\sqrt{2}}{4}$

d) $\Delta\frac{1}{2}(1+\sqrt{3})2 \cdot \frac{\sqrt{3}}{2} = \frac{3+\sqrt{3}}{2}$

5. INEQUALITIES

WORKSHEET – 1

LEVEL – I

1. B	2. D	3. D	4. A	5. A
6. B	7. C	8. C	9. A	10. D

LEVEL – II

1. B	2. C	3. B	4. A	5. B
6. 4	7. D	8. A	9. B	10. 3
11. D	12. C	13. A	14. C	15. C
16. B	17. C			

LEVEL – III

18. A, C	19. B	20. A	21. C
22. a-r; b-p; c-q; d-t		23. 1, 2	24. 3
25. A	26. B	27. A	

SOLUTIONS

Level – I

1. Transitive property.
2. $a \leq b$ represent $a < b$ and $a = b$.
3. $a > b > 0$ and $n > 0$. So, $a^n > b^n$
4. As $0 < a < 1, \log_a^x < \log_b^y \Rightarrow x > y$
5. As, $a > b$, if $c < 0$, then $ac < bc$ and if $c > 0$, then $ac > bc$.
6. $a > b \Rightarrow \frac{1}{a} < \frac{1}{b}$
7. A.M $\Rightarrow \geq$ G · M $\Rightarrow \frac{x+\frac{1}{x}}{2} \geq \sqrt{x \times \frac{1}{x}} \Rightarrow x + \frac{1}{x} \geq 2$
8. $a \geq b$ represent a > b and $a = b$
9. A.M $= \frac{x_1 + x_2 \ldots\ldots\ldots x_n}{n}$
10. G · M $= (x_1 \cdot x_2 \cdot x_3 \ldots\ldots\ldots x_n)^{\frac{1}{n}}$

LEVEL – II

1. If $a, b > 0$, then $(a-b)^2 \geq 0 \Rightarrow a^2 + b^2 - 2ab \geq 0$

   $\Rightarrow a^2 + b^2 + 2ab \geq 4ab \Rightarrow (a+b)^2 \geq 4ab \Rightarrow 1 \geq 4ab$ (given $a + b = 1$) $\Rightarrow ab \leq \frac{1}{4}$.
2. $(\sqrt{x} - \sqrt{y})^2 \geq 0 \Rightarrow x + y \geq 2\sqrt{xy} \Rightarrow$ Similarly $y + z \geq 2\sqrt{yz}$ and $z + x \geq 2\sqrt{zx}$

   Multiplying above inequations $(x+y)(y+z)(z+x) \geq 8xyz$

3. $(x-y)^2 \geq 0 \Rightarrow x^2+y^2 \geq 2xy \Rightarrow$ Similarly $y^2+z^2 \geq 2yz$ and $x^2+z^2 \geq 2zx$

   Adding the above equations, we get

   $2(x^2+y^2+z^2) \geq 2(xy+yz+zx) \Rightarrow xy+yz+zx < x^2+y^2+z^2$.

4. $(x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx) \leq x^2+y^2+z^2+2(x^2+y^2+z^2) \leq 3(x^2+y^2+z^2)$

   $\Rightarrow \frac{x+y+z}{3} \leq \left(\frac{x^2+y^2+z^2}{3}\right)^{1/2}$.

5. We know that, $32 > 30 \Rightarrow 2^5 > 30$

   $$(2^5)^{100} > (30)^{100} \Rightarrow (2)^{500} > (30)^{100} \Rightarrow (2)^{567} > (30)^{100}$$

6. In triangle difference of two sides is less than third side.

   $a > b-c \Rightarrow a^2 > b^2+c^2-2bc; b > c-a \Rightarrow b^2 > a^2+c^2-2ac$
   $c > a-b \Rightarrow c^2 > a^2+b^2-2ab$ on adding, we get
   $a^2+b^2+c^2 > 2(a^2+b^2+c^2)-2(ab+bc+ca)$
   $2(ab+bc+ca) > 2(a^2+b^2+c^2)-(a^2+b^2+c^2) \Rightarrow 2(ab+bc+ca) > a^2+b^2+c^2$
   $2(ab+bc+ca)+2(ab+bc+ca) > a^2+b^2+c^2+2ab+2bc+2ca$
   $\Rightarrow (a+b+c)^2 < 4(ab+bc+ca)$.

7. A. M $>$ G. M $\Rightarrow \frac{1+a}{2} \geq \sqrt{1 \cdot a}$

   (1); Similarly, $1+b \geq 2\sqrt{b}$

   $1+c \geq 2\sqrt{c}$

   (3); $1+d \geq 2\sqrt{d}$

   Multiplying four equations, we get $(1+a)(1+b)(1+c)(1+d) > 16\sqrt{abcd}$

   $\Rightarrow (1+a)(1+b)(1+c)(1+d) > 16\sqrt{1} > 16$.

8. $(150^3)^{100}$ or $(20000 \times 100)^{100} \Rightarrow (3375000)^{100}$ or $(20{,}00{,}000)^{100}$

   $\Rightarrow 3375000 > 20{,}00{,}000 \Rightarrow (3375000)^{100} > (2000000)^{100} \Rightarrow (150)^{300} > (20000 \times 100)^{100}$.

9. On applying A.M $\geq$ G.M for six numbers $a^2b, a^2c, b^2c, b^2a, c^2a, c^2b$

   we get $\frac{a^2b+a^2c+b^2c+b^2a+c^2a+c^2b}{6} \geq (a^2b \cdot a^2c \cdot b^2c \cdot c^2a \cdot c^2b)^{\frac{1}{6}}$

   $a^2(b+c)+b^2(c+a)+c^2(a+b) \geq 6abc$.

10. Using the harmonic mean inequality, $\frac{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}{3} \geq \frac{3}{a+b+c}$ $\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \geq \frac{9}{6} \Rightarrow \frac{1}{a}+\frac{1}{b}+\frac{1}{c} \geq \frac{3}{2}$.

    Now $\sqrt{\frac{\left(a+\frac{1}{b}\right)^2+\left(b+\frac{1}{c}\right)^2+\left(c+\frac{1}{a}\right)^2}{3}} \geq \frac{a+\frac{1}{b}+b+\frac{1}{c}+c\frac{1}{a}}{3}$

$$\sqrt{\frac{\left(a+\frac{1}{b}\right)^2+\left(b+\frac{1}{c}\right)^2+\left(c+\frac{1}{a}\right)^2}{3}} \geq \frac{6+\frac{3}{2}}{3} \geq \frac{5}{2}$$

squaring on both sides,

$\frac{\left(a+\frac{1}{b}\right)^2+\left(b+\frac{1}{c}\right)^2+\left(c+\frac{1}{a}\right)^2}{3} \geq \frac{25}{4} \Rightarrow \left(a+\frac{1}{b}\right)^2+\left(b+\frac{1}{c}\right)^2+\left(c+\frac{1}{a}\right)^2 \geq \frac{75}{4}.$

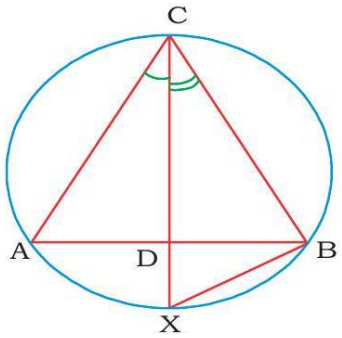
11. A.M > G.M

$\frac{\tan A+\tan B+\tan C}{3} \geq \sqrt[3]{\tan A\tan B\tan C} \Rightarrow \frac{\tan A\tan B\tan C}{3} \geq \sqrt[3]{\tan A\tan B\tan C}$

$(\tan A\tan B\tan C)^3 \geq 27(\tan A\tan B\tan C) \Rightarrow (\tan A\tan B\tan C)^2 \geq 27$

$\tan A\tan B\tan C \geq 3\sqrt{3} \Rightarrow \cot A\cot B\cot C \leq \frac{1}{3\sqrt{3}}.$

12.



In $\triangle ACD, \triangle BCX \Rightarrow \underline{ACD} = \underline{BCX} = \theta; \underline{CAD} = \underline{CXB}$

By AA similarity $\triangle$ ACD $\sim\triangle$ XCD $\Rightarrow \frac{AC}{XC} = \frac{CD}{CB} = \frac{AD}{XB}; AC \cdot CB = CD \cdot CX$

In fig $CD < CX; CD^2 < CD.CX \Rightarrow CD^2 < AC.CB.$

13. $abc(a+b+c) = (ab)(ca)(ab)(bc) + (bc)(ac)$

$(ab)(ac) + (ab)(bc) + (bc)(ac) \leq (ab)^2 + (bc)^2 + (ac)^2 \Rightarrow abc(a+b+c) \leq a^2b^2 + b^2c^2 + a^2c^2.$

14. By Bernoulis inequality, $(1+0.01)^8 > 1+0.08$

$(1+0.01)^{1000} > (1+0.08)^{125} = [(1+0.08)^5]^{25} > (1+0.4)^{25} > (1+04)^{24}$
$= [(1.4)^3]^8 > (2.7)^8 > 7^4 = 2401 > 1000.$

15. We know that, $(b-c)^2 \geq 0 \Rightarrow b^2 + c^2 - 2bc \geq 0 \Rightarrow b^2 + c^2 \geq 2bc$

$2(b^2+c^2) \geq b^2 + c^2 + 2bc \Rightarrow 2(b^2+c^2) \geq (b+c)^2 \Rightarrow \frac{b^2+c^2}{b+c} \geq \frac{b+c}{2}$

Similarly, $\frac{c^2+a^2}{c+a} \geq \frac{c+a}{2}; \frac{a^2+b^2}{a+b} \geq \frac{a+b}{2}$

on adding three inequalities,

$$\frac{b^2+c^2}{b+c}+\frac{c^2+a^2}{c+a}+\frac{a^2+b^2}{a+b} \geq \frac{a+b+b+c+c+a}{2} \Rightarrow \frac{b^2+c^2}{b+c}+\frac{c^2+a^2}{c+a}+\frac{a^2+b^2}{a+b} \geq a+b+c.$$

16. On applying A.M ≥ G.M for four numbers $a, a, a, 3b$,

we get, $\frac{a+a+a+3b}{4} \geq (3a^3b)^{1/4} \Rightarrow \frac{3(a+b)}{4} \geq (3a^3b)^{1/4} \Rightarrow \frac{3a^3b}{(a+b)^4} \leq \left(\frac{3}{4}\right)^4 = \frac{27}{256}$.

17. As a,b,c are the sides of a triangle, we have a, b, c are all > 0 and also $c + a - b, a + b - c$, b + c − a are all > 0. Now by A.M-G.M inequality.

$$\frac{a}{c+a-b}+\frac{b}{a+b-c}+\frac{c}{b+c-a} \geq 3\left[\frac{abc}{(c+a-b)(a+b-c)(b+c-a)}\right]^{1/3}$$

Now we have

$(1) \Rightarrow a^2 \geq a^2-(b-c)^2$

$a^2 \geq (a+b-c)(a-b+c) \ldots\ldots(2) \Rightarrow b^2 \geq b^2-(c-a)^2$

$b^2 \geq (b+c-a)(b-c+a) \ldots\ldots(3) \Rightarrow c^2 \geq (c+a-b)(c-a+b)$

Multiplying (2),(3) , (4) we have $a^2b^2c^2 \geq (a+b-c)^2(b+c-a)^2(c+a-b)^2 \Rightarrow$ $abc \geq (a+b-c)(b+c-a)(c+a-b). \Rightarrow \left[\frac{abc}{(a+b-c)(b+c-a)(c+a-b)}\right]^{1/3} \geq 1 \Rightarrow$ from $\ldots..(1) \Rightarrow \frac{a}{c+a-b}+\frac{b}{a+b-c}+\frac{c}{b+c-a} \geq 3$

18. $a > b \Rightarrow$ if $c > 0 \Rightarrow ac > bc \Rightarrow$ if $c < 0 \Rightarrow ac < bc$.

LEVEL – III

19. $(x-y)^2 > 0 \Rightarrow x^2+y^2 \geq 2xy$.

Similarly, $x^2+z^2 \geq 2xz;\ y^2+z^2 \geq 2yz$.

By adding inequalities, $2(x^2+y^2+z^2) \geq 2(xy+yz+xz)$.

20. $xyz(x+y+z) = (xy)(zx)(xy)(yz)+(yz)(xz) \leq x^2y^2+y^2z^2+z^2x^2$.

21. $(x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx) \leq x^2+y^2+z^2+2(x^2+y^2+z^2) =$ $3(x^2+y^2+z^2)\ \frac{x+y+z}{3} \leq \left(\frac{x^2+y^2+z^2}{3}\right)^{1/2}$.

22. (a) $a^2+b^2 = (a+b)^2-2ab = 1-2ab \Rightarrow ab \leq \frac{1}{4} \Rightarrow -2ab \geq \frac{-1}{2} \Rightarrow 1-2ab \geq 1-\frac{1}{2}$.

$1-2ab \geq \frac{1}{2} \Rightarrow a^2+b^2 \geq \frac{1}{2}$

(b) $a^4+b^4 = (a^2+b^2)^2-2a^2b^2 \Rightarrow (a^2+b^2)^2 \geq \frac{1}{4}$ and $ab \leq \frac{1}{4} \Rightarrow -2a^2b^2 \geq -\frac{1}{8}$

$a^4+b^4 = (a^2+b^2)^2+(-2a^2b^2) \geq \frac{1}{4}-\frac{1}{8} = \frac{1}{8}$.

(c) $\left(a+\frac{1}{a}\right)^2+\left(b+\frac{1}{b}\right)^2=a^2+\frac{1}{a^2}+b^2+\frac{1}{b^2}+4=a^2+b^2+\frac{a^2+b^2}{a^2b^2}+4\geq\frac{1}{2}+\frac{16}{2}+4=12\frac{1}{2}$.

(d) $a,b>0$ then $(a-b)^2\geq 0\Rightarrow a^2+b^2-2ab\geq 0\Rightarrow a^2+b^2+2ab\geq 4ab$

$(a^2+b)^2\geq 4ab\Rightarrow ab\leq\frac{1}{4}$.

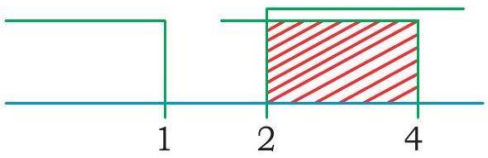
23. $\frac{a^3}{b}+\frac{b^3}{c}+bc\geq 3ab$

$\frac{b^3}{c}+\frac{c^3}{a}+ca\geq 3bc$

$\frac{c^3}{a}+\frac{a^3}{b}+ab\geq 3ca$

$(1)+(2)+(3)\Rightarrow\frac{a^3}{b}+\frac{b^3}{c}+\frac{c^3}{a}\geq ab+bc+ca.$

24.



$x\in(2,4)$

$(x+1)^2>5x-1\Rightarrow x^2+2x+1-5x+1>0\Rightarrow x^2-3x+2>0\Rightarrow x^2-2x-x+2>0$

$(x-1)(x-2)>0\Rightarrow(x+1)^2<7x-3\Rightarrow x^2+2x+1-7x+3<0\Rightarrow x^2-5x+4<0$

$(x-1)(x-4)<0\Rightarrow x\in(2,4).$

25. Let $f(x)=\tan x\Rightarrow f^1(x)=\sec^2 x; f^{11}(x)=2\sec^2 x\tan x>0$, in $\left(0,\frac{\pi}{2}\right)$.

By using Jensen's inequality, $f\left(\frac{x_1+x_2+x_3+\cdots x_n}{n}\right)\leq\frac{f(x_1)+f(x_2)+\cdots.+f(x_n)}{n}$

$\Rightarrow\tan\left(\frac{A+B+C}{3}\right)\leq\frac{\tan A+\tan B+\tan C}{3}$

$\Rightarrow\tan\left(\frac{\pi}{3}\right)\leq\frac{\tan A+\tan B+\tan C}{3}\Rightarrow\tan A+\tan B+\tan C\geq 3\sqrt{3}.$

26. $f(x)=\cos x\Rightarrow f^1(x)=-\sin x; f^{11}(x)=-\cos x<0$, in $\left(0,\frac{\pi}{2}\right)$

By using Jensen's inequality,

$$f\left(\frac{x_1+x_2+x_3+\cdots x_n}{n}\right)\geq\frac{f(x_1)+f(x_2)+\cdots.+f(x_n)}{n}$$

$$\Rightarrow\cos\left(\frac{A+B+C}{3}\right)\geq\frac{\cos A+\cos B+\cos C}{3}$$

$\Rightarrow\cos\left(\frac{\pi}{3}\right)\geq\frac{\cos A+\cos B+\cos C}{3}\Rightarrow\cos A+\cos B+\cos C\leq\frac{3}{2}.$

27. $f(x) = \sin x \Rightarrow f^1(x) = \cos x; f^{11}(x) = -\sin x < 0$, in $\left(0, \frac{\pi}{2}\right)$

By using Jensen's inequality,

$f\left(\frac{x_1 + x_2 + x_3 + \cdots x_n}{n}\right) \geq \frac{f(x_1) + f(x_2) + \cdots . + f(x_n)}{n}$

$\Rightarrow \sin\left(\frac{A + B + C}{3}\right) \geq \frac{\sin A + \sin B + \sin C}{3}$

$\Rightarrow \sin\left(\frac{\pi}{3}\right) \geq \frac{\sin A + \sin B + \sin C}{3} \Rightarrow \sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$.

WORKSHEET – 2

LEVEL – I

1. B	2. C	3. D	4. D	5. A
6. A	7. B	8. B	9. D	10. D

1. C	2. B	3. B	4. C	5. B
6. 8	7. A	8. C	9. A	10. A
11. 10	12. C	13. D	14. A	15. A
16. C	17. B, C	18. 9	19. A	20. B
21. D	22. B, C, D	23. a- q ,b-p, c-s,d-r		24. B, C
25. 1				

SOLUTIONS

LEVEL – II

1. Weighted A.M > weighted G.M $\frac{a \cdot a + b \cdot b}{a+b} > (a^a \cdot b^b)^{\frac{1}{a+b}} \Rightarrow \left(\frac{a^2+b^2}{a+b}\right)^{a+b} > a^a \cdot b^b$.

2. By power mean theorem,

$\frac{a^3 + b^3}{2} \geq \left(\frac{a+b}{2}\right)^3 \Rightarrow \frac{8}{2} \geq \left(\frac{a+b}{2}\right)^3 \Rightarrow 4 \geq \left(\frac{a+b}{2}\right)^3 \Rightarrow \sqrt[3]{4} \geq \frac{a+b}{2} \Rightarrow a + b \leq 2\sqrt[3]{4}$

3. Given n is a positive integer. By power mean theorem,

$\frac{1^{1/2} + 2^{1/2} + 3^{1/2} + \cdots .. n^{1/2}}{n} < \left(\frac{1 + 2 + 3 + \cdots . + n}{n}\right)^{1/2}$

$\Rightarrow \sqrt{1} + \sqrt{2} + \sqrt{3} + \cdots . \sqrt{n} < n \cdot \left(\frac{n(n+1)}{2n}\right)^{1/2} \Rightarrow < n\left(\frac{n+1}{2}\right)^{1/2}$.

4. By AM-GM Inequality,

$\frac{bcx+cay+abz}{3} \geq (bcx \times cay \times abz)^{1/3} \Rightarrow bcx+cay+abz \geq 3(a^3b^3c^3)^{1/3}$. [$\because xyz = abc$]

$\therefore bcx+cay+abz \geq 3abc$.

5. Given $x+y+z=1$.

By cauchy schwartz inequality,

$(1x+1y+1z)^2 \leq (1^2+1^2+1^2)(x^2+y^2+z^2) \Rightarrow \frac{1}{3} \leq x^2+y^2+z^2$.

6. By A.M.-G.M. Inequality,

$$\frac{xy+yz+zx}{3} \geq \sqrt[3]{x^2y^2z^2} \Rightarrow \frac{12}{3} \geq \sqrt[3]{x^2y^2z^2} \Rightarrow x^2y^2z^2 \leq 64 \ \therefore xyz \leq 8$$

The greatest value of xyz is 8 .

7. Given $m>1, n \in N$; LHS $= 1^m+2^m+2^{2m}+2^{3m}+\cdots.+(2^{n-1})^m$

$\Rightarrow (2^0)^m+(2^1)^m+(2^2)^m+(2^3)^m+\cdots.+(2^{n-1})^m$ By power mean theorem,

$$\Rightarrow (2^0)^m+(2^1)^m+(2^2)^m+\cdots.+(2^{n-1})^m \geq \left(\frac{2^0+2^1+2^2+2^3+\cdots.2^{n-1}}{n}\right)^m$$

$$\Rightarrow 1^m+2^m+2^{2m}+\cdots.+2^{mn-m} \geq n\left(\frac{\frac{2^n-1}{2-1}}{n}\right)^m \Rightarrow 1^m+2^m+2^{2m}+\cdots.+2^{mn-m} \geq n\left(\frac{2^n-1}{n}\right)^m$$

$\Rightarrow 1^m+2^m+2^{2m}+\cdots.+2^{mn-m} \geq n^{1-m}(2^n-1)^m$.

8. By power mean theorem $\Rightarrow \frac{a^2+b^2+c^2+d^2}{4} \geq \left(\frac{a+b+c+d}{4}\right)^2$

$$\Rightarrow \frac{16-e^2}{4} \geq \left(\frac{8-e}{4}\right)^2 \Rightarrow 4-\frac{e^2}{4} \geq \left(2-\frac{e}{4}\right)^2 \Rightarrow 4-\frac{e^2}{4} \geq 4+\frac{e^2}{16}-e \Rightarrow 0 \geq \frac{e^2}{16}+\frac{e^2}{4}-e$$

$$\Rightarrow \frac{5e^2}{16}-e \leq 0 \Rightarrow 5e^2-16e \leq 0 \Rightarrow e(5e-16) \leq 0 \Rightarrow 0 \leq e \leq \frac{16}{5}.$$

9. We have AM > GM $\Rightarrow \frac{\frac{b}{a}+\frac{c}{a}+\frac{c}{b}+\frac{a}{b}+\frac{a}{c}+\frac{b}{c}}{6} \geq \sqrt[66]{\frac{b}{a}\times\frac{c}{a}\times\frac{c}{b}\times\frac{c}{b}\times\frac{a}{c}\times\frac{b}{c}} \Rightarrow \frac{b+c}{a}+\frac{c+a}{b}+\frac{a+b}{c} \geq 6.$

10. We know that by CS inequality

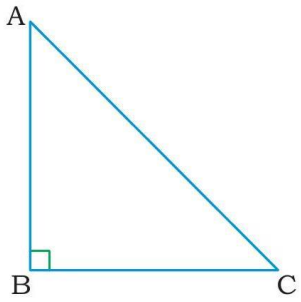
$$\Rightarrow (ax+by)^2 \leq (a^2+b^2)(x^2+y^2) \Rightarrow ax+by \leq \sqrt{a^2+b^2}\sqrt{x^2+y^2}$$

11. By weighted $AM-GM$ Inequality $\Rightarrow \frac{2\left(\frac{3x}{2}\right)+3\left(\frac{4y}{3}\right)}{2+3} \geq \left(\left(\frac{3x}{2}\right)^2 \cdot \left(\frac{4y}{3}\right)^3\right)^{\frac{1}{2+3}} \Rightarrow \frac{3x+4y}{5} \geq \left(\frac{9x}{4} \cdot \frac{64y^3}{27}\right)^{1/5}$

$$\Rightarrow 3x+4y \geq 5\left(\frac{16}{3}\times x^2y^3\right)^{1/5} \Rightarrow 3x+4y \geq 5\left(\frac{16}{3}\times 6\right)^{1/5} \ \therefore 3x+4y \geq 10.$$

12. In triangle ABC$\angle B = 90°$

∴ AC is longest side.



$\Rightarrow AC > AB, AC > BC$

By phthogurus theorem, $AC^2 = AB^2 + BC^2$.

Multiplying with AC $\Rightarrow AC^3 = AC \cdot AB^2 + AC \cdot BC^2 > AB^3 + BC^3$.

13. $n^4 < 10^n \Rightarrow n \geq 2$

$\Rightarrow \frac{n+1}{n} \leq \frac{3}{2} \Rightarrow \frac{(n+1)^4}{n^4} \leq \frac{81}{16}(n+1)^4 \leq \frac{81}{16}n^4 \Rightarrow (n+1)^4 \leq \frac{81}{60}(10^n) \Rightarrow 10.10^n \Rightarrow (n+1)^4 < 10^{n+1}$.

14. $\frac{x^2}{(x+y)(x+z)} + \frac{y^2}{(y+z)(y+x)} + \frac{z^2}{(z+x)(z+y)}$

$\Rightarrow \frac{x^2}{x^2+xz+yz+xy} + \frac{y^2}{y^2+yx+yz+zx} + \frac{z^2}{z^2+yz+xz+xy} = \frac{(x+y+z)^2}{x^2+y^2+z^2+3(xy+zy+zx)}$

By Cauchy-Schwarz Revisited,

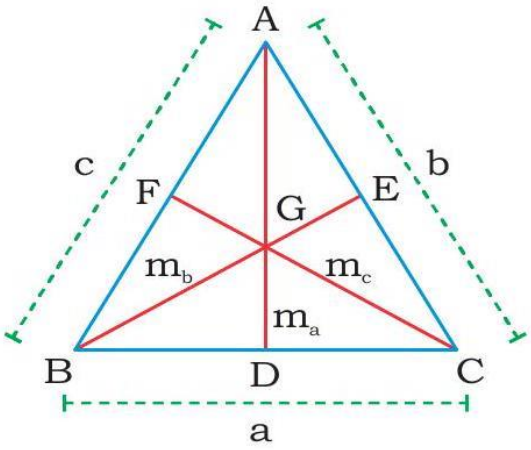
$\frac{a^2}{x^2} + \frac{b^2}{y^2} = \frac{(a+b)^2}{x^2+y^2} \Rightarrow x^2 + y^2 + z^2 \geq xy + yz + zx$

$\Rightarrow x^2 + y^2 + z^2 + 2(xy + yz + zx) \geq 3(xy + yz + zx) \Rightarrow (x + y + z)^2 \geq 3(xy + yz + zx)$

from (1)

$$\Rightarrow \geq \frac{3(xy + yz + zx)}{xy + yz + zx + 3(xy + yz + zx)} \Rightarrow \geq \frac{3(xy + yz + zx)}{4(xy + yz + zx)} \geq \frac{3}{4}.$$

15. In triangle BGC,



$\Rightarrow BG + GC > BC \Rightarrow \frac{2}{3}BE + \frac{2}{3}CF > BC \Rightarrow \frac{2}{3}m_b + \frac{2}{3}m_c > BC \Rightarrow 2m_b + 2m_c > 3a$

Similarly, $2(m_c + m_a) > 3b \Rightarrow 2(m_a + m_b) > 3c$

Add all the equations $\Rightarrow 4(m_a + m_b + m_c) > 3(a + b + c)$

In triangle ABD $\Rightarrow AB + BD > AD \Rightarrow c + \frac{a}{2} > m_a$

Similarly $b + \frac{c}{2} > m_c \Rightarrow a + \frac{b}{2} > m_b$

on adding $\Rightarrow a + b + c + \frac{1}{2}(a + b + c) > m_a + m_b + m_c \Rightarrow 3(a + b + c) >$

$2(m_a + m_b + m_c)$

$\therefore 2(m_a + m_b + m_c) < 3(a + b + c) < 4(m_a + m_b + m_c).$

16. Given,

a, b, c, d are the positive real numbers. By AM-GM inequality

$\Rightarrow \frac{a + b + c + d}{4} \geq \sqrt[4]{abcd} \Rightarrow \frac{12}{4} \geq \sqrt[4]{abcd} \Rightarrow 3 \geq \sqrt[4]{abcd}$

$\Rightarrow abcd \leq 81$

Given,

$abcd = 27 + ab + ac + ad + bc + dd + cd$

By AM-GM inequality.

$\Rightarrow \frac{(abcd-27)}{6} \geq \sqrt{abcd} \Rightarrow (abcd - 27)^2 \geq 36(abcd)$

$\Rightarrow \frac{ab+ac+ad+bc+db+cd}{6} \geq \sqrt[6]{a^3b^3c^3d^3}$

$\Rightarrow (t - 27)^2 \geq 36t \Rightarrow t^2 - 54t + 729 - 36t \geq 0$

$\Rightarrow t^2 - 90t + 729 \geq 0 \Rightarrow (t - 81)(t - 9) \geq 0 \Rightarrow t \leq 9$ or $t \geq 81$

$abcd \leq 9$ or $abcd \geq 81$

From (1) & (2) $\Rightarrow abcd = 81 \Rightarrow a = b = c = d =$ k

$\Rightarrow k^4 = 81 \Rightarrow k = 3 \Rightarrow \therefore a = 3, b = 3, c = 3, d = 3.$

17. By power mean theorem,

If a, b are positive reals and $m < 0$ (or) $m > 1$, then $\frac{a^m + b^m}{2} \geq \left(\frac{a+b}{2}\right)^m$.

The, equality holds when $a = b \Rightarrow AM \geq GM$

18. By power mean theorem

$$\frac{x^3 + y^3 + z^3}{3} \geq \left(\frac{x + y + z}{3}\right)^3 \Rightarrow \frac{81}{3} \geq \left(\frac{x + y + z}{3}\right)^3$$

$$\Rightarrow \frac{x + y + z}{3} \leq (27)^{1/3} \Rightarrow x + y + z \leq 9$$

19. $AM \geq GM \Rightarrow \frac{x^3 + \frac{1}{x^3}}{2} \geq \sqrt{x^3 \times \frac{1}{x^3}} \Rightarrow x^3 + \frac{1}{x^3} \geq 2.$

20. $AM \geq GM \Rightarrow \frac{x+\frac{1}{x}}{2} \geq \sqrt{x \times \frac{1}{x}} \Rightarrow x + \frac{1}{x} \geq 2.$

21. $AM \geq GM \Rightarrow \frac{x^2+\frac{1}{x^2}}{2} \geq \sqrt{x^2 \cdot \frac{1}{x^2}} \Rightarrow x^2 + \frac{1}{x^2} \geq 2.$

22. [2,3,4] By verification.

23. a − q, b − p, c − s, d − r

a) $\frac{x+y+z}{3} \geq \sqrt[3]{xyz} \Rightarrow x + y + z \geq 3 \cdot \sqrt[3]{10}.$

b) $abc(a + b + c).$

c) $\lambda > 2.$

d) $a, b > 0, a + b = 1 \Rightarrow \frac{a^{\frac{1}{3}}+b^{\frac{1}{3}}}{2} \leq \left(\frac{a+b}{2}\right)^{\frac{1}{3}} \Rightarrow \sqrt[3]{a} + \sqrt[3]{b} \leq 2\left(\frac{1}{2}\right)^{\frac{1}{3}} \leq 2^{\frac{2}{3}}.$

24. By Cauchy-Schwartz inequality,

$(x + y + z)^2 \leq (1^2 + 1^2 + 1^2)(x^2 + y + z^2) \Rightarrow (x + y + z)^2 \leq 3(27) \Rightarrow x + y + z \leq 9$

We know that,

$(x^2 + y^2 + z^2) \leq (x^3 + y^3 + z^3)(x + y + z)$
$\Rightarrow (27)^2 \leq (x^3 + y^3 + z^3)(9)$
$\Rightarrow x^3 + y^3 + z^3 \geq 81.$

25. AM ≥ HM

$$\frac{1 + x + 1 + y + 1 + z}{3} \geq H$$

$$\frac{x + y + z}{3} \geq h$$

From (1) & (2)

$H \geq 1 + h.$

6. STRAIGHT LINES

WORKSHEET – 1

LEVEL – I

1. B	2. B	3. C	4.	5. B
6. A	7. C	8. B	9. A	10. B

LEVEL – II

1. B	2. B	3. A	4. B	5. A
6. A	7. A, B, C	8. C	9. D	10. B
11. C	12. A	13. B	14. 5	15. B
16. C	17. D	18. C	19. C	20. B
21. A				

LEVEL – II

22. A, B, C	23. B, C	24.A	25. A	
26. B	27. D	28. A	29. a-r; b-q; c-t; d-p ;	
30.a-r; b-p; c-s; d-q		31.3	32. 0	33. A, B
34. a-r; b-q; c-s; d-p				

SOLUTIONS

LEVEL – I

2. $\frac{y-7}{3-2} = \frac{4-6}{-1-0} \therefore y = 9$

6. $y = \left(\frac{-8}{6}\right)x + \left(\frac{15}{6}\right) \Rightarrow 6y = -8x + 15$

7. Equation of line in intercept form is $\frac{x}{a} + \frac{y}{b} = 1$

8. Equation of line joining the points is $y - 1 = \frac{2-1}{-1-2}(x - 2) \Rightarrow x + 3y = 5$

9. $y = mx + c, m = \frac{1}{2}, c = 2$

LEVEL – II

1. Slope of the line is $\tan 45^\circ = 1$.
   Equation of the line passing through (1,0) and having slope 1 , is $(y - 0) = 1(x - 1) \Rightarrow x - y - 1 = 0$

2. $y = mx + c$

3. $y-7=\frac{5-7}{2-4}(x-4) \Rightarrow x-y+3=0$

4. Slope of bisector is '- 1 '

   $\therefore$ Equation of bisector is $y-3=-1(x-2)$

5. $(0,-4)$ lies on perpendicular bisector $\overline{PQ}$

6. Given equation is $4x-3y+12=0 \Rightarrow \frac{-4}{\sqrt{4^2+3^2}}x+\frac{3}{\sqrt{4^2+3^2}}y=\frac{12}{\sqrt{4^2+3^2}}$

   $\Rightarrow \frac{-4}{5}x+\frac{3}{5}y=\frac{12}{5}$

7. Normal form of line is $\frac{2}{\sqrt{13}}x+\frac{3}{13}y=\frac{6}{\sqrt{13}}$

8. Line equation is $\sqrt{3}x+y-4=0 \Rightarrow \frac{\sqrt{3}}{2}x+\frac{1}{2}y=2 \Rightarrow x\cos\frac{\pi}{6}+y\sin\frac{\pi}{6}=2$

9. Slope $=-\frac{1}{\sqrt{3}}=\text{Tan}\,\theta \Rightarrow \theta=150^\circ \therefore$ Symmetric form is $\frac{x+2}{-\frac{\sqrt{3}}{2}}=\frac{y-0}{\frac{1}{2}}$

10. $(x_1,y_1)=(3,2); \tan\theta=\frac{3}{4}\ \sin\theta=\frac{3}{5}, \cos\theta=\frac{4}{5}$

    $(x,y)=(x_1\pm r\cos\theta, y_1\pm r\sin\theta)$

11. Given points are $\text{A}(\text{am}_1^2, 2\text{am}_1), \text{B}(\text{am}_2^2, 2\text{am}_2)$

    Slope of AB(m) $=\frac{\text{y}_2-\text{y}_1}{\text{x}_2-\text{x}_1} \Rightarrow \text{m}=\frac{2\text{am}_2-2\text{am}_1}{\text{am}_2^2-\text{am}_1^2}=\frac{2(\text{m}_2-\text{m}_1)}{\text{m}_2^2-\text{m}_1^2}=\frac{2}{\text{m}_1+\text{m}_2}$

    $\therefore$ The required equation is $\text{y}-\text{y}_1=\text{m}(\text{x}-\text{x}_1) \Rightarrow \text{y}-2\text{am}_1=\frac{2}{\text{m}_1+\text{m}_2}(\text{x}-\text{am}_1^2)$

    $\Rightarrow \text{y}(\text{m}_1+\text{m}_2)-2\text{am}_1(\text{m}_1+\text{m}_2)=2\text{x}-2\text{am}_1^2$

    $\therefore 2\text{x}-\text{y}(\text{m}_1+\text{m}_2)-2\text{am}_1^2-2\text{am}_1\,\text{m}_2+2\text{am}_1^2=0$

    $\Rightarrow 2\text{x}-\text{y}(\text{m}_1+\text{m}_2)-2\text{am}_1\,\text{m}_2=0$, is the required equation.

12. Area of Δ le is $\left|\frac{c^2}{2ab}\right|$ sq. units

13. Equation line is $x-y=a$ or $-x+y=a$

14. $a_1a_2+b_1b_2=0 \Rightarrow \text{K}=-5$

15. Equation of line in intercept form is $\frac{x}{a}+\frac{y}{b}=1$

16. $2\sqrt{3}x+2y=-5$; Divide it by 4

    $$\frac{\sqrt{3}}{2}x+\frac{1}{2}y=\frac{-5}{4} \Rightarrow \left(-\frac{\sqrt{3}}{2}\right)x+\left(-\frac{1}{2}\right)y=\frac{5}{4} \ \therefore x\cos 7\pi/6+y\sin 7\pi/6=\frac{5}{4}$$

17. sum of the intercepts of the line $2x+3y+k=0$ is 15. $\frac{-k}{2}+\frac{-k}{3}=15$

18. Given line is $\left(\frac{1}{a}+\frac{k}{b}\right)x+\left(\frac{1}{b}+\frac{k}{a}\right)y-(1+k)=0$

    $\therefore$ Slope $=-\frac{\left(\frac{1}{\text{a}}+\frac{\text{k}}{\text{b}}\right)}{\frac{1}{\text{b}}+\frac{\text{k}}{\text{a}}}$. But given that the slope is -1

$$\therefore \frac{\left(\frac{1}{a}+\frac{k}{b}\right)}{\left(\frac{1}{b}+\frac{k}{a}\right)} = 1 \Rightarrow \frac{1}{a}+\frac{k}{b} = \frac{1}{b}+\frac{k}{a} \Rightarrow \frac{1}{a}-\frac{1}{b} = \frac{k}{a}-\frac{k}{b} \Rightarrow \left(\frac{1}{a}-\frac{1}{b}\right) = k\left(\frac{1}{a}-\frac{1}{b}\right) \Rightarrow k = 1$$

19. Given $P(\alpha, \beta)$ lies on $y = 6x - 1 \Rightarrow \beta = 6\alpha - 1 \Rightarrow 6\alpha - \beta - 1 = 0$

Also $Q(\beta, \alpha)$ lies on $2x - 5y = 5 \Rightarrow 2\beta - 5\alpha = 5 \Rightarrow 5\alpha - 2\beta + 5 = 0$

Solving (1) and (2), we have $\alpha = 1, \beta = 5$

$$\therefore P = (1,5), Q = (5,1) \text{ Slope of } PQ = \frac{5-1}{1-5} = -1$$

Equation of PQ is $y - 5 = -1(x - 1) \Rightarrow y - 5 = -x + 1 \Rightarrow x + y = 6$

20. Let p be the length of the perpendicular drawn from the origin to the required line. The perpendicular makes an angle of 60° with the x-axis. So, the equation of the line is $x\cos 60° + y\sin 60° = p \Rightarrow x + \sqrt{3}y = 2p$

This cuts the coordinate axes at $A(2p, 0)$ and $B\left(0, \frac{2p}{\sqrt{3}}\right)$.

It is given that Area of $\triangle OAB = 54\sqrt{3}$ sq.units.

$$\Rightarrow \frac{1}{2}(OA)(OB) = 54\sqrt{3} \Rightarrow \frac{1}{2} \times 2p \times \frac{2p}{\sqrt{3}} = 54\sqrt{3} \Rightarrow p^2 = 81 \Rightarrow p = 9$$

Hence the equation the line is $x + \sqrt{3}y = 18$.

21. The equation of a line passing through the point of intersection of $x - \sqrt{3}y + \sqrt{3} - 1 = 0$ and $x + y - 2 = 0$ is $(x - \sqrt{3}y + \sqrt{3} - 1) + \lambda(x + y - 2) = 0$

$\Rightarrow x(1 + \lambda) + \lambda(x + y - 2) = 0 \Rightarrow x(1 + \lambda) + y(\lambda - \sqrt{3}) + \sqrt{3} - 1 - 2\lambda = 0$

Now, the line $x - \sqrt{3}y + \sqrt{3} - 1 = 0$ makes 30° angle with x-axis.

Since, the line makes an angle of 15° with this line.

Hence, the line will make 45° angle with x-axis.

$\therefore$ Its slope is $1 \Rightarrow -\left(\frac{1+\lambda}{\lambda-\sqrt{3}}\right) = 1 \Rightarrow \lambda = \frac{\sqrt{3}-1}{2}$

Putting the value of λ in (1), we get $x - y = 0$

LEVEL – III

22. Divide equation by $\sqrt{13} \Rightarrow \frac{2}{\sqrt{13}}x + \frac{3}{\sqrt{13}}y = \frac{6}{\sqrt{13}}$

23. Conceptual

24. Statement I: Normal equation is $\frac{-4}{5}x + \frac{3}{5}y = \frac{12}{5}$

Statement II: $\frac{|c|}{\sqrt{a^2+b^2}} = \frac{|-12|}{\sqrt{4^2+3^2}} = \frac{12}{5}$.

25. Conceptual.

26. Given lines are parallel $\Rightarrow \frac{2}{k} = \frac{3}{6} \Rightarrow k = 4$

27. Given lines are perpendicular $\Rightarrow 3(2) + (-k)(1) = 0 \Rightarrow k = 6$

28. Given lines are perpendicular $\Rightarrow 5(k) + 3(-7) = 0 \Rightarrow k = \frac{21}{5}$

29. a-r; b-q; c-t; d-p ;

30. Slope of a line $ax + by + c = 0$ is $\frac{-a}{b}$

31. $\frac{-3}{k} = -1 \therefore k = 3$

32. Slope of a Horizontal line is 'zero'

33. given $3x + 4y = 5$, dividing both sides with $\sqrt{3^2 + 4^2} = 5 \Rightarrow \frac{3x}{5} - \left(\frac{-4}{5}\right)y = 1$

34. a) $\frac{x}{5k} + \frac{y}{3k} = 1 \Rightarrow \frac{-4}{5k} + \frac{3}{3k} = 1 \Rightarrow k = \frac{1}{5}$. The equation is $3x + 5y = 3$.

WORKSHEET – 2

LEVEL – I

1. C	2. B	3. D	4. C	5. D
6. D	7. D	8. B	9. A	10. C

LEVEL – II

1. B	2. B	3. C	4. A	5. D
6. B	7. A	8. B	9. C	10. B
11. A	12. D	13. C	14. D	15. C
16. A	17. B	18. D	19. C	20. A

LEVEL – III

21. B, C	22. A, B, C, D	23. A, C	24. A, B
25. 45^0 *or* 135^0	26. a-r; b-q; c-s; d-t	27. A	28. A
29. 10	30. 4	31. 3	32. a-r; b-s; c-q; d-p
33. – 3	34. B, D	35. ABCD	

SOLUTIONS

LEVEL – I

4. $d = \frac{|7(0)+24(0)+15|}{\sqrt{(7)^2+(24)^2}}$

5. $d = \frac{|c_1 - c_2|}{\sqrt{a^2+b^2}} = \frac{|7-5|}{\sqrt{1+1}}$

LEVEL – II

1. Direct formula

2. Line joining points $A(-1,-1) \in B(2,1)$ is $2x - 3y - 1 = 0$

   $\therefore$ Ratio is $-\frac{(2(3)-3(4)-1)}{2(1)-3(2)-1} = \frac{-7}{5}$

3. $\alpha_{11} = 2(2) + 3(3) - 20 = -7 \Rightarrow \alpha_{22} = 2(7) + 3(10) - 20 = 24 \Rightarrow \therefore \alpha_{11} \times \alpha_{22} < 0$

4. Ratio $= \frac{|m_1 - m_2|}{1 + m_1 m_2}$ or $\cos\theta = \frac{|a_1 a_2 + b_1 b_2|}{\sqrt{a_1^2 + b_1^2}\sqrt{a_2^2 + b_2^2}}$

5. The perpendicular distance from (x_1, y_1) to the st line $ax + by + c = 0$ is $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

6. Equation of line joing points $(1, -2), (3,2)$ is $2x - y - 4 = 0$

$$\therefore a_1 = 2 : a_2 = 1; b_1 = -1, b_2 = 2$$

8. Let the equation of the line be $ax + by + c = 0$.

Then, according to the given condition, we have

$$\frac{2a + 0b + c}{\sqrt{a^2 + b^2}} + \frac{0 + 2b + c}{\sqrt{a^2 + b^2}} + \frac{a + b + c}{\sqrt{a^2 + b^2}} = 0 \Rightarrow 3a + 3b + 3c = 0 \Rightarrow a + b + c = 0$$

This shows that $ax + by + c = 0$ passes through the fixed point (1,1)

9. Required line is the perpendicular bisector of given points.

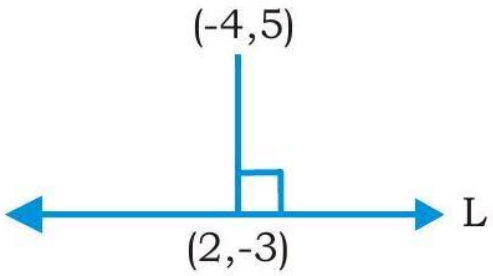
10. Image of A w.r.to diagonal BD is C

11. $\alpha_{11} = 2(3) + 3(-2) - 5 = -5, \alpha_{22} = 2(1) + 3(2) - 5 = 3, \alpha_{33} = 2(1) + 3(1) - 5 = 0$

12. $\alpha_{11} = 3(1) - 5(2) + a = -7 + a, \alpha_{22} = 3(3) - 5(4) + a = -11 + a$

Points were to be same side $\alpha_{11} \times \alpha_{22} > 0 \Rightarrow \therefore (a - 7)(a - 11) > 0$

13. Slope of the line perpendicular to $L = \frac{5+3}{-4-2} = \frac{-4}{3}$ $\therefore$ Slope of $L = 3/4$



Equation of the line L is $y + 3 = \frac{3}{4}(x - 2) \Rightarrow 3x - 4y = 18$

14. $\cos 45° = \frac{[3(2) - k]}{\sqrt{10} \times \sqrt{4 + k^2}}$

$$\frac{1}{2} = \frac{(6 - k)^2}{10(4 + k^2)} \Rightarrow k^2 + 3k - 4 = 0 \Rightarrow (\mathrm{k} - 1)(\mathrm{k} + 4) = 0$$

15. Required line is the perpendicular bisector of given points.

16. We have, p = Length of the perpendicular from (0,0) on $\mathrm{xSec}\,\theta + \mathrm{yCosec}\,\theta = \mathrm{a} \Rightarrow$

$$p = \frac{a}{\sqrt{\mathrm{Sec}^2\,\theta + \mathrm{Cosec}^2\,\theta}} = \frac{a\mathrm{Sin}\,2\theta}{2} \Rightarrow 2p = a\mathrm{Sin}\,2\theta$$

p' = Length of the perpendicular from (0,0) on $\mathrm{xCos}\,\theta - \mathrm{ySin}\,\theta = \mathrm{aCos}\,2\theta$

$$\Rightarrow p' = \frac{a\mathrm{Cos}\,2\theta}{\sqrt{\mathrm{Cos}^2\,\theta + \mathrm{Sin}^2\,\theta}} = a\mathrm{Cos}\,2\theta$$

Now, $(2p)^2 + (p')^2 = a^2\text{Sin}^2\,2\theta + a^2\text{Cos}^2\,2\theta (\because$ from (i) and (ii)) $\Rightarrow 4p^2 + (p')^2 = a^2$

17. $\text{p} = \frac{\text{ab}}{\sqrt{\text{a}^2+\text{b}^2}} \& 2\text{p}^2 = \text{a}^2 + \text{b}^2 \Rightarrow \text{a}^4 - 2\text{p}^2\text{a}^2 + 2\text{p}^4 = 0$

18. Putting $x = t$, in $x + y + 3 = 0$, we get $y = -3 - t$. So let the required point be $(\text{t}, -3 - \text{t})$. This point is at a distance of $\sqrt{5}$ units from $\text{x} + 2\text{y} + 2 = 0$.

Therefore, $\left|\frac{\text{t}-6-2\text{t}+2}{\sqrt{\text{t}^2+2^2}}\right| = \sqrt{5} \Rightarrow \left|\frac{-\text{t}-4}{\sqrt{5}}\right| = \sqrt{5} \Rightarrow \text{t} + 4 = \pm 5 \Rightarrow \text{t} = 1, -9.$

Hence, the required points are $(1, -4)$ and $(-9,6)$.

19. Equate the distance from (0,0) to the line $\frac{x}{x_1} + \frac{y}{y_1} = 2$

20. Let slope of another side is $\text{mtan}\,60° = \left|\frac{m-1}{1+m}\right|; m = -2 \pm \sqrt{3}$

21. For every pair calculate value of $\alpha_{11} \times \alpha_{22}$, where product is > 0

22. Point must satisfy the region $2x - 3y + 5 > 0$ if it lies on origin side

23. $\cos\theta = \frac{|a_1a_2+b_1b_2|}{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}}$

24. Let the point $= (x, 0)$, $\therefore \frac{|x-1|}{\sqrt{2}} = \sqrt{2} \Rightarrow |x - 1| = 2 \therefore x - 1 = \pm 2$

25. $\cos\theta = \frac{\pi}{2}\ m_1 \times m_2 = -1$

If $\theta = 0^0\ m_1 = m_2$

26. a-r; b-q; c-s; d-t ; If L is line, then $\frac{-\text{L}_{11}}{\text{L}_{22}}$ in ratio.

27. We have $\text{Cos}\,\theta = \frac{|a_1a_2+b_1b_2|}{\sqrt{(a_1^2+b_1^2)(a_2^2+b_2^2)}} \Rightarrow 0 = \frac{6-3\text{k}}{\sqrt{(4+9)(9+\text{k}^2)}} \Rightarrow \text{k} = 2.$

28. R is the correct explanation of A.

29. Co-ordinates of $c = \left(\frac{3\lambda}{10}, \frac{\lambda}{10}\right)$. If c dives AB in ratio p: q then p: q = 9: 1

30. Any point on the line $x + y = 4$ is $(t, 4 - t)$, where $t \in R$.

Now, the distance of this point from the line $4x + 3y - 10 = 0$ is 1 . Therefore,

$\left|\frac{4\text{t}+3(4-6)-10}{5}\right| = 1 \Rightarrow |\text{t} + 2| = 5 \Rightarrow \text{t} = 3$ or -7 .

Therefore, the sum of values is -4 .

31. $\frac{\left|k-\frac{2}{3}\right|}{\sqrt{5}} = \frac{7}{3\sqrt{5}} \therefore k = 3$

32. $\cos\theta = \frac{|a_1a_2+b_1b_2|}{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}} \therefore \cos\theta = \frac{2ab}{a^2+b^2}, \sin\theta = \frac{a^2-b^2}{a^2+b^2}, \tan\theta = \frac{a^2-b^2}{2ab}, \cot\theta = \frac{2ab}{a^2-b^2}$

33. Equation of the line midway between the line is $2x - y + \frac{(-9+3)}{2} = 0$

$$\therefore 2x - y - 3 = 0$$

34. [2,4] Slope of line is $\text{m} = \tan(\alpha \pm \theta); \tan\alpha = \frac{1}{2}, \tan\theta = 2$

$$m = \frac{\frac{1}{2}+2}{1-\frac{1}{2}\cdot 2} = \infty; m = \frac{\frac{1}{2}-2}{1+\frac{1}{2}\cdot 2} = -\frac{3}{4}$$

Equation of lines $x = 2$ and $y - 3 = -\frac{3}{4}(x-2)$ or $x = 2$ and $4y + 3x = 18$

35. Let the slope of ' $u = o$ ' be ' m ', then the slope of ' $v = 0$ ' is $\frac{9m}{2}$.

$\frac{7}{9} = \left|\frac{m-\frac{9m}{2}}{1+m\left(\frac{9m}{2}\right)}\right| = \left|\frac{-7m}{2+9m^2}\right| \Rightarrow 9m^2 - 9m + 2 = 0$ or $9m^2 + 9m + 2 = 0$

$\Rightarrow m = \frac{2}{3}, \frac{1}{3}$ or $m = -\frac{2}{3}, -\frac{1}{3}$. Therefore, the equations of the lines are

i. $3y = x$ and $2y = 3x$,

ii. $3y = 2x$ and $y = 3x$,

iii. $x + 3y = 0$ and $3x + 2y = 0$

iv. $2x + 3y = 0$ and $3x + y = 0$.

WORKSHEET – 3

LEVEL – I

1. C	2. B	3. C	4. A	5. D
6. C	7. D	8. B	9. A	10. B
11. C	12. D	13. B		

LEVEL – II

1. B	2. A	3. A	4. B	5. B
6. A	7. B	8. A	9. A	10. C
11. D	12. A	13. A	14. A	15. A
16. B	17. A	18. A	19. A	20. B

LEVEL – III

21. A, C, D	22. D	23. D	24. A	25. D
26. a-q, b-p, c-s, d-r		27. A, B, C	28. A	29. C
30. C	31. B	32. D	33. A	34. A, B, C, D

SOLUTIONS

LEVEL – II

1. [2] 2. [1]

2. [1] Let $S(x, y)$ be the circumcentre $SA = SB = SC \Rightarrow SA^2 = SB^2 = SC^2$

$\Rightarrow (x+2)^2 + (y-3)^2 = (x-2)^2 + (y+1)^2 = (x-4)^2 + (y-0)^2 \Rightarrow (x, y) = \left(\frac{3}{2}, \frac{5}{2}\right)$.

4. [2] $\begin{vmatrix} 3 & 2 & -5 \\ 2 & -5 & 3 \\ 5 & b & c \end{vmatrix} = 0$

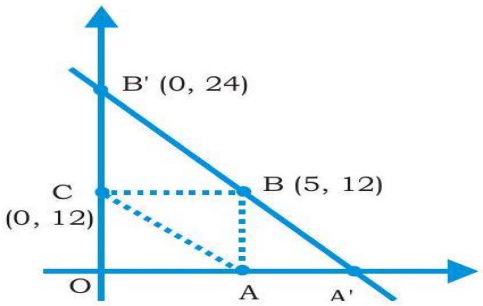
5. [2] $\frac{3x-4y+7}{5} = \pm\frac{12x+5y-2}{13} \Rightarrow 39x - 52y + 91 = -(60x + 25y - 10)$
and $39x - 52y + 91 = +(60x + 25y - 10)$; One of the bisectors is $11x - 3y + 9 = 0$.

6. [1] Solve $3x - 2y - 1 = 0, x - 3y + 9 = 0$

7. [2] Intersecting point $(14, -9)$ $m = \frac{-2}{5}$

8. [1] $\frac{q^3-r^3}{q-r} = \frac{r^3-p^3}{r-p} = \frac{p^3-q^3}{p-q}$

9. [1]AB ⊥ BC.



The vertices of the triangle are $O(0,0), A'(10,0), B'(0,24)$.
∴ Orthocentre of $\Delta A'OB'$ is $O(0,0)$.

10. [3] $a + b + c = 0$ $a^3 + b^3 + c^3 = 3abc$

11. [4] Clearly the line $2x - y + 5 = 0$ doesnot contain $(5, -2)$ so $(x + y + 1) + \lambda(2x - y + 5) = 0$
passes through $(5, -2) \Rightarrow 4 + \lambda(17) = 0 \Rightarrow \lambda = \frac{-4}{17}$
∴ The equation of the required line is $17(x + y + 1) - 4(2x - y + 5) = 0$
$\Rightarrow 3x + 7y - 1 = 0$.

12. [1] The three lines are concurrent if $\begin{vmatrix} 1 & 2 & -9 \\ 3 & 5 & -5 \\ a & b & -1 \end{vmatrix} = 0$
$\Rightarrow 35a - 22\,b + 1 = 0 \Rightarrow 35x - 22y + 1 = 0$ passes through (a, b)

13. [1] $I_1 = \left(\frac{-ax_1+bx_2+cx_3}{-a+b+c}, \frac{-ay_1+by_2+cy_3}{-a+b+c}\right)$

14. [1] $a_1a_2 + b_1b_2 < 0$; $\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}} = -\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$

15. [1] Foot of the perpendicular $D = \left(\frac{1}{2}, \frac{1}{2}\right)$; G(= 0) divides median in the ratio 2: 1

16. [2] Draw the diagram

17. [1] Verify angular bisector formula

18. [1] Point on two lines are $B(\alpha, 1-\alpha), C(\alpha, 2\alpha+4)$ and two lines intersected at $A(-1,2), S(2,1)$ is circumsentre $SA = SB = SC$

19. [1] Given $S = (0,0), G =$ Mid point of $(2,3)$ and $(4,7) = (3,5)$.

Let $O = (a, b)$ We know that O, G, S lie on the same line and G divides OS in the ratio $2:1 \Rightarrow 0 = (9,15)$ which lies on $5x - 3y = 0$.

20. [2] Given $A(4,0), B(0,3)$ and $C(6,1)$.

We have the Slope of $ACm_1 = \frac{1}{2}$, Slope of $BCm_2 = \frac{-1}{3}$.

Let ' m ' be the Sope of Bisector of $\angle C$, which makes equal angles with AC and BC.

$$\Rightarrow \left|\frac{m-\frac{1}{2}}{1+\frac{m}{2}}\right| = \left|\frac{m+\frac{1}{3}}{1-\frac{m}{3}}\right| \Rightarrow (2\ m-1)(3-m) = \pm(3\ m+1)(2+m)$$

$$\Rightarrow m^2 + 14\ m - 1 = 0 \Rightarrow m = -7 \pm 5\sqrt{2}.$$

21. [1,3,4] $\begin{vmatrix} 3 & 1 & 2 \\ 2 & -1 & 3 \\ a^2 & 2a & 6 \end{vmatrix} = 0$

22. [4] $x - y - 2 = 0$ is perpendicular to $x + y + 4 = 0$ but, $x - y - 2 = 0$ is not bisector

23. [4] Third vertex $= \left(\frac{x_1+x_2 \pm \sqrt{3}(y_1-y_2)}{2}, \frac{y_1+y_2 \mp \sqrt{3}(x_1-x_2)}{2}\right)$

$$= \left[\frac{2+2 \pm \sqrt{3}(4-6)}{2}, \frac{(4+6) \mp \sqrt{3}(0)}{2}\right] = \left(\frac{4 \pm \sqrt{3}(-2)}{2}, \frac{10}{2}\right)$$

$$= \left(\frac{4 \mp 2\sqrt{3}}{2}, 5\right) = (2 \mp \sqrt{3}, 5) = [2-\sqrt{3}, 5] \text{ and } (2+\sqrt{3}, 5)$$

24. [1] Orthocentre = centroid $= \left(\frac{6-\sqrt{3}}{3}, 5\right)$ or $\left(\frac{6+\sqrt{3}}{3}, 5\right)$

25. [4] As the circumcentre and orthocentre coincides, distance between them is ' O ' 26. a-q; b-p; c-s; d-r

(a) $(x - 3y + 9) + k(3x - 2y - 1) = 0 \Rightarrow$ solve equations

$x - 3y + 9 = 0, 3x - 2y - 1 = 0$

(b) $a(x + y + 1) + b(2x - y + 5) = 0$

solve equations $x + y + 1 = 0, 2x - y + 5 = 0$

(c) Solve equations $2x + 3y + 1 = 0,\ 3x - 2y - 5 = 0$

(d) Solve equations $x + y - 4 = 0, 2x - y - 2 = 0$

27. [1,2,3] ; for concurrency of three lines $px + qy + r = 0; qx + ry + p = 0; rx + py + q = 0$

we must have, $\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} = 0$

Operate $C_1 + C_2 + C_3 \begin{vmatrix} p+q+r & q & r \\ p+q+r & r & p \\ p+q+r & p & q \end{vmatrix} = 0 \Rightarrow (p+q+r)\begin{vmatrix} 1 & q & r \\ 1 & r & p \\ 1 & p & q \end{vmatrix} = 0$

Operate $C_1 - C_2 - C_3 \Rightarrow (p+q+r)\begin{vmatrix} 0 & q-r & r-p \\ 0 & r-p & p-q \\ 1 & p & q \end{vmatrix} = 0$

$\Rightarrow (p+q+r)(pq - q^2 - rp + rq - r^2 + pr + pr - p^2) = 0$

$\Rightarrow (p+q+r)(p^2+q^2+r^2-pq-pr-rp) = 0 \Rightarrow p^3+q^3+r^3-3pqr$

It is clear that 1,2,3 are correct options.

28, 29, 30 Equations of bisector of the angles between the given lines are

$$\frac{4x+3y-6}{\sqrt{4^2+3^2}} = \pm\frac{5x+12y+9}{\sqrt{5^2+12^2}} \Rightarrow \frac{4x+3y-6}{5} = \pm\frac{5x+12y+9}{13}$$

$\Rightarrow 52x+39y-78 = \pm(25x+60y+45) \Rightarrow 27x-21y-123 = 0, 77x+99y-33 = 0$

$\Rightarrow 9x-7y-41 = 0, 7x+9y-3 = 0$

Let the angle between the line $4x+3y-6=0$ and bisector $9x-7y-41=0$ be ' θ ', then $\text{Tan}\,\theta = \left|\frac{-\frac{4}{3}-\frac{9}{7}}{1+\left(\frac{-4}{3}\right)\frac{9}{7}}\right| = \frac{11}{3} > 1$

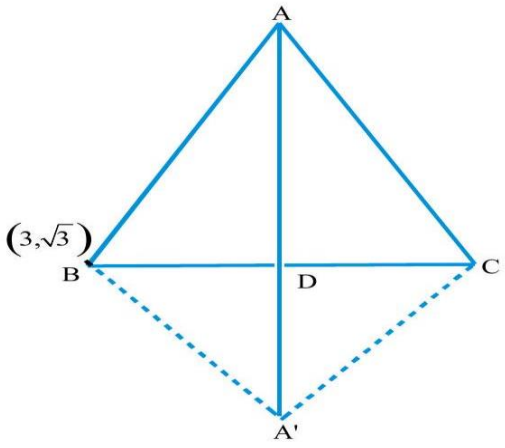
28. [1] The bisector of the obtuse angle is $9x-7y-41=0$

29. [3] The bisector of the acute angle is $7x+9y-3=0$

30. [3] The bisector of the angle containing the origin

$$\frac{-4x-3y+6}{\sqrt{(-4)^2+(-3)^2}} = \frac{5x+12y+9}{\sqrt{5^2+12^2}} \Rightarrow 7x+9y-3=0$$

31. [2]



32. [4] Equation of altitude AD is $\sqrt{3}x + y - 2\sqrt{3} = 0$

C = The image of point B about the line AD is $(0,0)$ $\therefore$ BC $= 2\sqrt{3}$

Find A = Third Vertex of the Equilateral triangle

i.e., $A = (3, -\sqrt{3}), B = (3, \sqrt{3})$ and $C = (0,0)$

33. [1] In Equilateral triangle Orthocentre = Centroid = $(2,0)$ or $(1, \sqrt{3})$

34. [1,2,3,4] Side BC will be perpendicular to the bisector of the angle BAC. Now equations of the bisectors of lines AB and AC are $\frac{(x+y-5)}{\sqrt{2}} = \pm\frac{(7x-y-3)}{5\sqrt{2}} \Rightarrow x - 3y + 11 = 0$ or $3x + y - 7 = 0$

Let equation of side BC be $x - 3y + \lambda = 0$ and altitude through vertex A be AD. Then equation of AD is $3x + y - 7 = 0$. If AD = λ, then $\triangle ABC = \frac{1}{2} \times \lambda \times 2\lambda\,|\tan\theta| = \lambda^2|\tan\theta|$.

Hence $\lambda^2 \times \frac{1}{2} = 5 \Rightarrow \lambda^2 = 10 \Rightarrow (11-\lambda)^2 = 100 \Rightarrow 11 - \lambda = \pm 10 \Rightarrow \lambda = 1, 21$

Hence, equation of BC is $x - 3y + 1 = 0$ or $x - 3y + 21 = 0$. Similarly, if equation of BC is $3x + y + \lambda = 0$, then equation of AD will be $x - 3y + 11 = 0$. Therefore,

$$|\tan\theta| = \left|\frac{7 - \frac{1}{3}}{1 + \frac{7}{3}}\right| = 2 \Rightarrow \lambda^2|\tan\theta| = 2\lambda^2 = 5 \Rightarrow \lambda^2 = \frac{5}{2}$$

$$\frac{5}{2} = \frac{(3+4+\lambda)^2}{10} \Rightarrow 7 + \lambda = \pm 5 \Rightarrow \lambda = 2, -12$$

Hence, equation of BC is $3x + y - 2 = 0$ or $3x + y - 12 = 0$. Finally, there are four possible equations of side BC, viz.,

$x - 3y + 1 = 0, x - 3y + 21 = 0, 3x + y - 2 = 0$ or $3x + y - 12 = 0$.

WORKSHEET – 4

LEVEL – I

1. C	2. D	3. B	4. C	5. A
6. D	7. C	8. B	9. B	10. D

LEVEL – II

1. C	2. D	3. A	4. A	5. A
6. A	7. A	8. A	9. C	10. A
11. C	12. B	13. A	14. C	15. B
16. C	17. A	18. C	19. D	20. D
21. A				

LEVEL – III

22. A, C, D	23. D	24. B	25. A
26. C	27. 3	28. A, B	29. a-s; b-r; c-q; d-p
30. C	31. B	32. D	

SOLUTIONS

LEVEL – I

7. Let P be $(4,-13)$ and let $Q=(x_2,y_2)$ be the mirror image P with respect to line mirror $5x+y+6=0$.

Let $M(\alpha,\beta)$ be the foot of perpendicular from $P(4,-13)$ on the line mirror $5x+y+6=0$, then $\frac{\alpha-4}{5}=\frac{\beta+13}{1}=\frac{-(5(4)-13+6)}{5^2+1^2}$

$\Rightarrow \frac{\alpha-4}{5}=\frac{\beta+13}{1}=-\frac{1}{2} \Rightarrow M=\left(\frac{3}{2},\frac{-27}{2}\right)$

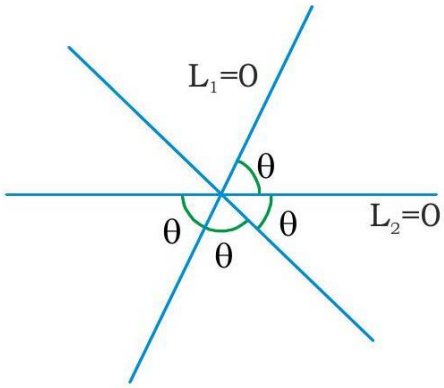
$\because$ M is the mid point of P and Q, then

$Q=(x_2,y_2)=\left(2\times\frac{3}{2}-4,2\times\left(\frac{-27}{2}\right)+13\right)=(-1,-14)$

8. Image line is $(3^2+2^2)(2x-y-1)-2[2(3)+(-1)(-2)](3x-2y+4)=0$

$\Rightarrow 13(2x-y-1)-16(3x-2y+4)=0 \Rightarrow 22x-19y+77=0.$

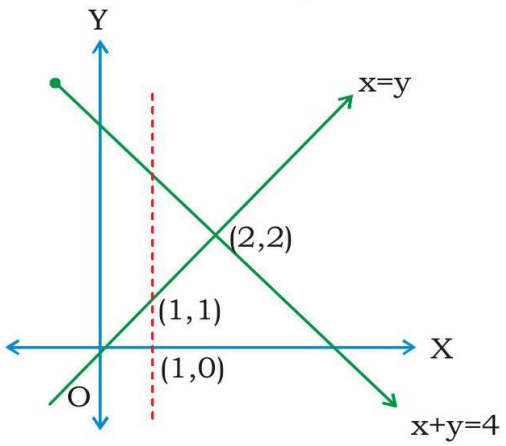
9.



from the diagram, we have $3\theta=180 \Rightarrow \theta=60^\circ$

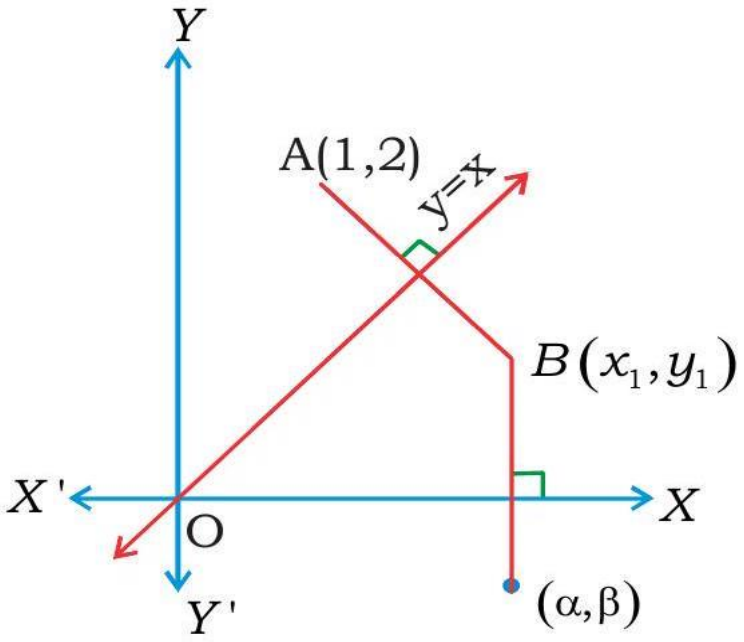
LEVEL – II

1. From the figure, the y-coordinates of P vary from 0 to 1.



So, to be an interiorpoint. $\Rightarrow$ 0 < Ordinate of P < 1

2.



Let (x_1, y_1) be the image of the point (1,2) about the line y = x.

Then $x_1 = 2, y_1 = 1$

Also given image of $B(x_1, y_1)$ by the line mirror y = 0 is (α, β). Then $\alpha = x_1 = 2$ and $\beta = -y_1 = -1$ { from (1)}. Hence $\alpha = 2$ and $\beta = -1$.

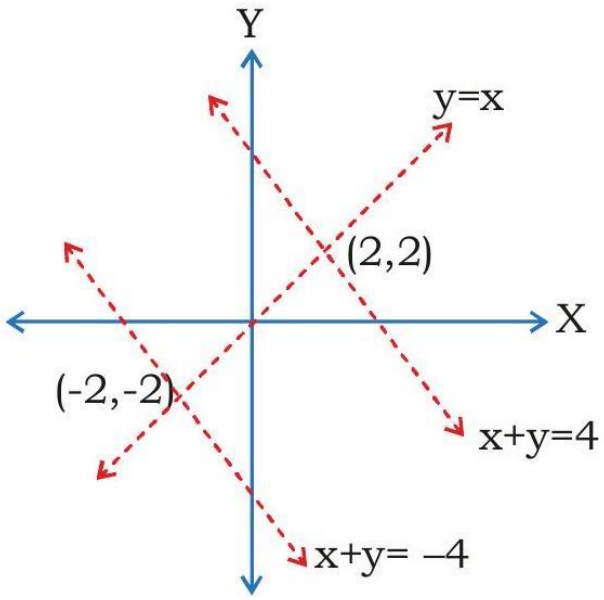
3. The given lines are Parallel and here the Image Equation will be $ax + by + 2c_2 - c_1 = 0 \Rightarrow 2x + 3y + 2(9) - 4 = 0 \Rightarrow 2x + 3y + 14 = 0$.

4. (1) Lines $x + y = 4$ and $x + y = -4$ are parallel and points (2,2) and $(-2, -2)$ lie on these lines.

If the point (a, a) lies between the lines, then $a > -2$ and $a < 2$, i.e.,

$-2 < a < 2$

or $|a| < 2$



5. Let (x_1, y_1) be the image of the point (1,2) about the line y = x.

   Then $(x_1, y_1) = (2,1)$

   Also given the image of $B(x_1, y_1)$ by the line mirror y = 0 is (α, β).

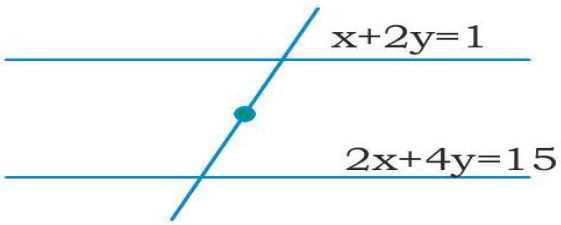
   Then $\alpha = x_1 = 2$ and $\beta = -y_1 = -1$ from (1)

   Hence $(\alpha, \beta) = (2, -1)$.

6. $ax + by + c = 0$ w.r.to $lx + my + n = 0$ is

   $(l^2 + m^2)(ax + by + c) = 2(a1 + bm)(lx + my + n)$

7.



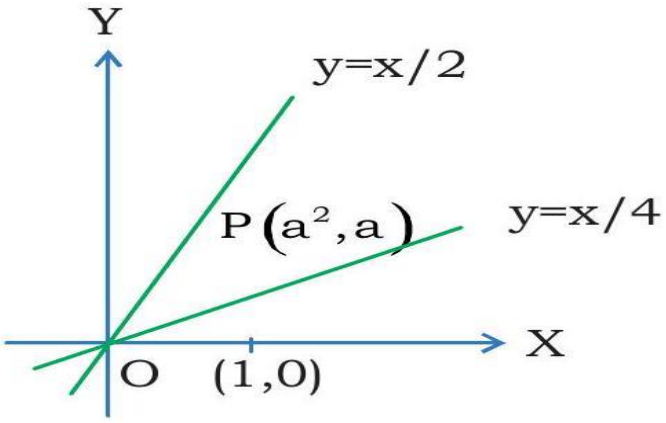
   Let P be on $x + 2y = 1$, then

   $1 + \frac{t}{\sqrt{2}} + 2\left(2 + \frac{t}{\sqrt{2}}\right) = 1 \Rightarrow t = \frac{-4\sqrt{2}}{3}$

   Let P be on $2x + 4y = 15$, then $2\left(1 + \frac{t}{\sqrt{2}}\right) + 4\left(2 + \frac{t}{\sqrt{2}}\right) = 15 \Rightarrow t = \frac{5\sqrt{2}}{6}$

   Since the point lies between the lines and $x = t, t \in \left(\frac{-4\sqrt{2}}{3}, \frac{5\sqrt{2}}{3}\right)$.

8. Acute angle is formed by the lines in the first and third quadrants. But $a^2 > 0$. Hence, point $P(a^2, a)$ lies in the first quadrant.



Since (1,0) and P lie on the same side of $x - 2y = 0$ and (1,0) and P lie on the opposite sides of $x - 4y = 0$, we have

$a - (a^2/4) > 0$ and $a - (a^2/2) < 0$

or $0 < a < 4$ and $a \in (-\infty, 0) \cup (2, \infty)$

or $a \in (2,4)$

9. We have to find the locus of the point (h, k) whose image on the line $2x - y - 1 = 0$ lies on the line y = x,

Now, the image of (h, k) on the line $2x - y - 1 = 0$ is given by

$\frac{x_2 - h}{2} = \frac{y_2 - k}{-1} = -\frac{2(2h - k - 1)}{5}$ or $x_2 = \frac{-3h + 4k + 4}{5}$ and $y_2 = \frac{4h + 3k - 2}{5}$.

This point lies on y = x. Then, $\frac{-3h + 4k + 4}{5} = \frac{4h + 3k - 2}{5}$ or $7h - k = 6$

10. Let $L(x, y) = 3x + 2y + 10$

$\therefore L(4,2) = 12 + 4 + 10 = 26$ and

$L(2,4) = 6 + 8 + 10 = 24$

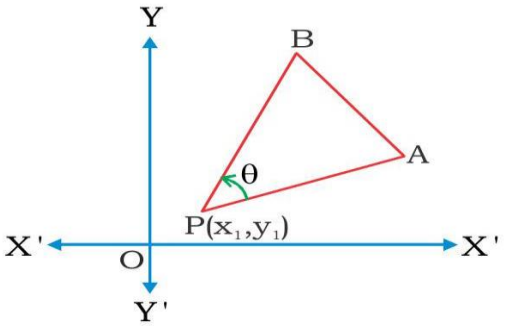
$\therefore$ A and B lie on the same side of the line $3x + 2y + 10 = 0$

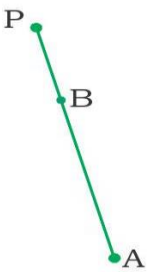
Equation of line AB is $y - 2 = \frac{4-2}{2-4}(x - 4)$

$\Rightarrow x + y - 6 = 0$

This line (2) meets (1) at $P \equiv (-22,28)$ which is the required point.

Alternative Method : Let P be (x_1, y_1)





and $\angle APB = \theta$ then $\cos\theta = \frac{(PA)^2+(PB)^2-(AB)^2}{2PA\cdot PB}$

since $\cos\theta \le 1 \Rightarrow \frac{(PA)^2+(PB)^2-(AB)^2}{2PA\cdot PB} \le 1$

$\Rightarrow (PA-PB)^2 \le (AB)^2 \Rightarrow |PA-PB| \le |AB| \Rightarrow |PA-PB| \le 2\sqrt{2}$

Maximum value of $|PA-PB|$ is $2\sqrt{2}$ when $\theta = 0$.

i.e., P lies on the line AB as well as on the given line.

$\therefore$ Equation of AB is $y-2 = \frac{4-2}{2-4}(x-4) \Rightarrow y-2 = -x+4 \Rightarrow x+y = 6$ and given line $3x+2y+10 = 0$

Solving (1) and (2), we get $P(-22,28)$.

11. Required equation is $(l^2+m^2)(ax+by+c) = 2(al+bm)(lx+my+n)$

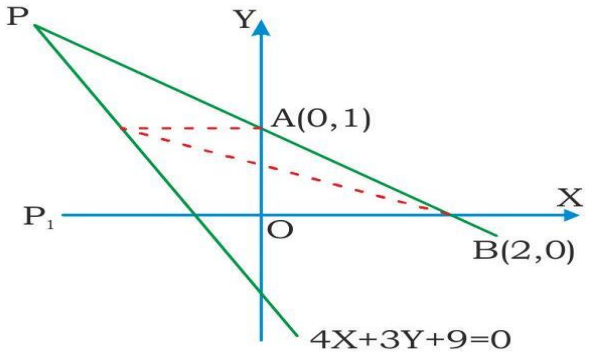
$\Rightarrow 5(3x+4y+5) = 2(6+4)(2x+y+1) \Rightarrow 5x-1 = 0.$

12. Here $P(a,a^2)$ lies in the first quadrant. $\therefore a > 0$.

Also the line for $L = x+y-2 = 0$, $L(0,0) = 0+0-2 < 0 \therefore a+a^2-2 < 0$

$\Rightarrow (a-1)(a+2) < 0 \Rightarrow -2 < a < 1 \Rightarrow 0 < a < 1 \ (\because a > 0)$

13.



Let $L(x,y) = 4x+3y+9$

$\therefore$ $L(0,1) = 3+9 = 12$ and $L(2,0) = 8+9 = 17$

$\therefore$ A and B lie on the same side of the line $4x+3y+9 = 0$

Equation of the line AB is $y-1 = \frac{0-1}{2-0}(x-0) \Rightarrow x+2y-2 = 0$

Thus, $|PA-PB| \le AB$ is maximum if the points A, B, P are collinear.

Solving (1) and (2) the point P is $\left(\frac{-24}{5}, \frac{17}{5}\right)$.

14. Sol: (3) Consider a point A', the image of A in $y = x$.

Therefore the coordinates of A' are (4,3). (or)

(Notice that A and B lie on the same side with respect to $x = y$).

Then $PA = PA'$. Thus, $PA + PB$ is minimum, if $PA' + PB$ is minimum, i.e., if P, A', B are collinear. Now, the equation of AB is

$y-3=\frac{13-3}{7-4}(x-4) \Rightarrow -10x+3y+31=0$

It intersects y = x at $\left(\frac{31}{7},\frac{31}{7}\right)$, which is the required point P.

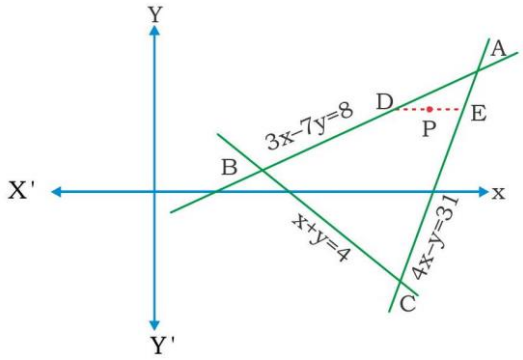
15. Let the equation of the line be $(y-2)=m(x-8)$, where $m<0$

or $P \equiv \left(8-\frac{2}{m},0\right)$ and $Q \equiv (0,2-8\ m)$

Now, $OP+OQ=\left|8-\frac{2}{m}\right|+|2-8\ m|=10+\frac{2}{-m}+(-8\ m)$

$\geq 10+2\sqrt{\frac{2}{-m}\times(-8\ m)} \geq 18\ (\because AM \geq GM)$

16. Let $P \equiv (\lambda,2)$. First draw the exact diagram of $\triangle ABC$ the point $P(\lambda,2)$ move on the line y = 2 for all λ.



Now D and E are the intersection of $3x-7y=8, y=2$ and $4x-y=31, y=2$.

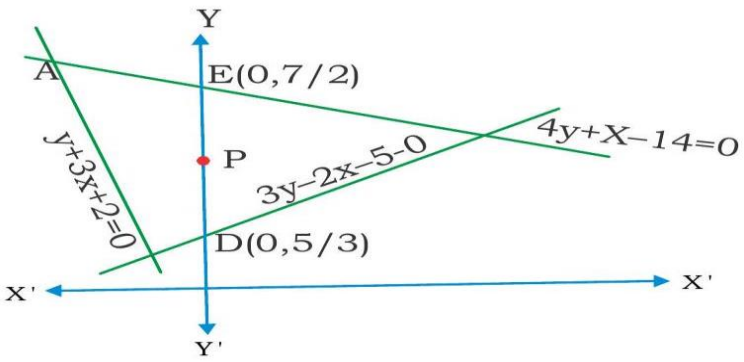
$\therefore\ D \equiv \left(\frac{22}{3},2\right)$ and $E \equiv \left(\frac{33}{4},2\right)$.

Thus the points on the line $y=2$ whose x-co-ordinates lies between $\frac{22}{3}$ and $\frac{33}{4}$ lie within the $\triangle ABC$. Hence, $\frac{22}{3}<\lambda<\frac{33}{4}$ i.e., $\lambda \in \left(\frac{22}{3},\frac{33}{4}\right)$.

17. Let $P \equiv (0,\beta)$, first draw the exact diagram of the triangle.

The point $P(0,\beta)$ move on the line $x=0$(y-axis) for all β.

Now, D and E are the intersection of $3y-2x-5=0, x=0$ and $4y+x-14=0, x=0$.



$\therefore D \equiv (0,5/3)$ and $E \equiv (0,7/2)$

Thus the points on the line $x = 0$ whose y-co-ordinates lies between $5/3$ and $7/2$ lie on or inside the triangle ABC.

Hence $\frac{5}{3} \leq \beta \leq \frac{7}{2}$ i.e., $\beta \in \left[\frac{5}{3}, \frac{7}{2}\right]$.

18. Given lines are $L_1 \equiv 3x - y + 1 = 0$ ..... (i) and $L_2 \equiv x + 2y - 5 = 0$ .... (ii)

Now, $P(a^2, a+1)$ lies in the angle containing O(0,0). Then O and P must lie on the same side of both the lines. Now, $L_1(0,0) = 1 > 0$

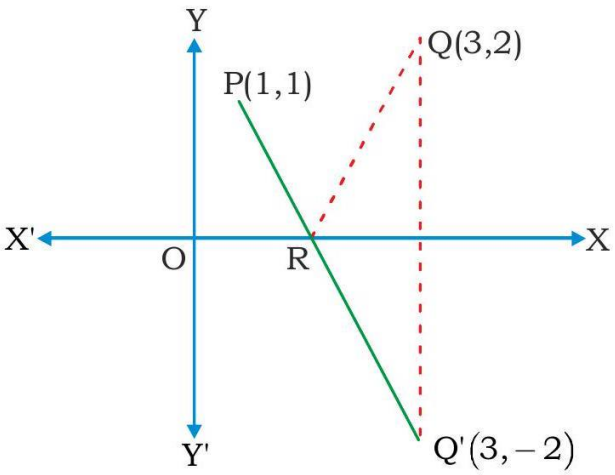
$\therefore L_1(a^2, a+1) = 3a^2 - (a+1) + 1 > 0$ or $a(3a-1) > 0$ or $a \in (-\infty, 0) \cup \left(\frac{1}{3}, \infty\right)$ ...

Now, $L_2(0,0) = -5 < 0$

$\therefore L_2(a^2, a+1) = a^2 + 2(a+1) - 5 < 0$ or $a^2 + 2a - 3 < 0$ or $(a-1)(a+3) < 0$ or $a \in (-3,1)$ ... (iv)

From (iii) and (iv), we have $a \in (-3,0) \cup \left(\frac{1}{3}, 1\right)$.

19. Since P and Q lie on the same side of x-axis.



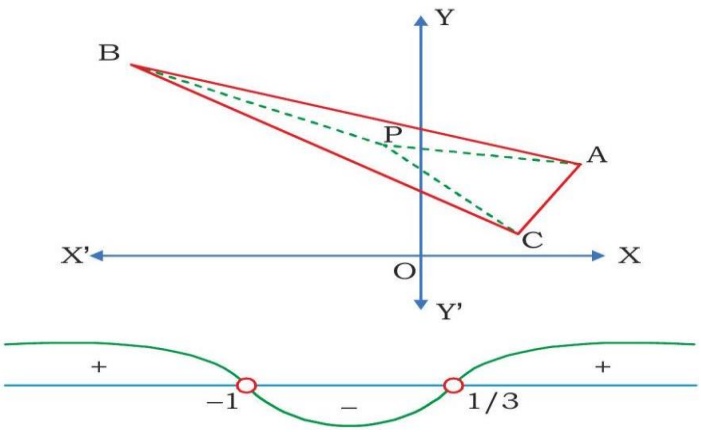
The image of $Q(3,2)$ about x-axis is $Q'(3,-2)$ then the equation of line PQ' is

$$y - 1 = \frac{-2-1}{3-1}(x-1) \Rightarrow 3x + 2y - 5 = 0$$

This line meets x-axis at $R\left(\frac{5}{3}, 0\right)$ which is the required point.

20. The co-ordinates of the vertices are

$A\left(\frac{5}{4}, \frac{7}{8}\right), B(-7,5)$ and $C\left(\frac{1}{3}, \frac{1}{9}\right)$



$\because P(\alpha, \alpha^2)$ lies inside the $\triangle ABC$ then

(i) A and P must lie on the same side of BC

(ii) B and P must lie on the same side of CA

(iii) C and P must lie on the same side of AB, then

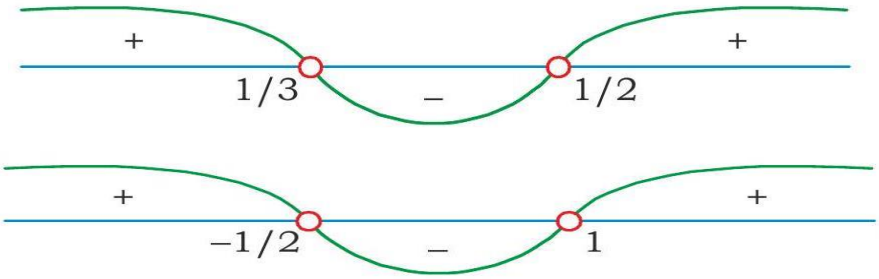
$$\frac{\frac{5}{2}+\frac{21}{8}-1}{2\alpha+3\alpha^2-1}>0 \Rightarrow \frac{33}{2\alpha+3\alpha^2-1}>0 \Rightarrow 3\alpha^2+2\alpha-1>0 \Rightarrow (\alpha+1)\left(\alpha-\frac{1}{3}\right)>0$$

$$\Rightarrow \alpha \in (-\infty,-1)\cup\left(\frac{1}{3},\infty\right)$$

and $\frac{-35-30-1}{5\alpha-6\alpha^2-1}>0 \Rightarrow 5\alpha-6\alpha^2-1<0 \Rightarrow \left(\alpha-\frac{1}{3}\right)\left(\alpha-\frac{1}{2}\right)>0$

$\therefore \alpha \in (-\infty,1/3)\cup(1/2,\infty)$ and $\frac{\frac{1}{3}+\frac{2}{9}-3}{\alpha+2\alpha^2-3}>0 \Rightarrow \alpha+2\alpha^2-3<0$

$\Rightarrow (2\alpha+3)(\alpha-1)<0$



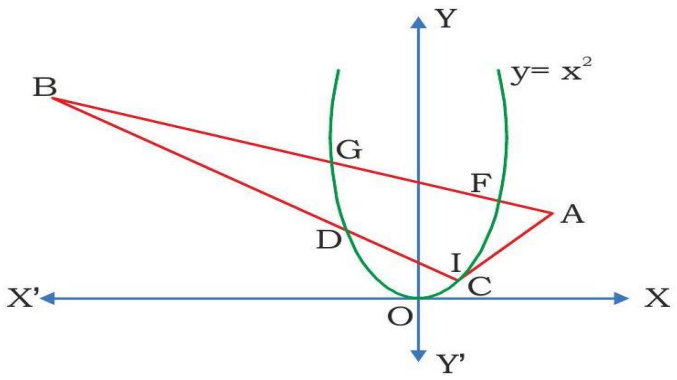
$\therefore \alpha \in (-3/2,1)$

From (1), (2) and (3), we get $\alpha \in (-3/2,-1)\cup(1/2,1)$

: Let $P(\alpha,\alpha^2)$ first draw the exact diagram of $\triangle ABC$. The point $P(\alpha,\alpha^2)$ move on the curve $y=x^2$ for all α.

Now intersection of $y=x^2$ and $2x+3y-1=0 \Rightarrow 2x+3x^2-1=0 \therefore x=-1, x=\frac{1}{3}$

Let intersection points $D \equiv (-1,1)$ and $E \equiv \left(\frac{1}{3},\frac{1}{9}\right)$ intersection of $y=x^2$ and $x+2y-3=0$



or $x + 2x^2 - 3 = 0 \ \therefore x = 1, x = -3/2$

Let intersection points $F \equiv (1,1)$ and $G \equiv \left(-\frac{3}{2}, \frac{9}{4}\right)$ and intersection of $y = x^2$ and $5x - 6y - 1 = 0 \Rightarrow 5x - 6x^2 - 1 = 0 \ \therefore \ x = \frac{1}{3}, x = \frac{1}{2}$

Let intersection points $H \equiv \left(\frac{1}{3}, \frac{1}{9}\right)$ and $I \equiv \left(\frac{1}{2}, \frac{1}{4}\right)$.

Thus the points on the curve $y = x^2$ whose x-co-ordinate lies between $-3/2$ & -1 and $\frac{1}{2}$&1 lies within the triangle ABC.

Hence, $-\frac{3}{2} < \alpha < -1$ and $\frac{1}{2} < \alpha < 1$

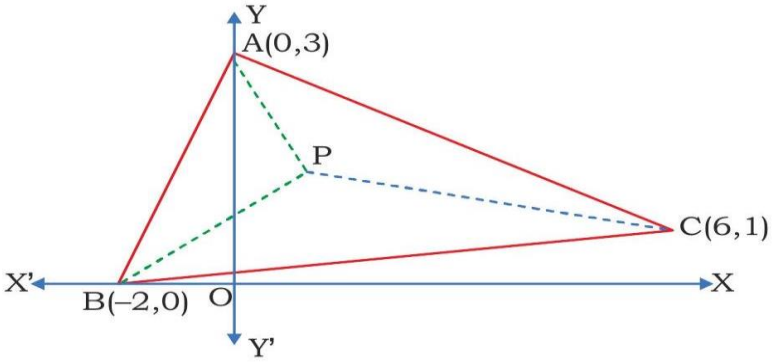
21. Equations of sides

$BC: x - 8y + 2 = 0, CA: x + 3y - 9 = 0$ and $AB: 3x - 2y + 6 = 0$

Since $P(r, t+1)$ lies inside the triangle ABC, then P and A must be on the same side of BC

$$\therefore \frac{\text{value of } (x - 8y + 2) \text{ at } P(r, t+1)}{\text{value of } (x - 8y + 2) \text{ at } A(0,3)}$$

i.e., $\frac{t-8(t+1)+2}{0-24+2} > 0$ or $\frac{-7t-6}{-22} > 0$ or $7t + 6 > 0 \ \therefore \ t > -\frac{6}{7}$



and P, B must be on the same side of CA

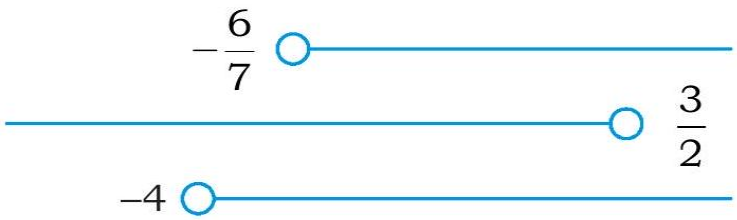
$$\frac{\text{value of } (x + 3y - 9) \text{ atP } (t, t+1)}{\text{value of } (x + 3y - 9) \text{ at } B(-2,0)}$$

i.e., $\frac{t+3(t+1)-9}{-2+0-9} > 0$ or $\frac{4t-6}{-11} > 0$ or $4t-6<0 \ \therefore \ t<\frac{3}{2}$

and P, C must be on the same side of AB

$\therefore \ \frac{\text{value of } (3x-2y+6) \text{ at } P(t,t+1)}{\text{value of } (3x-2y+6) \text{ at } C(6,1)} > 0$

i.e., $\frac{3t-2(t+1)+6}{18-2+6} > 0$ or $\frac{t+4}{22} > 0$ or $t+4>0 \ \therefore \ t>-4$



From equations (1), (2) and (3), we get $-\frac{6}{7}<t<\frac{3}{2}$ i.e., $t \in \left(-\frac{6}{7},\frac{3}{2}\right)$

22. 0 and the point (α,α^2) lie on the opposite sides with respect to $2x+3y-1=0$, Hence $2\alpha+3\alpha^2-1>0$

0 and the point (α,α^2) lie on the same sides with respect to

$x+2y-3=0$. Hence, $\alpha+2\alpha^2-3<0$

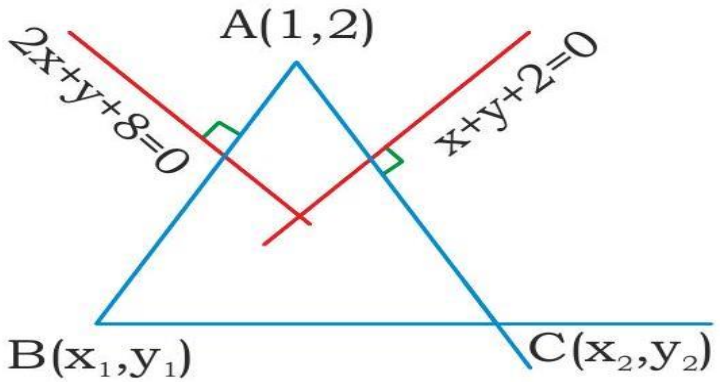
Again 0 and the point (α,α^2) lie on the opposite sides with respect to

$5x-6y-1=0$. Hence, $5\alpha-6\alpha^2-1<0$ or $6\alpha^2-5\alpha+1>0$

23. According to the given data

$2a-5+a^2-3<0$ or $a^2+2a-8<0$ or $(a-2)(a+4)<0$ or $a \in (-4,2)$

24. Let the given vertex be $A(1,2), B, C$ are the image of A with respect to the lines $2x+y+8=0$ and $x+y+2=0$.



Let the $B(x_1,y_1)$ and $C(x_2,y_2)$ be the co-ordinates.

$\therefore$ Co-ordinates of B are

$$\frac{x_1-1}{2}=\frac{y_1-2}{1}=\frac{-2(2\times1+2+8)}{2^2+1} \Rightarrow B(x_1,y_1)=\left(\frac{-43}{5},\frac{-14}{5}\right)$$

25. $\therefore$ Co-ordinates of C are

$$\frac{x_2-1}{1}=\frac{y_2-2}{1}=\frac{-2(1+2+2)}{1^2+1}\Rightarrow C(x_2,y_2)=(-4,-3)$$

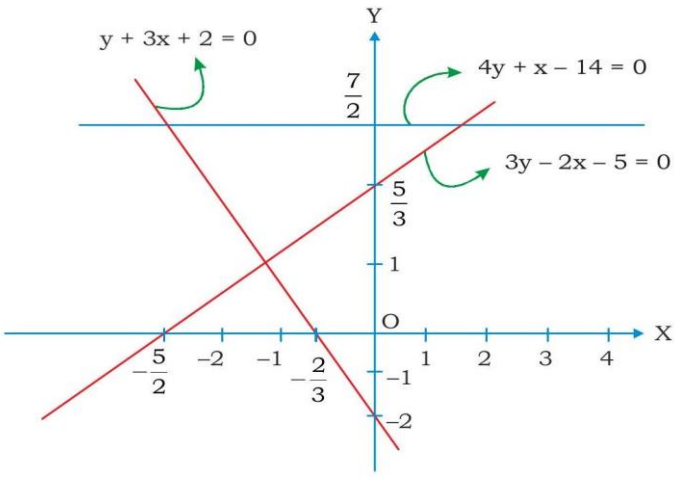
26. $\therefore$ Centroid $G=\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)=\left(\frac{-58}{15},\frac{-19}{15}\right)$

27. Image of $ax+by+c=0$ w.r.t $lx+my+n=0$ is

$$(l^2+m^2)(ax+by+c)=2(al+bm)(lx+my+n)$$
$$\therefore((2)^2+(-1)^2)[3x-4y+11]=2[(3)(2)+(-4)(-1)][2x-y-1]$$
$$\therefore x=3$$

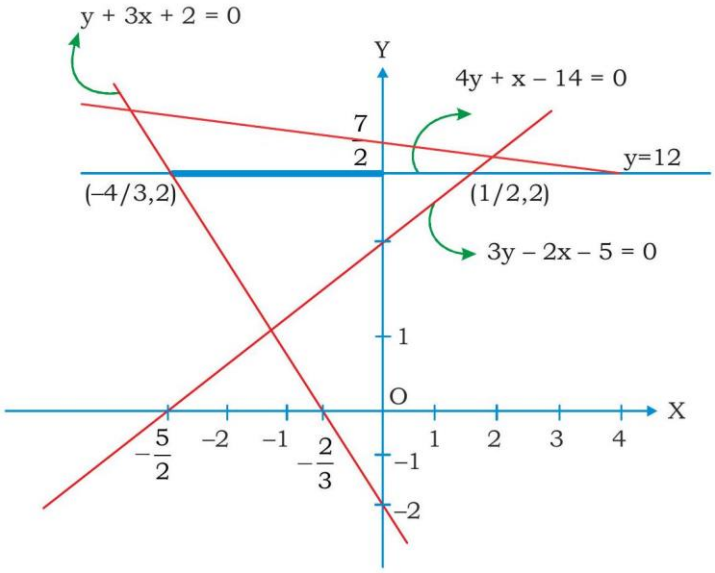
28. The equation of the lines along OA, OB, and AB are $y=0, x=0$, and $x+y=\sqrt{3/2}$, respectively. Now, P and B will lie on the same side of $x=0$ if $\cos\theta>0$. Similarly, P and A will lie on the same side of $x=0$ if $\sin\theta>0$, and P and O will lie on the same side of $x+y=\sqrt{\frac{3}{2}}$ if $\sin\theta+\cos\theta<\sqrt{\frac{3}{2}}$. Hence, P will lie inside $\triangle ABC$ if $\sin\theta>0,\cos\theta>0$, and $\sin\theta+\cos\theta<\sqrt{\frac{3}{2}}$. Now, $\sin\theta+\cos\theta<\sqrt{\frac{3}{2}}$ or $\sin\left(\theta+\frac{\pi}{4}\right)<\sqrt{\frac{3}{4}}$. Since $\sin\theta>0$ and $\cos\theta>0, 0<\theta<\pi/12$ or $5\pi/12<\theta<\pi/2$.

29. a) $a\rightarrow s; b\rightarrow r; c\rightarrow q; d\rightarrow p$.

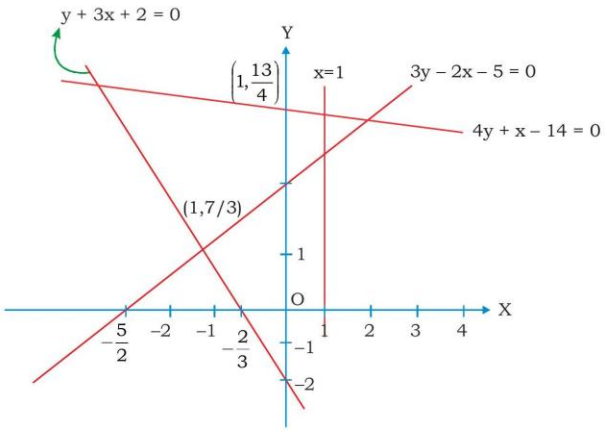


b) Clearly, point $(\alpha,0)$ lies on the x-axis, which is not intersecting any side of triangle. Hence, no such α exists.

c)



d)



30. Sol: Let BE be the angle bisector and CF be the median. Given equations of BE and CF are $x - 4y + 10 = 0$ and $6x + 10y - 59 = 0$.

Since image of A with respect to BE lie on BC. If image of A is $L(h,k)$ then

$$\frac{h-3}{1} = \frac{k+1}{-4} = \frac{-2(3+4+10)}{1^2+(-4)^2} = -2 \Rightarrow h = 1, k = 7 \ \therefore L = (1,7)$$

$\because$ F be the mid point of AB. Let $F = (\alpha, \beta)$ then $B = (2\alpha - 3, 2\beta + 1)$

$\because$ B lies on BE, then $(2\alpha - 3) - 4(2\beta + 1) + 10 = 0$

i.e., $2\alpha - 8\beta + 3 = 0$

and F lies on CF, then

$$6\alpha + 10\beta - 59 = 0$$

Solving (1) and (2), we get $\alpha = \frac{13}{2}, \beta = 2$

then $F = \left(\frac{13}{2}, 2\right)$ and $B = (10,5)$.

Equation of AB is $y + 1 = \frac{2+1}{\frac{13}{2}-3}(x - 3)$

$\Rightarrow\ 6x - 7y - 25 = 0.$

31. Equation of BC is $y - 5 = \frac{7-5}{1-10}(x - 10)$

$\Rightarrow\ 2x + 9y - 65 = 0.$

32. $\because$ CA is the family of lines of CB and CF then

$(2x + 9y - 65) + \lambda(6x - 7y - 25) = 0$

it passes through $A(3, -1)$ then $\lambda = \frac{-4}{3}$

from (3), we get the equation of AC is $18x + 13y - 41 = 0.$

7. PAIR OF STRAIGHT LINES

WORKSHEET – 1

LEVEL – I

1) B	2) A	3) C	4) A	5) A

LEVEL – II

1) D	2) B	3) C	4) A	5) A
6) D	7) B	8) C	9) C	10) B
11) D	12) D	13) C	14) A	15) B
16) D	17) A	18) A	19) B	20) B
21) A				

LEVEL – III

22) A, C	23) A	24) A	25) D	26) A
27) a-q; b-r; c-s; d-p		28) A, B, C	29) a-q; b-r; c-s; d-p	
30) A	31) D	32) A, B	33) A, D	

SOLUTIONS

LEVEL – I

1. $6x^2 + 9xy - 4xy - 6y^2 = 0 \Rightarrow 3x(2x + 3y) - 2y(2x + 3y) = 0 \Rightarrow (3x - 2y)(2x + 3y) = 0$

2. $h = -\frac{5}{2}, a = 3, b = 2 \Rightarrow h^2 - ab = +\frac{25}{4} - 6 = \frac{1}{4} > 0 \Rightarrow \therefore$ Real and distinct

3. $\tan\theta = \frac{2\sqrt{4-1}}{2} = \sqrt{3} \Rightarrow \therefore \theta = 60°$

4. $m_1 + m_2 = \frac{-2h}{b} = \frac{-2\times\frac{-5}{2}}{1} = 5$

5. $(2x + y)(x + 2y) = 0 \Rightarrow 2x^2 + 5xy + 2y^2 = 0$

LEVEL – II

1. $\cos^2\theta(y^2 - xy - x^2) = 0 \Rightarrow y^2 - xy - x^2 = 0$

$$\cos\theta = \frac{|a + b|}{\sqrt{(a - b)^2 + 4h^2}} = \frac{|-1 + 1|}{\sqrt{(-1 - 1)^2 + 1}} = 0 \Rightarrow \theta = \frac{\pi}{2}$$

2. $\frac{|ax_1^2 + 2hx_1y_1 + by_1^2|}{\sqrt{(a-b)^2 + 4h^2}} = \frac{3}{2} \Rightarrow \frac{|1 - 4k + k^2|}{\sqrt{16}} = \frac{3}{2} \Rightarrow |k^2 - 4k + 1| = 6 \Rightarrow k^2 - 4k + 1 = \pm 6$

$\Rightarrow k^2 - 4k - 5 = 0$ or $k^2 - 4k + 7 = 0 \Rightarrow k = -1$ or $5 \therefore k = 5$

3. $\sin\theta = 2\sqrt{\frac{h^2 - ab}{(a-b)^2 + 4h^2}}$

$$= 2\sqrt{\frac{\frac{9}{4}+4}{(1+4)^2+4\times\frac{9}{4}}} = 2\sqrt{\frac{25}{100+36}} = 2\sqrt{\frac{25}{4\times 34}} = 2\frac{5}{2\times\sqrt{34}} = \frac{5}{\sqrt{34}}$$

4. $bx^2 - 2hxy + ay^2 = 0 \Rightarrow 4x^2 - 5xy + 2y^2 = 0$

5. $\frac{|1^2-m^2|}{\sqrt{(1-(-1))^2}} = \frac{|1^2-m^2|}{2}$ 6. $\frac{n^2\sqrt{h^2-ab}}{|am^2-2hlm+bl^2|} = \frac{1\sqrt{4-1}}{|1-4+1|} = \frac{\sqrt{3}}{2}$

6. Combined form of co-ordinate axes is $xy = 0$,
   on comparing with standard form we have a = 0, h = 1/2, b = 0
   Combined equation of angle bisectors is $h(x^2 - y^2) = (a - b)xy$
   $\Rightarrow x^2 - y^2 = 0 \Rightarrow x + y = 0$ (or) $x - y = 0 \Rightarrow x \pm y = 0$

8. $m_1 + m_2 = \frac{-2h}{b} = \frac{2c}{-7}$ and $m_1 m_2 = \frac{-1}{7} \Rightarrow \frac{-2c}{7} = 4 \times \frac{-1}{7} \ \therefore c = 2$

9. $(6x^2 - xy - 4cy^2) = (3x + 4y)(2x - cy) = 6x^2 - (3c - 8)xy - 4cy^2 \Rightarrow 3c - 8 = 1 \Rightarrow c = 3$

10. $3x^2 - 2xy - 8y^2 = (x - 2y)(3x + 4y) \Rightarrow m_1 = \frac{1}{2}$ and $m_2 = \frac{-3}{4}$
    given $lx + y + 4 = 0 \Rightarrow m_3 = -l \Rightarrow m_1 m_3 = -1 \Rightarrow \frac{1}{2}(-l) = -1 \Rightarrow l = 2$
    $m_2 m_3 = -1 \Rightarrow \frac{-3}{4}(-l) = -1 \Rightarrow l = -\frac{4}{3}$

11. $h(x^2 - y^2) = xy(a - b) \Rightarrow \frac{-3}{2}(x^2 - y^2) = xy(2 - 1) \Rightarrow 3x^2 + 2xy - 3y^2 = 0$

12. $(\tan\alpha - 1)x^2 + 2xy\tan\alpha + (\tan\alpha - \tan^2\alpha)y^2 = 0$
    $a + b = 0 \Rightarrow \tan\alpha - 1 + \tan\alpha - \tan^2\alpha = 0 \Rightarrow 2\tan\alpha = 1 + \tan^2\alpha$
    $\frac{2\tan\alpha}{1+\tan^2\alpha} = 1 \Rightarrow \sin 2\alpha = 1 \Rightarrow 2\alpha == \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4}$

13. $2x^2 + 2xy + xy + y^2 = 0 \Rightarrow (x + y)(2x + y) = 0 \Rightarrow y = -x$ and $y = -2x$
    $\Rightarrow \tan\theta_1 = -1 \tan\theta_2 = -2 \Rightarrow \tan(\theta_1 - \theta_2) = \frac{\tan\theta_1 - \tan\theta_2}{1 + \tan\theta_1\tan\theta_2} = \frac{-1+2}{1+2} = \frac{1}{3}$

14. $\frac{n^2\sqrt{(h^2-ab)}}{|(am^2-2hlm+b1^2)|} = \frac{k^2\sqrt{1+24}}{|3+4-32|} = 5 \Rightarrow \frac{5k^2}{25} = 5 \Rightarrow k^2 = 25 \Rightarrow k = 5$

15. The condition that $ax^2 + 2hxy + by^2 = 0$ to form an isosceles triangle with $lx + my + n = 0$ is $h(1^2 - m^2) = (a - b)lm$

16. $[y - \tan 30x][y - \tan 60°x] = 0 \Rightarrow \left[y - \frac{x}{\sqrt{3}}\right][y - \sqrt{3}x] = 0$
    $y^2 - \sqrt{3}xy - \frac{xy}{\sqrt{3}} + x^2 = 0 \Rightarrow (x^2 + y^2)\sqrt{3} = 4xy$

17. Given $ax^2 - 2hxy + by^2 = 0$ and $y = mx$
    $\Rightarrow m_1 + m_2 = \frac{2h}{b} \ldots . (1),\ m_1 m_2 = \frac{a}{b} \ldots \ldots$

$\Rightarrow m = \tan\theta, m_1 = \tan\theta_1, m_2 = \tan\theta_2 \Rightarrow \theta_1 - \theta = \theta - \theta_2 \Rightarrow \theta_1 + \theta_2 = 2\theta$

$\tan(\theta_1 + \theta_2) = \tan 2\theta \Rightarrow \dfrac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1\tan\theta_2} = \dfrac{2\tan\theta}{1 - \tan^2\theta} \Rightarrow \dfrac{2h/b}{1 - a/b} = \dfrac{2m}{1 - m^2}$

$h(1 - m^2) + m(a - b) = 0$

18. $\mathrm{m} + 2\,\mathrm{m} = \dfrac{-2\times\frac{1}{\mathrm{h}}}{\frac{1}{\mathrm{b}}} = \dfrac{-2\,\mathrm{b}}{\mathrm{h}}$, $\mathrm{m} = \dfrac{-2\,\mathrm{b}}{3\,\mathrm{h}}$

$2\,\mathrm{m}^2 = \dfrac{\frac{1}{\mathrm{a}}}{\frac{1}{\mathrm{b}}} = \dfrac{\mathrm{b}}{\mathrm{a}} \Rightarrow 2 \times \dfrac{4\,\mathrm{b}^2}{9\,\mathrm{h}^2} = \dfrac{\mathrm{b}}{\mathrm{a}} \Rightarrow 8\mathrm{ab} = 9\,\mathrm{h}^2 \Rightarrow \mathrm{ab}:\mathrm{h}^2 = 9:8$

19. Let y = mx be a common to the given pairs of lines

$\mathrm{m}^2 + 2\,\mathrm{m} + \mathrm{a} = 0$ and $\mathrm{am}^2 + 2\,\mathrm{m} + 1 = 0 \Rightarrow \dfrac{\mathrm{m}^2}{2(1-\mathrm{a})} = \dfrac{\mathrm{m}}{\mathrm{a}^2 - 1} = \dfrac{1}{2(1-\mathrm{a})}$

$\Rightarrow \mathrm{m}^2 = 1$ and $\mathrm{m} = -\dfrac{\mathrm{a}+1}{2} \Rightarrow (\mathrm{a}+1)^2 = 4 \Rightarrow \mathrm{a} = 1$ or $-3 \;\therefore \mathrm{a} = -3$

$\mathrm{ax}^2 + 2\mathrm{xy} + \mathrm{y}^2 = -3\mathrm{x}^2 + 2\mathrm{xy} + \mathrm{y}^2 = 0 \Rightarrow (\mathrm{x} - \mathrm{y})(3\mathrm{x} + \mathrm{y}) = 0$

$\mathrm{x}^2 + 2\mathrm{xy} + \mathrm{ay}^2 = 0 \Rightarrow \mathrm{x}^2 + 2\mathrm{xy} - 3\mathrm{y}^2 = 0 \Rightarrow (\mathrm{x} - \mathrm{y})(\mathrm{x} + 3\mathrm{y}) = 0$

Therefore joint equation is $(3x + y)(x + 3y) = 0 \Rightarrow 3x^2 + 10xy + 3y^2 = 0$

20. $x^2 - y^2 = 0\; x + 2y = 2 \Rightarrow \dfrac{4\sqrt{1}}{|4-1|} = \dfrac{4}{3}$ sq.units

21. $\mathrm{h}^2 > \mathrm{ab} \Rightarrow 1^2 > 4\mathrm{a}^2 \Rightarrow \mathrm{a}^2 < \frac{1}{4} \Rightarrow -\frac{1}{2} < \mathrm{a} < \frac{1}{2}$

LEVEL – III

22. $6x^2 - xy + 4cy^2 = (3x + 4y)(kx - 3y) = 3kx^2 - 9xy + 4kxy - 12y^2$

$\therefore 9 - 4\mathrm{k} = 1 \Rightarrow \mathrm{k} = 2; 4\mathrm{c} = -12 \Rightarrow \mathrm{c} = -3$

23. $\cos\theta = \dfrac{|6-6|}{\sqrt{\frac{25}{4}}} = 0 \Rightarrow \theta = \dfrac{\pi}{2}$

24. coefficient x^2 + coefficient of $\mathrm{y}^2 = \sin^2\theta + \cos^2\theta - 1 = 0 \Rightarrow\therefore \theta = \dfrac{\pi}{2}$

25. $\sin^2\theta - 1 + \cos^2\theta \Rightarrow 1 - 1 = 0 \Rightarrow\therefore \theta = \dfrac{\pi}{2}$

26. $a + b = 0 \Rightarrow 6 - k = 0 \Rightarrow k = 6$

a) $\mathrm{m}_1 + \mathrm{m}_2 = \dfrac{-2\,\mathrm{h}}{\mathrm{b}}$

b) $\dfrac{\mathrm{m}_1 + \mathrm{m}_2}{\mathrm{m}_1\,\mathrm{m}_2} = \dfrac{\frac{-2\,\mathrm{h}}{\mathrm{b}}}{\frac{\mathrm{a}}{\mathrm{b}}} = \dfrac{-2\,\mathrm{h}}{\mathrm{a}}$

c) $\mathrm{m}_1\,\mathrm{m}_2 = \dfrac{\mathrm{a}}{\mathrm{b}}$

d) $(\mathrm{m}_1 - \mathrm{m}_2)^2 = (\mathrm{m}_1 + \mathrm{m}_2)^2 - 4\,\mathrm{m}_1\,\mathrm{m}_2 = 4\left[\dfrac{\mathrm{h}^2 - \mathrm{ab}}{\mathrm{b}^2}\right] \Rightarrow \mathrm{m}_1 - \mathrm{m}_2 = \dfrac{2\sqrt{\mathrm{h}^2 - \mathrm{ab}}}{\mathrm{b}}$

28. $\mathrm{m}_1 + \mathrm{m}_2 = \dfrac{-8}{3}\;\; \mathrm{m}_1\,\mathrm{m}_2 = -1 \Rightarrow \mathrm{m}_1 - \mathrm{m}_2 = \dfrac{2\sqrt{16+9}}{-3} = \dfrac{-10}{3}$

29. a) $3x^2 - 5xy + 2y^2 = 0\ a = 3, h = \frac{-5}{2}, b = 2 \Rightarrow h(x^2 - y^2) = (a - b)xy$

$\frac{-5}{2}(x^2 - y^2) = (3 - 2)xy \Rightarrow 5x^2 + 2xy - 5y^2 = 0$

b) $x^2 + 4xy + y^2 = 0\ a = 1, b = 1, h = 2 \Rightarrow 2(x^2 - y^2) = (1 - 1)xy \Rightarrow x^2 - y^2 = 0$

c) $2x^2 + 6xy + y^2 = 0\ a = 2, h = 3, b = 1 \Rightarrow 3(x^2 - y^2) = (2 - 1)xy \Rightarrow 3x^2 - xy - 3y^2 = 0$

d) $6x^2 + 5xy - 6y^2 = 0\ a = 6, b = -6, h = \frac{5}{2}$

$\frac{5}{2}(x^2 - y^2) = (6 + 6)xy \Rightarrow 5x^2 - 24xy - 5y^2 = 0$

30. Concept

31. Equations AB, BC, CA are $3x + 4y - 6 = 0$

$12x - 5y - 3 = 0 \ldots.(2), 4x - 3y + 12 = 0$

solving (1) and (2) coordinates of B are $\left(\frac{2}{3}, 1\right)$

solving (2) and (3) coordinates of C are $\left(\frac{69}{16}, \frac{39}{4}\right)$

equation of the bisector of (A between 1 and 2 are

$$\frac{3x + 4y - 6}{\sqrt{3^2 + 4^2}} = \pm\frac{4x - 3y + 12}{\sqrt{4^2 + (-3)^2}} \text{ ie } 3x + 4y - 6 = \pm(4x - 3y + 12)$$

$\Rightarrow$ taking + in $4 \rightarrow x - 7y + 18 = 0$_(5)

taking − in $4 \rightarrow 7x + y + 6 = 0$ ...

Sub B in (5) it is $\frac{35}{3}$ which is + ve $\Rightarrow$ Sub C in (5) it is $\frac{-735}{16}$ which is - ve

B and C lie on opposite side of (5) $\Rightarrow\therefore$ x − 7y + 18 = 0 is the internal bisector [A

32. $y^2 + xy - 12x^2 = 0 \Rightarrow (y - 3x)(y + 4x) = 0$

Case-I : $y - 3x = 0 \Rightarrow y = 3x$

If $y = 3x$ then $ax^2 + 2hx(3x) + b(3x)^2 \Rightarrow a = -3(2h + 3b)$

Case-II: $y + 4x = 0 \Rightarrow y = -4x \Rightarrow ax^2 + 2hx(-4x) + b(-4x)^2 = 0$

$ax^2 - 8hxx^2 + 16bxx^2 = 0 \Rightarrow a = 8(h - 2b)$

33. $my(y - mx) + x(y - mx) = 0 \Rightarrow (y - mx)(my + x) = 0$

so, the lines are $y = mx$ or $y = -\frac{1}{m}x$

Bisectors between the lines $xy = 0$ are $y = x$ or $y = -x\ \therefore m = 1$ or $m = -1$

WORKSHEET – 2

LEVEL – I

1) D	2) B	3) C	4) C	5) B
6) A	7) C	8) C	9) C	10) C

LEVEL – II

1) C	2) B	3) D	4) C	5) B
6) A	7) A	8) A	9) B	10) C
11) B	12) D	13) B	14) A	15) C
16) A	17) C	18) B	19) 5	20) D
21) B	22) A			

LEVEL – III

23) A, B	24) A, D	25) A	26) D	27) A
28) D	29) a-r; b-p; c-s; d-q		30) A, B	31) A
32) A	33) A	34) C	35) a-s; b-r; c-q; d-p	
36) a-s; b-r; c-q; d-p				

SOLUTIONS

LEVEL – I

1. $\left(\frac{hf-bg}{ab-h^2},\frac{gh-af}{ab-h^2}\right)=(2,-1)$ 2.

2. $\sqrt{\frac{g^2-ac}{a(a+b)}}=\frac{2}{\sqrt{10}}$ where $a=9, g=9, c=8, b=1$

3. $\Delta=abc+2fgh-af^2-bg^2-ch^2=0\Rightarrow a=\lambda, b=12, g=\frac{5}{2}, f=-8, c=-3, h=-5$

4. $xy-x-y+1=0\Rightarrow x(y-1)-1(y-1)=0\Rightarrow(x-1)(y-1)=0$

   $x=1, y=1, ax+2y-3=0$ are concurrent

5. $2\sqrt{\frac{g^2-ac}{a(a+b)}}=2\sqrt{\frac{f^2-bc}{b(a+b)}}\Rightarrow\sqrt{\frac{g^2-ac}{f^2-bc}}=\sqrt{\frac{a(a+b)}{b(a+b)}}=\sqrt{\frac{a}{b}}$

6. [1]

7. Combined equation of the lines $x=a$ and $y=b$

8. [3]

9. $a+b=0$

10. $\cos\theta=\cos 60^\circ=\frac{1}{2}$

LEVEL – II

1. $x^2-6x+8=0\Rightarrow(x-2)(x-4)=0\Rightarrow x-2=0: x-4=0$

   $y^2-5y+6=0\Rightarrow(y-2)(y-3)=0\Rightarrow y-2=0; y-3=0$

   $x-2=0; x-4=0$ are vertical lines and the distance between them is 2 $y-2=0; y-3=0$ are horizontal lines and the distance between them is 1

   ∴ Given lines form a rectangle

2. Substitute $(1,k)$ in $2x^2+5xy+3y^2+6x+7y+4=0\Rightarrow k=-2$

3. $\begin{vmatrix} 0 & 2 & 1 \\ 2 & 0 & k \\ 1 & k & 3 \end{vmatrix} = 0 \Rightarrow -2(6-k)+(2k-0)=0 \Rightarrow k=3$

4. $x^2+y^2-\frac{4(\sqrt{3}x+y)^2}{4}=0 \Rightarrow 2x^2+2\sqrt{3}xy=0 \ \therefore \cos\theta=\cos\frac{\pi}{3} \Rightarrow \theta=\frac{\pi}{3}$

5. $a(x-\alpha)^2+2h(x-\alpha)(y-\beta)+b(y-\beta)^2=0$

   $(x+2)^2+4(x+2)(y-3)+(y-3)^2=0 \Rightarrow x^2+4xy+y^2-8x+2y-11=0$

6. $b(x-\alpha)^2-2\,h(x-\alpha)(y-\beta)+a(y-\beta)^2=0 \Rightarrow -2(x-1)^2+(x-1)(y+1)+(y+1)^2=0$ $-2x^2-2+4x+xy+x-y-1+y^2+1+2y=0 \Rightarrow 2x^2-xy-y^2-5x-y+2=0$

7. $9x^2-24xy+16y^2-12x+16y-12=0 \Rightarrow a=9, b=16, h=-12, g=-6, f=8, c=-12$

   $2\sqrt{\frac{g^2-ac}{a(a+b)}}=2\sqrt{\frac{36+108}{225}}=\frac{8}{5}$

8. Substitute $y=0$ in $ax^2+2hxy+by^2+2gx+2fy+c=0 \Rightarrow ax^2+2gx+c=0$

   equation (1) roots are real $\Rightarrow \Delta \geq 0 \Rightarrow 4\,g^2-4ac=0 \Rightarrow g^2=ac$

9. $bx^2-2hxy+ay^2=0 \Rightarrow 2x^2-5xy+2y^2=0$

10. $h[(x-\alpha)^2-(y-\beta)^2]=(a-b)(x-\alpha)(y-\beta)$

    $x(y+1)+1(y+1)=0 \Rightarrow (x+1)(y+1)=0 \ \therefore (\alpha,\beta)=(-1,-1)$

    $\frac{1}{2}[(x+1)^2-(y+1)^2]=0 \Rightarrow x^2+1+2\,h-y^2-1-2y=0 \Rightarrow x^2-y^2+2x-2y=0$

11. Given equation represents two parallel pairs of lines $a+b\neq 0$

    and $(a-b)fg+h(f^2-g^2)=0+\frac{5}{2}\left[\left(\frac{-15}{2}\right)^2-\left(\frac{-15}{2}\right)^2\right]=0$

12. 4

    $\therefore$ Given lines form a Rhombus

13. $x^2+2y^2-4(x+y)^2=0 \Rightarrow x^2+2y^2-4x^2-4y^2-8xy=0 \Rightarrow 3x^2+2y^2+8xy=0$

14. $2x^2+3xy-4x(3x+y)+(3x+y)^2=0$

    $2x^2+3xy-12x^2-4xy+9x^2+y^2+6xy=0 \Rightarrow x^2-5xy-y^2=0$

15. Equation of Y-axis is $x=0. \Rightarrow b^2+2fy+c=0 \Rightarrow y=\frac{-2f\pm\sqrt{4f^2-4bc}}{2b}$

    $\Rightarrow A=\left(0,\frac{-f+\sqrt{f^2-bc}}{b}\right);\ B=\left(0,\frac{-f-\sqrt{f^2-bc}}{b}\right)$

    AB = Length of the intercept $=|y_1-y_2|=\frac{2\sqrt{f^2-bc}}{|b|}$

16. Given pair of lines $ax^2+2hxy+by^2+2gx+2fy+c=0$

    equation to X-axis is $y=0$ solving we get $ax^2+2gx+c=0$

equation (1) equal roots $\Rightarrow 4\,g^2 - 4ac = 0 \Rightarrow g^2 = ac$

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \Rightarrow abc + 2fgh - af^2 - abc - ch^2 = 0 \Rightarrow 2fgh = af^2 + c^2$

17. $x^2(c^2 - a^2m^2) - 2ma^2xy + y^2(c^2 - a^2) = 0$ as angle is 90°

$a + b = 0 \Rightarrow c^2 - a^2\,m^2 + c^2 - a^2 = 0 \Rightarrow 2c^2 = a^2(1 + m^2)$

18. $2xy + gx + fy + 5 = 0$,

On comparing with standard form $a = 0, b = 0, h = 1, g = \frac{g}{2}, f = \frac{f}{2}, c = 5$

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \Rightarrow 0 + 2\frac{f}{2} \times \frac{g}{2} - 0 - 0 - 5(1)^2 = 0; \frac{2fg}{4} - 5 = 0 \Rightarrow fg = 10$

19. $a = 1, h = 1, b = 1, g = -4a, f = -4a, c = -9a^2, d = 25\sqrt{2}$

$25\sqrt{2} = 2\sqrt{\frac{16a^2 + 9a^2}{2}} \Rightarrow 25\sqrt{2} = 2\sqrt{\frac{25a^2}{2}} \Rightarrow 25\sqrt{2} = \frac{2 \times 5a}{\sqrt{2}} \Rightarrow 50 = 10a \Rightarrow a = 5$

20. $x^2 + y^2 - 4\left(\frac{y-3x}{c}\right)^2 = 0 \Rightarrow c^2x^2 + c^2y^2 - 4y^2 - 36x^2 + 24xy = 0$

$x^2(c^2 - 36) + y^2(c^2 - 4) + 24xy = 0$

coefficient of x^2 + coefficient of $y^2 = 0 \Rightarrow c^2 - 36 + c^2 - 4 = 0 \;\therefore c^2 = 20$

21. Find the point of Intersection of both pairs then find distance

22. $x^2 + y^2 - 2ax\left(\frac{x+y}{b}\right) - 4a^2\left(\frac{x+y}{b}\right)^2 = 0$

coefficient of x^2 + coefficient of $y^2 = 0 \Rightarrow 1 - \frac{2a}{b} - \frac{4a^2}{b^2} + 1 - \frac{4a^2}{b^2} = 0$

$2b^2 - 2ab - 8a^2 = 0 \Rightarrow b(b - a) = 4a^2$

23. $a = 4, b = 1, g = 3, c = -10 \Rightarrow 2\sqrt{\frac{g^2-ac}{a(a+b)}} = 2\sqrt{\frac{9+40}{4(4+1)}} = 2\sqrt{\frac{49}{20}} = 2 \times \frac{7}{2\sqrt{5}} = \frac{7}{\sqrt{5}}$

24. $x^2\left(2 + \frac{2}{k} - \frac{1}{k^2}\right) + xy\left(-2 + \frac{4}{k} - \frac{1}{k} - \frac{4}{k^2}\right) + y^2\left(3 - \frac{2}{k} - \frac{4}{k^2}\right) = 0$

coefficient of x^2 + coefficient of $y^2 = 0$

$2 + \frac{2}{k} - \frac{1}{k^2} + 3 - \frac{2}{k} - \frac{4}{k^2} = 0 \Rightarrow 5k^2 - 5 = 0 \Rightarrow k^2 = 1 \Rightarrow k = \pm 1$

25. $a = 1, b = -1, h = 0, g = 0, f = \frac{k}{2}, c = -a^2 \Rightarrow a^2 + 2 \times 0 \times \frac{k}{2} - \frac{k^2}{4} - 0 - 0 = 0$

$a^2 - \frac{k^2}{4} = 0 \Rightarrow a^2 = \frac{k^2}{4} \Rightarrow k = \pm 2a$

26. $xy + 2x + 2y + 4 = 0 \Rightarrow x(y + 2) + 2(y + 2) = 0$

$(x + 2)(y + 2) = 0 \Rightarrow x = y = -2 \Rightarrow (x, y) = (-2, -2).$

27. $a = 2, h = \frac{5}{2}, b = 2, g = \frac{-17}{2}, f = \frac{-19}{2}, c = 35 \Rightarrow \left(\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right) = (3,1)$

28. (1,2)

29. $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

a) if $a = 0 \Rightarrow 2fgh = bg^2 + ch^2$

b) if $b = 0 \Rightarrow 2fgh = af^2 + ch^2$

c) if $c = 0 \Rightarrow 2fgh = af^2 + bg^2$

d) if $f = 0 \Rightarrow abc = bg^2 + ch^2$

30. $kx^2 - 10xy + 12kyy^2 + \frac{8}{\sqrt{3}}x - 16y - 3 = 0$

$a = k, h = -5,\ b = 12k, g = \frac{4}{\sqrt{3}}, f = -8, c = -3$ Use $h^2 = ab$

31. $g = 3, a = 2, c = 1 \Rightarrow \frac{2\sqrt{9-2}}{2} = \sqrt{7}$

32. $\frac{1}{\sqrt{2}} = \frac{2+3}{\sqrt{(2-3)^2+4\times\frac{\lambda^2}{4}}} \Rightarrow \frac{1}{\sqrt{2}} = \frac{5}{\sqrt{1+\lambda^2}} \Rightarrow 1 + \lambda^2 = 50 \ \therefore \lambda^2 = 49 \Rightarrow \lambda = \pm 7$

33. $\Delta = 0$ from this we get $p = 6$

$$\sin\theta = 2\sqrt{\frac{h^2 - ab}{(a-b) + 4h^2}} = 2\sqrt{\frac{\frac{25}{4} - 6}{(1-6)^2 + \frac{100}{4}}} = 2\sqrt{\frac{1}{200}} = \frac{1}{\sqrt{50}}$$

34. $\Delta = 0$ from this we get $\lambda = 2$

$$\sin\theta = 2\sqrt{\frac{h^2 - ab}{(a-b) + 4h^2}} = 2\sqrt{\frac{\frac{9}{4} - 2}{(1-2)^2 + 4\times\frac{9}{4}}} = 2\sqrt{\frac{1}{40}} = \frac{1}{\sqrt{10}}$$

$\Rightarrow \operatorname{cosec}\theta = \sqrt{10} \Rightarrow \operatorname{cosec}^2\theta = 10$

35. Conceptual

36. Conceptual a-s; $-r; c - q; d - p$